Unchanging Core Syntax: Gender in Heritage Tamil

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Unchanging Core Syntax: Gender in Heritage Tamil

by

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Madhusmitha Venkatesan

Abstract

Heritage languages are sites of rapid language change, as evidenced by the innovations observed in multiple grammatical domains such as inflectional morphology, case marking, discourse phenomena, etc. However, one domain forms an exception to this by exhibiting stability - core syntax. Ongoing research in the field is focused on defining the bounds of what constitutes core syntax for heritage grammars. This dissertation takes the idea forward by placing core syntax in a grammatical domain that has been established as vulnerable across language acquisition contexts - gender features and gender agreement.

Such a juxtaposition of the reported stability of core syntax and the volatility of gender features in heritage grammars allows this dissertation to explore the ramifications for the structural representation of gender features in heritage grammars. The question is addressed in three ways: (i) Gender assignment to nouns (ii) Dependent marking of gender, or gender agreement (iii) Gender features interacting with structural aspects like possession.

The empirical core of the dissertation is formed by Tamil (Dravidian) acquired as a heritage language in New Delhi. Through a series of picture-description and sentence elicitation tasks, this dissertation examines if heritage Tamil converges to or diverges from the baseline in terms of the structural representations of gender in these domains.

In the domain of gender assignment to nouns, the approach adopted to gender is a configurational one; this dissertation shows that gender features encode representational differences in terms of their location (lexical vs functional heads) and assignment strategies (inherent vs discourse-obtained). Novel empirical evidence from heritage Tamil points towards sensitivity towards these: lexical gender assignment remains stable whereas gender assignment that involves discourse elements undergoes variation in heritage Tamil.

In the domain of gender agreement in adjectives, the dissertation first lays out the structural representation of adjectives, which are also configurational in Tamil. The structure of adjectives requires the application of two mutually exclusive strategies of adjective formation, thus requiring intensive, case-by-case learning. Further, predicative adjectives, to the exclusion of attributive adjectives, participate in agreement relations with nouns. Data from heritage Tamil shows two broad trends: the mechanisms underlying gender agreement on adjectives remain stable, whereas adjective formation shows variation.

The third domain explored in this dissertation is gender and its interaction with possessives. Nouns carrying gender features are disallowed from occurring in the configuration of predicative possession. This constraint is explained as a result of the functional heads involved in possession and their incompatibility with gender features in Tamil. Evidence from heritage Tamil shows that the core structural properties of gender remain stable. The structural constraint is retained in heritage Tamil.

To conclude, this dissertation identifies three grammatical domains where the core-syntactic aspects of gender features remain stable in heritage Tamil. The broad implication of these findings is that we can argue for syntax emerging as a resilient grammatical phenomenon despite the presence of reduced/ restricted primary linguistic data and potential transfer effects from other languages in the environment.

सारांश

हेरिटेज भाषाओं में व्याकरण का बहुत शीघ्रता से बदलाव देखा जाता है जैसे कि -इन्फ्लेक्शनल मॉर्फोलॉजी, केस मार्किंग, डिस्कोर्स फेनोमेना। इसके विपरीत, एक अनुक्षेत्र है जिसमे स्थिरता देखी जाती है - कोर सिंटैक्स। हेरिटेज भाषाओं में चल रही रिसर्च इसको परिभाषित करने पर केंद्रित है कि हेरिटेज ग्रामर के लिए कोर सिंटेक्स क्या है। इस शोध प्रबंध में कोर सिंटेक्स को एक ऐसे ग्रामर के क्षेत्र में रखा गया है जिसे भाषा सीखने के संदर्भ में बदलाव के आधीन माना जाता है - जेंडर फीचर्स और जेंडर अग्रीमेंट।

कोर सिंटैक्स की स्थिरता और जेंडर फीचर्स की अस्थिरता के संयोजन का प्रयोग करके यह शोध प्रबंध हेरिटेज ग्रामर में जेंडर फीचर्स के संरचनात्मक प्रतिनिधित्व का परीक्षण करता है। इस प्रश्न को तीन दृष्टिकोण से देखा गया है: (i) संज्ञाओं का जेंडर असाइनमेंट (ii) जेंडर अग्रीमेंट (iii) जेंडर फीचर्स का अधिकार जैसे संरचनात्मक पहलुओं पर परस्पर प्रभाव

इस शोध प्रबंध का अनुभवजन्य भाग नई दिल्ली में हेरिटेज भाषा के रूप में सीखी गई तमिल (द्रविड़) से बना है। चित्रों के वर्णन करने और वाक्य बनाने के माध्यम से, यह शोध प्रबंध जाँच करता है कि क्या हेरिटेज तमिल इन अनुक्षेत्रों में जेंडर के संरचनात्मक प्रतिनिधित्व के मामले में बेसलाइन जैसी है या उससे विपरीत है।

इस शोध प्रबंध में जेंडर असाइनमेंट के लिए कॉन्फ़िगरेशनल एप्रोच अपनाया गया है। यह देखा गया है कि जेंडर फीचर्स अपने स्थान (लेक्सिकल बनाम फंक्शनल हेड्स) और असाइनमेंट रणनीतियों (आंतरिक बनाम डिस्कोर्स से प्राप्त) के सन्दर्भ में पृथक हैं। हेरिटेज तमिल से आते हुए नए अनुभवजन्य प्रमाण इन बातों के प्रति संवेदनशीलता की ओर इशारा करते है कि लेक्सिकल जेंडर असाइनमेंट स्थिर रहता है और डिस्कोर्स तत्वों वाले जेंडर असाइनमेंट में हेरिटेज तमिल में बदलाव आता है।

विशेषण में जेंडर एग्रीमेंट के क्षेत्र में यह शोध प्रबंध पहले विशेषण के संरचनात्मक प्रतिनिधित्व का वर्णन करता है। इसके अतिरिक्त , विशेषणों की संरचना के लिए विशेषण निर्माण की दो रणनीतियों को विषयानुसार सीखने की आवश्यकता होती है। विशेषणात्मक विशेषण में जेंडर एग्रीमेंट नहीं पाया जाता है लेकिन विधेय विशेषणों में एग्रीमेंट पाया जाता है। हेरिटेज तमिल का डेटा दो व्यापक रुझान दिखाता है: विशेषणों पर जेंडर एग्रीमेंट के अंतर्निहित तंत्र स्थिर रहते हैं, जबकि विशेषण निर्माण में भिन्नता दिखाई देती है।

इस शोध प्रबंध में जेंडर फीचर्स और उनका अधिकारवाचक शब्दों के साथ संबंध भी देखा गया है। जिन संज्ञा में जेंडर फीचर्स होते हैं, वे विधेय अधिकार के विन्यास में उपयोग नहीं किये जाते हैं। इस प्रतिबन्ध को अधिकार में शामिल फंक्शनल हेड्स और तमिल में जेंडर फीचर्स के साथ उनकी असंगति के रूप में समझाया गया है। हेरिटेज तमिल के डेटा से पता चलता है कि जेंडर की मुख्य स्ट्रक्चरल प्रॉपर्टीज़ स्थिर रहती हैं।

निष्कर्ष के तौर पर, यह शोध प्रबंध तीन ग्रामेटिकल डोमेन की पहचान कराता जिसमे हेरिटेज तमिल में जेंडर फीचर्स के मुख्य सिंटैक्टिक पहलू स्थिर रहते हैं। इसके आधार पर हम यह तर्क दे सकते हैं कि भाषा अधिग्रहण के प्रसंग में सीमित प्राइमरी लिंगविस्टिक डेटा और दूसरी भाषाओं से संभावित ट्रांसफर प्रभावों के होते हुए भी सिंटैक्स सुदृढ़ रहता है।

Contents

List of Figures

List of Tables

List of Abbreviations

ACC	Accusative Case
ADJR	Adjectivizer
CLI	Cross-Linguistic Influence
COP	Copula
CPM	Conjunctive Participial Marker
DAT	Dative Case
ERG	Ergative Case
FSG	Feminine Singular
FUT	Future
GEN	Genitive Case
HL	Heritage Language
INF	Infinitive
INTF	Intensifier
LOC	Locative Case
MSG	Masculine Singular
NEG	Negation
NMLZ	Nominalizer
NOM	Nominative Case
PERF	Perfective Aspect
PL	Plural
PLD	Primary Linguistic Data
POSS	Possessive
PST	Past
REL	Relativizer

Chapter 1

Introduction

1.1 Introduction

Heritage languages (HL) are acquired under asymmetrically bilingual contexts, such as emigrant families when surrounded by a different language in the larger society (Polinsky, 2018). HLs are sites of rapid and systematic language change as they arise from situations of continuous language contact (Kupisch and Polinsky, 2022). There is empirical evidence from several grammatical domains such as inflectional morphology, case systems, etc. (Polinsky, 2018; Montrul, 2010; Rothman, 2009) illustrating variation and change in heritage grammars. However, there is a domain of grammar that is striking in its resistance to change: core syntax.

Polinsky (2018, p. 288) notes that core syntax (operationalized as knowledge of A and A-bar dependencies, binding, word order, etc.) remains fairly stable in heritage grammars; these are areas where heritage grammars show minimal variation from their baseline counterparts. This idea continues to find acceptance in more recent accounts of heritage language syntax (D'Alessandro et al., 2025; Lohndal and Putnam, 2024; Lohndal, 2021; Lohndal et al., 2019). Such a distinction between core syntax and other parts of grammar has been a fruitful one, and it has yielded consistent empirical attestations across different language combinations. However, there is a caveat here: The studies mentioned above all use different definitions of syntax, especially to distinguish it from morpho-syntax. One reliable metric used to define the bounds of core syntax is based on Chomsky (1981), based on which core syntax is taken to mean properties that are part of Universal Grammar, i.e., syntactic features and, more recently, operations following Chomsky (2000, 2001) (Merge, Agree, Spell-Out). Nevertheless, a concrete definition of core syntax is yet to be reached despite this useful guideline.

This dissertation aims to contribute to the discussion by starting out from the idea that core syntax does not undergo change. While it is difficult to define exactly what core syntax is, HLs can be of help here. By identifying more grammatical domains that remain stable (and resist change) in HL bilingualism, we can get closer to an appraisal of what constitutes core syntax. The empirical testing ground employed is gender in heritage Tamil (Dravidian). There are two major reasons for choosing gender as the phenomenon to observe resilience and change.

- Across different languages as well as modules—processing, acquisition, diachronic development—gender has been established as a vulnerable grammatical category (Igartua, 2019; Kramer, 2014; Hulk and van der Linden, 2010). HL studies are no exception. They too show that gender is more prone to variation in asymmetrically bilingual grammmars (Polinsky, 2018; Lohndal and Westergaard, 2016). Given such a setup, this dissertation seeks to look deeper into the internal structure of gender in order to uncover if there are aspects to it that remain stable relative to others that lend themselves to change easily. In other words, such an exercise would mark a shift from considering gender a monolithic category. Instead, we present a more nuanced understanding of its underlying representation. Empirically this can be translated as facets of gender that remain stable in heritage Tamil in accordance with the axiom that core syntax is less prone to change.

- The languages studied so far in HL syntax canon are those with grammatical/ arbitrary gender in them - Russian (Laleko, 2018), Norwegian (Lohndal and Westergaard, 2016), Italian (Bianchi, 2013) - a learning-intensive domain. The central discussion in these texts is therefore mostly concerned with the feasibility of acquiring such an input-dependent phenomenon in a HL context, typically characterized by impoverished linguistic input. As a consequence, the empirical questions raised in these focus on whether different gender values ([MASC], [FEM], [NEUT]) have been learnt appropriately in comparison to the baseline. While conclusions drawn in this regard inform the theory about the nature of learning gender values in an atypical acquisition context, they do not directly address the structural representation of gender features. By electing Tamil as the empirical core of this dissertation, we are able to access a more informative vantage point. Tamil has only semantic/ biological gender (Corbett, 1991). There are two major implications of a language having a fully semantic system of gender assignment: (i) Gender assignment is not arbitrary in Tamil, requiring case-by-case learning. It is rooted in certain semantic properties of the noun, making its learning rule-derivable. (ii) As with all semantic systems of gender assignment, gender in Tamil crucially hinges on a connection with the [+/- HUMAN] feature. Such a connection is a part of the underlying representation of gender in this grammar - arguably a part of its core syntactic structure. As such, it is reasonable to expect this feature to resist change in HL bilingualism. A combination of (i) and (ii) essentially means that Tamil provides a novel empirical ground to study the structural properties of gender in a HL context.

Thus, semantic/ biological gender in Tamil and its representations in heritage Tamil will be the main focus of this dissertation. There are three major ways in which we can break this objective down. Each of the points mentioned below feeds into a chapter of the dissertation, and is explained in greater detail in Section 1.6.1.

Gender assignment to nouns: Nouns carry the first-order information about gender in any grammar. In this regard, we focus on the various types of gender features found on nouns and the mechanisms by which they are assigned, resulting in their ultimate structural representation.

Dependent marking of gender: Gender values carried on nouns are also expressed on certain functional heads in the structure, as a reflex of agreement relations. This dissertation takes up the case of how gender agreement on predicative adjectives is represented in Tamil.

Gender and its interaction with other elements: There are principled ways in which gender features interact with the structure. This could manifest as the presence of gender features in a noun precluding the noun from occurring in certain structural positions. This dissertation uses the domain of possessives to illustrate a constraint where gendered nouns are disallowed in definite predicative possession.

Overall, this dissertation aims to understand how the grammatical construct of gender develops when Tamil is acquired as a heritage language with Hindi-Urdu (Indo-Aryan) as L2. The following sections offer a detailed overview of heritage languages, gender as a feature in grammar, and insights into the stability and variability of gender in heritage grammars. This will pave the way for a discussion of the relevant empirical facts of the Tamil gender system, and how the dissertation investigates each of its facets. This chapter also provides an overview into the population studied in this dissertation: heritage speakers of Tamil in New Delhi, India. Following this, Section 1.6 presents the core research questions of the dissertation.

1.2 About Heritage Languages

Heritage languages are the result of a specific type of language contact with a prominent asymmetry between the two (or more) languages involved. A language is defined as a HL when it is acquired naturalistically in a person's home domain or smaller community while the larger, societal language is a different one (Fridman et al., 2024; Montrul, 2023; Lohndal, 2021; Lohndal et al., 2019; Polinsky, 2018, 2015; Rothman, 2009). Typically, migration creates the context for HLs. When the speakers of Language A migrate to a region where Language B is the most-used language, the children of Speaker A will typically grow up as near-simultaneous/sequential bilinguals, speaking Language A at home and with close family while using Language B in all other spheres of their lives, such as formal education, mass media consumption, social exchanges in the public domain, and many other such instances. In this situation, Language A is identified as their HL. As an extension, this population qualifies as Heritage Speakers (HSs henceforth) of Language A.

In terms of the acquisition and development of the grammar of HLs, the linguistic input received by HSs is qualitatively and/or quantitatively less than what is typically available to monolingual speakers of the same language (Montrul, 2023; Polinsky and Scontras, 2020; Meisel, 2020; Scontras et al., 2015). This is a direct consequence of the HL being confined to the home domain. Additionally, the prevalence of the societal language in so many domains of their lives also indicates that, with the passage of time, the societal language will become dominant over the HL, leading to attrition in different domains of their heritage grammars.

There is general consensus within the subfield of HL that heritage grammars diverge from their monolingual baseline counterparts. Over the last two decades, research in this field has been able to characterize the observed divergences into different types, and explain them as arising in different circumstances. Broadly, the idea is that HLs are constrained by the limited input they receive, the relatively lesser usage they are subject to, and potentially heavy influence from the dominant language of the society. In an attempt to adapt to such operating conditions, HLs are expected to undergo systematic restructuring. Predictions about the shape and structure of HLs also stem from the assumption that, since the speakers cannot rely too much on input, the resulting grammar will disprefer any phenomena that necessitate large amounts of lexical learning, idiosyncrasies and irregularities (Aalberse et al., 2019; Polinsky, 2018). With this context in place, we can now proceed to look into the grammatical category of gender, and how it has been explored in HLs.

1.3 Gender and Gender Agreement

Gender is primarily a system of nominal categorization; it is a way of organising nouns into different classes based on some inherent semantic or formal attributes that the items may possess (Audring, 2014; Corbett, 2006, 1991). Several accounts of gender have labeled it a "puzzling" grammatical category (Spencer, 2002; Corbett, 1991). This is because gender is sometimes formal, sometimes semantic and formal, is subjected to considerable cross-linguistic variation, and participates in agreement relations. These aspects of gender make its study essential. Additionally, we conceive of gender as a feature on nouns, but it is also expressed on other lexical as well as functional categories in the form of agreement.

Gender in languages is broadly divided into two kinds: Natural/Biological or semantic gender, and grammatical gender. The former is a system in which meaning plays a central role. Only animate nouns are allotted a gender, and their gender value correlates with the biological sex of the referent (1). Thus, nouns denoting male humans or animals are assigned [MASCULINE] gender, and nouns denoting female humans or animals are assigned [FEMININE] gender. Crucially, these are nouns where the gender value of the noun is grounded in its meaning.

(1) a. *boy*
boy.MSG

b. *girl*
girl.FSG

Grammatical Gender, on the other hand, should be understood as a purely formal feature; meaning does not have a role to play in this case. Grammatical gender may or may not be congruent with natural/biological gender, animacy or other related semantic properties. It is often referred to as arbitrary gender (Corbett, 1991), as neither is there consistency in the assignment of gender to inanimate objects nor do the intrinsic properties of the noun have any role to play in its gender assignment. The lack of consistency in grammatical gender is illustrated in (2) and (3): 'moon' is assigned [MASC] in German and [FEM] in Spanish. Similarly, the lexical item 'sun' is assigned [FEM] in German and [MASC] in Spanish. These examples convey that grammatical gender assignment to nouns is largely independent of any semantic properties of the nouns. They may be idiosyncratic to different language systems, or contingent upon certain morpho-phonological characteristics of the nouns (Corbett, 1991, p. 33).

(2) German

a. *der* *mond*
the.MASC moon.MASC
'the moon'

b. *die* *sonne*
the.FEM sun.FEM
'the sun'

(3) Spanish

a. *la* *luna*
the.FEM moon.FEM
'the moon'

b. *el* *sol*
the.MASC sun.MASC
'the sun'

Another aspect of gender features is that even when present in the system, there is considerable variation in how many different parts of speech gender may be expressed on (Longobardi, 2018). The variation ranges from gender being marked on just the head to being marked on (some or) all dependent markings (Nichols, 1986).

It is crucial to note that regardless of whether the gender feature being marked is grammatical or semantic, it is visible most saliently on the functional heads surrounding the noun, i.e., determiners, adjectives, verbs, etc. In (4a) and (4b), the determiner and predicative adjective inflect for the masculine and feminine gender features of the noun, respectively.

(4) Spanish (Ogneva, 2023)

a. ***el*** *libro* *es amarill**o***
the.MSG book.MSG is yellow.MSG
'The book is yellow.'

b. ***la*** *hoja* *es roj**a***
the.FEM sheet.FEM is red.FEM
'The sheet is red.'

In some languages, particularly Romance, gender cues are morphologically visible on nouns, too; the vowel endings of nouns may reveal the gender values of the noun to a considerable degree (Ritter, 1993). However, generally, nouns in isolation are not entirely sufficient to understand their gender values. Thus, any study of gender must involve a close look at multiple functional heads and, by extension, the structure of the entire phrase/clause where the noun is housed.

According to the World Atlas of Language Structures (WALS), out of the 257 languages surveyed, gender is expressed in 145 of them (Corbett, 2013). Within these "gendered" languages, there is a further divide. Approximately half of them express grammatical gender along with biological gender, and the other half only have natural/biological gender. Another parameter of variation found among languages is the number of categories gender is divided into. A binary organization of [MASC] and [FEM] is the most common pattern, while a minority of languages has a third [NEUTER] category too. Additionally, there are some languages, such as Zulu and Swahili, with five or more gender classes (Corbett, 1991). This brief typological survey serves as a reminder of another aspect of gender systems that makes it interesting for study: the high degree of variation. Gender is one of the few syntactic categories that display cross-linguistic variation in such lengths. In addition to the common binary divide of a feature being present or absent in language X, there are several layers of complexity, all of which are subject to cross-linguistic variation with far-reaching structural ramifications. This dissertation does not address all these points of variation, as that would take us too far afield. Our focus will be on the structural representation of gender and how it changes under language contact.

1.4 Gender in Heritage Languages

The previous section echoes the well-accepted notion that gender is a complex grammatical category. We also introduced HL as languages spoken in situations of bilingualism with an inherent tilt in favour of one of the two languages. The HL, or L1, typically develops under conditions of restricted input and interference effects from the dominant language. Given this setup, we are poised to ask the question: How do we expect a category as complex as gender to develop under the constraints applying on HLs?

Polinsky (2018, p. 65) begins the discussion on gender by stating that it is one of the most vulnerable categories in heritage grammars. This generalization is obtained mostly from agreement studies on a diverse set of heritage languages: Dyirbal (Plaster and Polinsky, 2007), American Norwegian (Lohndal and Westergaard, 2016), Heritage Italian in Germany (Bianchi, 2013), and heritage Spanish (Hur et al., 2020)[1]. These studies cover considerable conceptual as well as empirical ground – they study a wide variety of bilingual contexts as well as multiple agreement relations. A thread of analysis common to all these studies is their focus on gender values. To elaborate, Lohndal and Westergaard (2016) notes that heritage Norwegian overgeneralizes the value [MASC] to nouns which are assigned [FEM] or [NEUT] in the baseline variety. Similarly, Bianchi (2013) reports that heritage Italian speakers produce target-inconsistent structures in gender assignment and agreement. Essentially, they assign [MASC] and [FEM] values to nouns in a manner that varies considerably from baseline Italian. While such conclusions offer insights into the internal restructuring seen within gender, they often sidestep the semantic / grammatical gender divide by grouping together [MASC] of both kinds, or [FEM] of both kinds together. As a consequence, we are unable to garner much about different gender assignment strategies fare in heritage grammars. In other words, it is not possible to make systematic generalizations about semantic gender in comparison to grammatical gender *vis a vis* their structural representation.

There is another set of studies which looks beyond the gender values, and takes the discussion forward into the domain of how gender is assigned to different nouns. Laleko (2018) presents a study on heritage Russian in the US with its focus on how different gender assignment strategies are treated when Russian is acquired as a heritage language. The study concluded that morphological gender marking is fairly stable in heritage Russian, whereas nouns where the gender values of a noun that are not inferable from its morphological shape tend to show more variation from the baseline. However, as all the items used in this study are animate nouns, we are unable to draw comparisons between semantic and formal types of gender here too. Mitrofanova et al. (2022) present a large-scale study on heritage Russian, focusing on the extent to which bilingual children are sensitive to morpho-phonological cues while assigning gender to nouns. The study concludes that bilingual children tend to rely more on such cues than monolingual

[1]The exact research questions, data and analyses provided in these studies are discussed in detail in forthcoming chapters.

ones (who are able to access the formal representation of gender). Additionally, they are able to point out that, while bilingual children do overgeneralize [MASC], they certainly maintain the three-way gender system found in baseline Russian. However, as the study focused solely on grammatical gender assignment, we are unable to extend the findings to other types of gender assignment.

Martinez-Nieto and Adelaida Restrepo (2024) and Pérez-Leroux et al. (2023) similarly present two studies on the expression of gender in heritage Spanish in the US. While these study offer insights into the difference between the modalities of production and comprehension in the domain of gender, they does not make distinctions between different types of gender. Nouns of all types (semantic gender 'girl', grammatical gender 'train') are grouped together for this study, making it nearly impossible to arrive at generalisations about the structural differences found within the category of gender.

A brief survey of the existing literature on gender in heritage grammars immediately leads us to two interconnected facts: Firstly, gender is unanimously identified as a vulnerable grammatical category. This finding is attested across different languages. Regardless of whether L2 also has gender marking, the gender system of L1 invariably seems to undergo variation. Secondly, the perspective adopted in these studies requires some attention. Gender has been understood as a monolithic category in the case studies considered here. While they address different facets of gender system - assignment strategies (Laleko, 2018; Mitrofanova et al., 2022), variation between different gender values (Lohndal and Westergaard, 2016; Bianchi, 2013), they do not comment about finer distinctions in the structural representation of gender. As a result, these studies are unable to check if there are core-syntactic aspects of gender that are more stable relative to others that undergo variation due to lack of sufficient input.

This point is particularly pertinent when we refer back to the idea that not all aspects of the grammar are equally affected when acquired as a HL. Core-syntactic, structural properties tend to be more stable, whereas domains with some idiosyncrasies in them, or phenomena that lie at the interface of two domains, eg. syntax and discourse, tend to undergo change.

In light of this finding, it immediately becomes apparent that the category of gender too should be dissected accordingly. Rather than considering it to be a monolithic entity, we need to focus on its structural underpinnings in an attempt to identify if there are layers in its representation.

In other words, it is important to investigate if there are core and "peripheral" aspects to the phenomenon of gender itself. One way of going about with this task is by differentiating between semantic and grammatical gender. The presence of semantic correlates in the former to the exclusion of the latter tells us that their underlying representations must be different. As an extension, semantic gender features can be generalized based on a rule that maps animate referents to their respective gender values. For instance, a male referent maps onto [MASC] feature. The same cannot be said of grammatical gender. Its arbitrary assignment to nouns means that the gender value of each inanimate noun must be learnt on a case-by-case basis during language acquisition. This property of grammatical gender features makes them heavily reliant on linguistic input in the environment and effective learning mechanisms. In this vein, it can be said that semantic gender is more intricately a part of the structural representation of nouns (with some connection to the semantics of the noun) than grammatical gender is. These points will emerge clearly when we discuss the empirical facts of gender in Section 1.5, as well as in Chapter 2 of this dissertation.

Such a structural difference between the two kinds of gender prompts us to probe whether heritage grammars systematically treat them differently, or not. This dissertation uses empirical data from Tamil to carry out this investigation. A precise characterization of the gender system of Tamil, and how it informs the current study on gender in heritage Tamil are laid out in the forthcoming sections. Section 1.5 provides an overview of the empirical ground covered in the study. Section 1.6 outlines the core research questions of the dissertation.

1.5 The Empirical Core of the Dissertation

The empirical core of this dissertation is formed by data elicited from HSs of Tamil (Dravidian) residing in New Delhi, India. The dominant language of New Delhi is Hindi-Urdu (Indo-Aryan), which is the second language (L2) of this population. Tamil is predominantly spoken in the southern state of Tamil Nadu, as shown in Fig. 1.1. This section provides an overview of the gender systems of the two languages and begins the discussion on their interaction in this particular context.

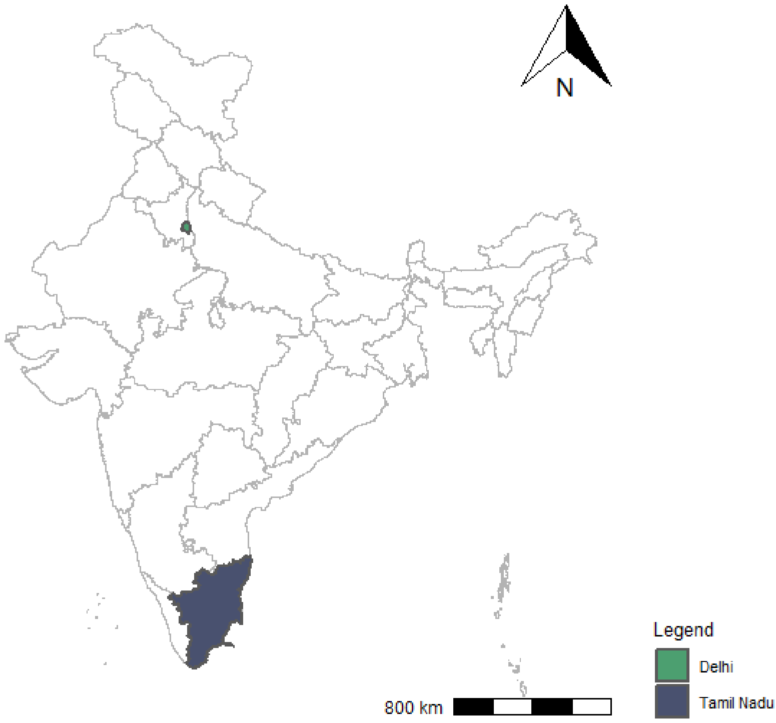


Figure 1.1: Map of India

1.5.1 Gender and Gender Agreement in Tamil

Tamil is a South Dravidian language with a gender system that is entirely semantic (Krishnamurti, 2003; Corbett, 1991; Pillai, 1964; Arden, 1891). Consequently, there exists a sharp divide between [+HUMAN] and [-HUMAN] nouns. Steever (2018) considers this to be a dyad of "rational" as opposed to "irrational" beings. Thus, all nouns denoting human beings are encoded with a gender value that corresponds to the (perceived/ conventionally understood) biological sex of the referent [MASCULINE] (5a) or [FEMININE] (5b). All nouns denoting animate non-human (5c), as well as inanimate entities (5d), are not marked for gender. As observed cross-linguistically, the presence of gender features on nominals is reflected in agreement relations. In Tamil too we see the tensed verbs in (5a) and (5b) inflecting differently for gender, whereas (5c) and (5d) do not inflect for gender values; these verbs express the same markers seen in the case of default agreement. As reported by Sarma (2013), the set of [III person singular] features is used to express default agreement in Tamil, as in (5e) where the experiencer subject is DAT-marked and unable to control agreement. The same inflection being present in (5c) and (5d) tells us that non-human denoting nouns in Tamil do not carry gender features; the verb expresses default agreement when these are present as the subject of the sentence.

(5) a. *anda paiyan va-nd-aan*
that boy.MSG come-PST-3MSG
'That boy came.'

b. *anda ponnu va-nd-aa*
that girl.FSG come-PST-3FSG
'That girl came.'

c. *anda maadu va-nd-udu*
that cow come-PST-3SG
'That cow came.'

d. *seidi va-nd-udu*
news come-PST-3SG
'News came.' (lit: the news reached us)

e. *ena-kku kuLur-udu.*
I-DAT cold-3SG
'I am (feeling) cold.'

The examples in (5) show that subject agreement is systematically expressed in Tamil. Object agreement takes place only in some restricted environments where the subject DP is overtly case marked, and a logical NOM object is present in the sentence (Murugesan, 2022) . In these cases, the tensed verb inflects for the features of the logical object, as shown in (6a) and (6b).

(6) a. *kohli-kku meena kidai-t-aaL*
Kohli.M-DAT Meena.F.NOM get-PST-3FSG
'Kohli got Meena.'

b. *leela-kku avan kannadi-la teri-nj-aan*
Leela.F-DAT he.M.NOM mirror-LOC appear-PST-3MSG
'Leela saw him in the mirror.' (lit: To Leela, he appeared in the mirror.)

Predicates are the only expressions of gender agreement in Tamil. Elements within the nominal domain do not carry the gender features of the noun. Determiners (7) and attributive adjectives (8) remain uninflected.

(7) a. *anda paiyan*
that boy.MSG
'that boy'

b. *anda poNNu*
that girl.FSG
'that girl'

c. *anda maadu-nga*
that cow-PL
'those cows'

(8) a. *periya paiyan*
big boy.MSG
'big boy'

b. *periya poNNu*
big girl.FSG
'big girl'

c. *periya maadu-nga*
big cow-PL
'big cows'

Along with the verbal predicates in (5), predicative adjectives participate in gender agreement with the subject of the predicate. (9) shows that the adjectives in predicative position, as well as when property indicating items occur with a copula, inflect for the gender features of the noun.

(9) a. *avan romba nalla-van*
he very good-MSG
'He is very good.'

b. *ava romba nalla-va*
she very good-FSG
'She is very good.'

c. *avan romba sandosham-aa iru-k-aan*
he very happiness-V be-PRS-MSG
'He is very happy.'

d. *ava romba sandosham-aa iru-k-aa*
she very happiness-V be-PRS-FSG
'She is very happy.'

A detailed analysis of the structural representations of these adjectives is presented in Chapter 3. For the purposes of the current discussion, it is relevant to note that they participate in gender agreement with the subject of the clause.

With regard to agreement, another point is worth mentioning here. Corbett (2006) and Baker (2008) discuss in their studies of agreement that there is considerable cross-linguistic diversity in terms of the targets of agreement. There are languages with a wide range of grammatical categories expressing agreement with nouns: verbs, auxiliaries, determiners, adjectives, and even complementizers. On the other hand, there are languages (Baker enumerates 9 out of a survey of 108) with only one expression of agreement - typically the tensed verb. For a closely related

Dravidian language, Kannada, Baker (2008, p. 249) notes the following properties: the language has only subject agreement morphologically realised on the main verb of a tensed construction. There is no agreement on the auxiliary verb, or on functional heads such as C, P, or D. The same can be said of Tamil. It is a system in which the subject of a tensed clause is the sole controller of agreement. The sole target of such agreement is the key verbal element of the clause - manifested as either main verbs or predicative adjectives.

To summarise, the gender system of Tamil is entirely semantic. Only [+HUMAN] nouns carry gender features, thus making gender connected with the structural representation of nouns. Additionally, gender agreement is seen in a restricted set of constructions: subject-verb agreement when the subject is not overtly case marked, predicative adjectives, and "object" agreement when the only DP available for agreement is the logical object of the predicate.

1.5.2 Gender and Gender Agreement in Hindi-Urdu

Hindi-Urdu is an Indo-Aryan language with semantic as well as formal systems of gender (Corbett, 2013, 1991). All nouns, regardless of their status with respect to animacy, are assigned a gender value. There are two gender categories in Hindi [MASCULINE] and [FEMININE]; every noun in the language is allotted one of the two. Additionally, Hindi-Urdu also has a rich agreement paradigm; gender (and other phi features) are marked on a range of functional heads: Possessive determiners (10a, 10b), Adjectives (11a, 11b), auxuliary/Tense (12a, 12b) and verbs (13a, 13b) .

(10) a. *mer-aa betaa*
my-MSG son.MSG
'my son'

b. *mer-ii kitaab*
my-FSG book.FSG
'my book'

(11) a. *baD-aa ghar*
big-MSG house.MSG
'big house'

b. *baD-ii gaaDi*
big-FSG car.FSG
'big car'

(12) a. *wah ladkaa roTii khaat-aa hai*
that boy.MSG bread.FSG eat.PRS-MSG be.PRS
'That boy eats bread.'

b. *wah ladkii ghar jaa rah-ii th-ii*
that girl.FSG house.MSG go.INF PROG-FSG be.PST-FSG
'That girl was going home.'

Case-marked DPs may not participate in agreement in Hindi-Urdu. Therefore, when the subject DP is overtly case marked, we see the object agreeing with the verb, (13a, 13b, 13c). When both the arguments are overtly case-marked, as in (13d), the verb displays default morphological marking, which is MASC in Hindi-Urdu.

(13) a. *vijay-ne roTii khaa-ii*
Vijay.MSG-ERG bread.FSG eat.PST-FSG/*MSG
'Vijay ate (the) bread.'

b. *vijay-ne roTii khaan-ii chaah-ii*
vijay.MSG-ERG bread.FSG eat.INF-FSG want-PST-FSG
'Vijay wanted to eat bread.'

c. *priya-ko yeh ghar dekh-naa/*ii hai*
Priya.FSG-DAT this house.MSG see-INF.MSG/*FSG be.PRS
'Priya wants to see this house.'

d. *priya-ne giita-ko dekh-aa/*ii*
Priya.FSG-ERG Geeta.FSG-DAT see.PST-MSG/*FSG
'Priya saw Geeta.'

To summarize, The differences between the gender systems of Tamil and Hindi are broadly two-fold:

1. Tamil has only semantic gender, while Hindi has both semantic and grammatical gender. Thus, the category/feature of gender is much more pervasive in Hindi-Urdu than in Tamil; only [+HUMAN] nouns are gender-marked in the latter, while all the nouns in Hindi-Urdu have a gender value assigned to them.
2. In Tamil, only predicates (verbal or adjectival) express agreement. None of the other functional categories participate in agreement. Hindi-Urdu displays gender agreement on multiple elements (V-T complex, adjectives, possessive determiners), suggesting that gender has an active role to play in the clausal domain as well.

Given such a configuration of the gender systems of the two languages, we now move on to elaborate on the bilingual context created by their contact.

1.5.3 Introducing Heritage Tamil in New Delhi

The empirical core of this dissertation is formed by the grammar of HSs of Tamil with Hindi-Urdu as their L2. This group of near-simultaneous/ sequential bilinguals were formed due to gradual but steady migration from Tamil Nadu to the national capital (New Delhi) throughout the 20th century. The second generation of these migrants constitute HSs of Tamil in New Delhi. These speakers acquire Tamil naturalistically in the home domain, and Hindi-Urdu through exposure in the immediate neighbourhood. Thus, Tamil is confined to the sphere of home and family, and Hindi-Urdu is present in almost all other spheres of their lives. As the *lingua franca* of New Delhi, it is used for all informal communication in the city, it is the language of mass media, and it is taught as a core subject in schools across New Delhi. Thus, as the HSs of Tamil grow up in New Delhi there is considerable disparity in the presence of the two languages in their life with Hindi-Urdu rapidly overtaking Tamil.

An observation worth making at this point is that heritage Tamil in New Delhi is the result of intra-country migration. This differs from the general tendency of heritage language studies to mostly focus on international migration. However, a change in national identity is not an essential requirement for defining a heritage community. Domestic migration, such as the case here, should be included while studying heritage languages, especially if there exists a significant disparity between the linguistic environments of the origin and destination of the migration. This fact is of particular relevance, given the socio-political landscape of India. In the 1950s, India undertook an exercise to re-organize its state boundaries on a linguistic basis; identity in terms of language was an important factor in the creation of states (Graziosi, 2017; Majeed, 2003; Mishra, 1972). Consequently, the larger, external environment that a language acquiring child is exposed to (Polinsky, 2015; Rothman, 2009) changes significantly across different parts of the same country, for example Tamil Nadu and New Delhi. Such a drastic change qualifies this migratory population as heritage speakers despite living in the same country.

The 23 participants (Age range= 18–30) in the studies covered in this dissertation are all self-reported L1 speakers of Tamil. They were all born and raised in New Delhi in Tamil speaking homes. The participants also report to using Tamil only with family members, and Hindi-Urdu in all other social spheres of their lives. In order to assess their linguistic proficiency in their HL, the participants were asked to complete a cloze passage test at the level of elementary school.

The sentences used in the passage were not directly related to any part of the current study; they aimed at capturing basic facts about Tamil. Successfully completing the task entails that the participant has working knowledge of Tamil word order, inflectional markers, a few frequently used lexical items, etc. 13 speakers opted out of the test in the Tamil script, as they were not literate in Tamil. On average, the remaining 10 speakers scored 65% in the test, indicating a fair amount of proficiency in their HL. Based on this pre-test, the main questionnaire only contained the roman script, so as to include all 23 speakers.

1.5.4 Methodology

As stated above, the empirical base of this dissertation is Heritage Tamil (HTamil) spoken in New Delhi. The focal topic of the dissertation is the structural representation of gender and the ways in which gender participates in relations such as agreement in HTamil. These issues are studied from the specific vantage point of a heritage grammar. Thus, in order to build an empirical corpus that can help attest or disprove the formulations made about gender, linguistic data was collected from 23 speakers of heritage Tamil and 12 monolingual speakers of Tamil forming the baseline group. This involved a detailed questionnaire focused on constructions with gender agreement or gendered lexical items present. Picture-description tasks were used to elicit sentences. The speakers were shown images that they had to describe using sentences or phrases in Tamil. The exact methods used in each study are outlined in the respective chapters.

1.6 Core Research Questions

We understand contexts of heritage bilingualism as being inherently asymmetrical. The HL is often in a disadvantageous position owing to two major reasons: reduced input in the HL itself, and increased influence from the societal language(s), especially as the speakers grow up and rapidly expand the domains in which they might use the societal language(s). Given such conditions, the main questions we ask are about how a grammar develops under such ostensibly unfavourable conditions. In this dissertation, the question is approached from the perspective of gender systems, which have been independently established as a challenging grammatical category for bilingual language acquirers.

Combining these two ideas, we narrow down our focus on how the mental representations of gender systems in HTamil develop. On the one hand, we have the empirical observation that gender invariably undergoes attrition in heritage grammars. On the other, we understand the gender system in Tamil as a structural phenomenon; gender features are strongly tied to the presence of [+HUMAN] features on nouns. Thus, the question becomes: Can the structural properties offer immunity to the gender system of Tamil even when it is acquired as a HL? In other words, can we expect that biological gender in heritage Tamil will remain impervious to the effects of asymmetrical bilingualism? Alternatively, will the conditions of acquisition override the support provided by the structural component and induce attrition in the gender system of heritage Tamil? Broadly, these questions frame the rationale of particular studies conducted in each chapter.

The second major issue under consideration is cross-linguistic influence of the societal language on the HL. Hindi-Urdu, as can be seen from the illustrations provided in this chapter, has a far-more elaborately organised gender system than Tamil. In fact, the gender system of Tamil can be understood as a sub-set of the types of gender and gender agreement operations present in Hindi-Urdu. Given a subset-superset relationship between L1 and L2, what kind of cross-linguistic transfer effects can we expect?

In other words, HTamil speakers might develop gender marking on elements that usually don't encode gender, i.e. transfer parts of the Hindi-Urdu system into Tamil. This would make gender in heritage Tamil acutely susceptible to the effects of asymmetrical bilingualism, an unlikely outcome if gender in heritage Tamil maintains its connection with the [HUMAN] feature.

In this dissertation, I explore the ramifications of these questions by considering the various venues where gender is manifest in Tamil. The dissertation is organised such that each chapter delves into the particulars of one construction where gender agreement takes place.

1.6.1 Dissertation Outline

Chapters 2, 3 and 4 address gender on nouns, gender agreement on predicative adjectives and gender as a feature constraining predicative possession, respectively. All three chapters are organized in a similar fashion. The expository section of each chapter is devoted to outlining a theoretical account of the phenomenon under discussion, followed by relevant empirical facts about Tamil and Hindi-Urdu. Based on these, questions are raised about how heritage Tamil can be expected to either align with or diverge from the baseline given restricted linguistic input in Tamil, and potential cross-linguistic influence from Hindi-Urdu. This sets the stage for discussing the novel studies and their results. An outline of each chapter is given below.

Chapter 2 presents gender assignment facts in Tamil and Hindi-Urdu along with various theoretical accounts of gender, and arrive at an understanding of how gender is assigned to nouns in Tamil and Hindi-Urdu. Tamil has semantic gender of two types: (i) inherently specified gender on N and (ii) gender obtained from a discourse referent through D. Both these are exclusively reserved for [+HUMAN] nouns. Hindi-Urdu has these two mechanisms in addition to arbitrary/ grammatical gender assignment. The empirical study in this chapter focuses on whether heritage Tamil retains both kinds of semantic gender, and whether it has added grammatical gender from Hindi-Urdu. The results of the study confirm the presence of semantic gender in Tamil with some gradation: inherently specified biological gender remains stable, whereas discourse-based biological gender undergoes variation. Additionally, we find no evidence for the inclusion of grammatical gender in heritage Tamil. These trends are consistent with our observation that semantic gender, with its [+HUMAN] correlate, is intricately connected to the structural representation of nouns in Tamil, and would consequently resist any change/ restructuring due to influence from the dominant language.

Chapter 3 delves into the domain of gender agreement. As stated in this chapter, predicates, i.e., verbs and predicative adjectives, are the only expressions of agreement in Tamil. This chapter mainly focuses on gender agreement on the latter, as adjectives are a derivationally formed category in Tamil, thus making them more informative in terms of how syntactic structures fare in heritage grammars. A review of empirical facts about Tamil and Hindi-Urdu reveals that the predicative agreement systems in Tamil are a subset of the ones in Hindi-Urdu. Hindi-Urdu expresses agreement on attributive as well as predicative adjectives, whereas Tamil does so only on predicative adjectives. The chapter therefore focuses on two aspects in heritage Tamil: (i) the formation of the adjectives and (ii) the execution of agreement on predicative adjectives. The results of the study show that while adjective formation strategies (which involve considerable learning) show some variation from the baseline, the mechanism of agreement as well as the attributive-predicative adjective distinction in terms of agreement remain intact in heritage Tamil.

From these, we can infer that the underlying structural representation of gender agreement resists any change/ restructuring in heritage Tamil. Additionally, there are no gender mismatches in the agreement data, reiterating the stability of semantic gender features. Once again, we find support for the resilience of structural phenomena, and the absence of significant cross-linguistic influence.

Chapter 4 presents yet another dimension of gender features in Tamil. Here, I defer to the well-accepted notion that features drive syntactic computation under the Minimalist Program. The chapter begins with the empirical observation that predicative possession in Tamil makes a distinction among nouns; not all nouns are allowed to occur as possessum of predicative possession. A probe into the nouns themselves as well as the structural representation of possessives makes it apparent that the gender features of the noun have a role to play in whether or not it may occur in the structural configuration of predicative possession. This constraint is present only in Tamil, and not in Hindi-Urdu. Another point of difference between the two languages is in terms of morphological agreement: Hindi-Urdu displays agreement between the POSS morpheme and the possessum, whereas Tamil does not. The current study was set up to cover both these aspects in heritage Tamil. The results of the study clearly indicate that heritage Tamil retains the gender feature-based constraint, speaking once again to the resilience of core structural properties of grammar. Additionally, heritage Tamil does not add gender agreement to the domain of possessives, avoiding yet another potential case of cross-linguistic transfer.

Chapter 5 concludes the dissertation by commenting on three major threads that are found across Chapters 2, 3 and 4. Firstly, it discusses the stability of core syntax in the domain of gender. Data from heritage Tamil shows a clear trend of the structural aspects of gender resisting change, either due to reduced PLD in Tamil or transfer from Hindi-Urdu. Secondly, it addresses the lack of cross-linguistic effects in the domains studied so far. Thirdly, it approaches the way forward for this study in terms of studying the variation found in heritage Tamil in greater detail. It outlines some plans for capturing inter-speaker, as well as intra-speaker variation in these and a few other empirical domains in heritage Tamil.

Chapter 2

Gender in Nouns

2.1 Introduction

This chapter presents a comprehensive picture of the structural representation of gender by looking at the different types of lexical items that express gender in Tamil and Hindi-Urdu. Specifically, this chapter addresses the exact mechanisms by which nouns are endowed with a gender value; it is an account of gender assignment to nouns. While looking at the structural representation of gender, it becomes evident that identifying different types of gender, such as biological gender, grammatical gender, discourse-based gender, etc., is important. These distinctions show a gradation in the nominal spine and should be understood as arising out of distinct underlying representations. In other words, biological and grammatical gender are located at different sites within the DP. The approach to gender adopted in this chapter is cognizant of these representations and the process of deriving these gender values on nouns.

The chapter then turns its focus to the question of stability and change in this domain in heritage Tamil. Thus, the current study investigates how different kinds of gendered NPs are represented in heritage Tamil and if there are any deviations from the baseline. The results of the study show different trajectories for biological gender and discourse-based gender in heritage Tamil. Based on these, we can form generalizations about aspects of gender assignment that remain stable and those that are prone to change in heritage grammars.

The chapter is organised as follows: Section 2.2 presents the gender systems in Tamil and Hindi-Urdu, and draws a few generalisations about the kinds of mechanisms that are needed in order to explain the entire range of gender found in these languages. Subsequently, section 2.3 covers some existing theoretical accounts of gender, so that the differences between the identified types of gender can be adequately explained. Based on these accounts, Section 2.4 presents a complete picture of the structural representation of gender in Tamil and Hindi-Urdu, while focusing on the connection between gender and [+/-HUMAN] features. Section 2.5 takes the point forward by formulating hypotheses and predictions about the ways in which Heritage Tamil could diverge from the baseline variety, considering the limited input available to heritage populations, and potential transfer effects from the dominant language Hindi-Urdu. Details of the current study and its results are also presented here. The results of the study show evidence for the stability of inherent biological gender and its connection with [+HUMAN], and variation in discourse-based gender in heritage Tamil. Section 2.6 concludes.

2.2 The Gender Systems of Tamil and Hindi-Urdu

In this section, we provide a brief sketch of the different types of nouns in Tamil and Hindi-Urdu that carry gender values.

2.2.1 Gender in Tamil - Empirical Facts

In Tamil, nouns that are specified for [+HUMAN] carry an inherent gender value. This is referred to as biological or semantic gender (Krishnamurti, 2003). Nouns such as (1) are either [MASC] or [FEM], the only two gender values available in Tamil. The gender value has correlates with the biological sex of the [+HUMAN] referent[1]. Additionally, III person pronouns are also morphologically marked for gender in Tamil (2).

(1) a. *paiyan*
boy.MSG
'boy'

b. *tambi*
younger brother.MSG
'younger brother'

c. *poNNu*
girl.FSG
'girl'

d. *akkaa*
older sister.FSG
'older sister'

[1] Steever (2018) notes that there are two basic genders in Tamil: 'rational' and 'irrational' corresponding roughly to human and non-human respectively.

(2) a. *avan*
'he (for human male nouns)'

b. *ava*
'she (for human female nouns)'

c. *adu*
'it (for all non-human nouns)'

Another instance of gender on nouns observed in Tamil is in the case of common nouns such as 'student', 'doctor' 'teacher', etc. When these terms refer to individual humans who are a part of the discourse, the gender value of the individual is attached to the common noun.

(3) a. *maanav-an*
student-MSG
'male student'

b. *aasiriy-ar*
teacher-MSG.HON
'male teacher (honorific by default)'

c. *maanav-ii*
student-FSG
'female student'

d. *aasiriy-ai*
teacher-FSG.HON
'female teacher (honorific by default)'

These examples indicate that the specification of [+HUMAN] is a necessary and sufficient condition for a noun to be assigned a gender value in Tamil. All non-human entities, including other animate beings, remain unspecified for gender.

2.2.2 Gender in Hindi-Urdu - Empirical Facts

All nouns in Hindi-Urdu, regardless of their animacy status, are specified for a gender value. Here, we need to make a distinction between biological gender (found on animate nouns, with biological correlates) and grammatical gender (found on inanimate objects). Biological gender is seen on [+ANIMATE] nouns such as (4) below:

(4) a. *aadmii*
man.MSG
'man'

b. *ladk-aa*
boy-MSG
'boy'

c. *sher*
lion.MSG
'lion'

d. *mor*
peacock.MSG
'peacock'

e. *aurat*
woman.FSG
'woman'

f. *ladk-ii*
girl-FSG
'girl'

g. *sher-nii*
lion-FSG
'lioness'

h. *mor-nii*
peacock-FSG
'peahen'

Grammatical gender is found on [-ANIMATE] nouns such as (5) below. The gender value of these nouns is not based on any semantic attribute of the noun.

(5) a. *ghar*
'house.MSG'

b. *darwaaza*
'door.MSG'

c. *aasmaan*
'sky.MSG'

d. *gaaDi*
'vehicle.FSG'

e. *kitaab*
'book.FSG'

f. *diiwaar*
'wall.FSG'

Additionally, we also observe gender values on common nouns when they refer to particular individuals.

(6) a. *chaatr*
student.MSG
'male student'

b. *adhyaapak*
teacher.MSG
'male teacher'

c. *chaatr-aa*
student-FSG
'female student'

d. *adhyaapika*
teacher.FSG
'female teacher'

One point of difference between Tamil and Hindi-Urdu can be seen in the domain of pronouns. In Tamil, pronouns morphologically encode the gender values of the referent (2). Hindi-Urdu, on the other hand, does not have exponence for the gender values of pronouns. The pronoun remains invariant across different genders; the gender values are seen on verbal agreement wherever necessary (7).

(7) a. *wah so ga-yaa*
3SG sleep.INF go-PST-MSG
'He slept.'

b. *wah so ga-yii*
3SG sleep.INF go-PST-FSG
'She slept.'

Thus, we see that every noun in Hindi-Urdu is assigned a gender value, either biological or grammatical, regardless of whether the gender value is semantically valid.

2.2.3 Generalisations

Table 2.1 presents a summary of the types of gendered nouns found in Tamil and Hindi-Urdu. It is evident that Tamil contains a subset of the types attested in Hindi-Urdu.

	Tamil	Hindi-Urdu	Basis for gender assignment
[+HUMAN] nouns	Yes	Yes	Semantic
Common nouns taking gender values	Yes	Yes	Semantic
Non-human animate nouns	No	Yes	Semantic
Grammatical gender for inanimate objects	No	Yes	No semantic basis

Table 2.1: Gender in Tamil and Hindi-Urdu

Given this distribution of gender in the two languages, we know proceed to uncover the underlying structural representation that could generate it.

2.3 The Role of Structure in Gender Assignment

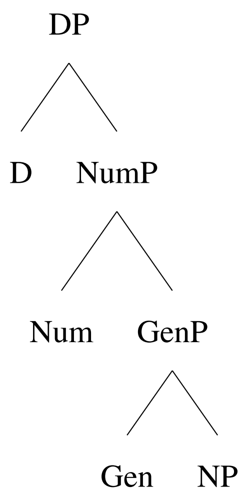
The survey of the gender systems of Tamil and Hindi-Urdu done previously indicates that there is some amount of variation in the kinds of gender present in the languages. The amount of idiosyncrasy in the distribution in Table 2.1 suggests that while acquiring the grammars of Tamil and Hindi-Urdu, speakers will have to learn the assigned gender value for each noun on a case by case basis. Therefore, we are inclined towards considering the lexicon as the ideal venue for gender assignment, as it allows for the idiosyncratic listing of gender values to each noun, while still remaining connected to the other features of that noun such as [+/-ANIMATE] or [+/-HUMAN].

This line of thought, known as the lexicalist approach, began from Chomsky (1965) and continues to be well accepted in many contemporary works (Chandra and Srishti, 2014; Chomsky, 1995; Hale and Keyser, 1993). According to this idea, all properties of a formative (lexical item, here) that are idiosyncratic will be specified in the lexicon. Essentially, any property that cannot be derived by a general rule will have to be supplied at the lexicon. Following from this, I adopt the stance that the gender value of a noun must be part of the lexical information of the noun; it is carried into the derivational space by any lexical/ functional head that is involved in the structural representation of that noun (Harris, 1991; Carstens, 2010). The role of lexical and functional heads in gender assignment becomes clear when we survey some existing theoretical accounts of gender assignment, and then build our mechanism from there.

2.3.1 Picallo and Longobardi's GenP System

Following the DP hypothesis proposed by Abney (1987), Picallo (2009) and Longobardi (2001) argue for an extended nominal domain where each property of the noun, such as number, gender, etc., heads its own functional projection, as shown in (8) below:

(8)



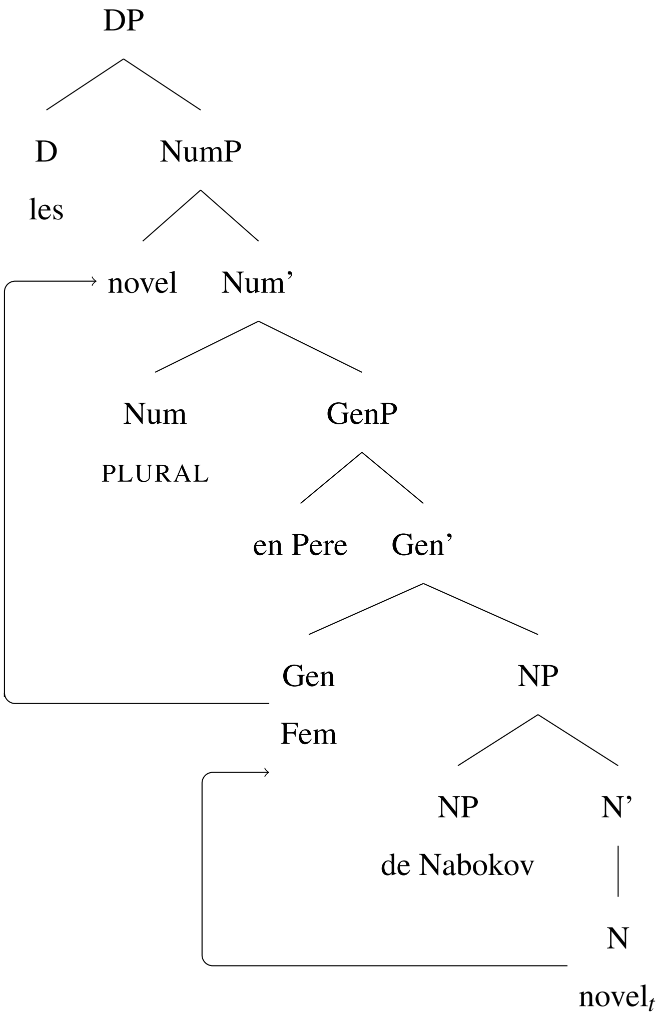
Although these accounts do not make explicit reference to gender assignment, they posit GenP to be the locus of the gender value of a noun. The implicit assumption is that gender is "assigned" to a noun by a functional head in syntax. Thus, the DP carrying the NP also carries its featural attributes on separate functional heads, all of them converging for the complete interpretation of the NP. Additional motivation for GenP comes from its specifier position serving as potential landing site for various DP internal movements. Using Catalan examples such as (9), Picallo proposes that GenP immediately dominates NP and that the head Gen is the source of gender inflection for all nominals. A crucial motivation for GenP is its specifier position, which is said to house the possessor argument *en Pere*, as in (10).

(9) from Picallo (2009)

les novel.les d'en Pere de Nabokov
the novels of Pere of Nabokov

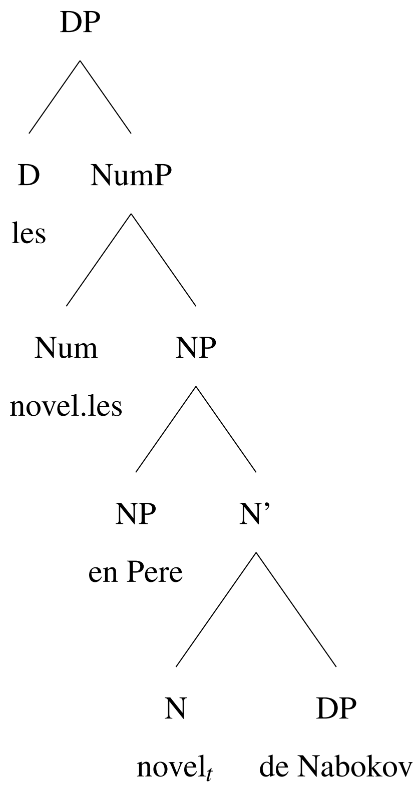
According to Picallo's analysis, in (10), the noun *novel* raises to GEN, is gender-valued as [FEM] and then ultimately moves to the Spec of NumP, where its plural attribute is given, arriving at the morphological form *novel.les* (F). However, this sequence of head-to-head movement (N → Gen) followed by head-to-spec movement (Gen → SpecNumP) is rather unusual for a nominal head to undertake. Nevertheless, these movements coupled with the possessor NP *en Pere* housed at Spec Gen P give rise to the appropriate word order.

(10)



Kramer (2015, p. 24) supersedes this analysis by demonstrating that the same word order can be attained without invoking GenP at all: *Nabokov* could be base generated as the complement of NP, and *en Pere* at Spec NP. The N *novel* would raise to NUM and get its plural inflection. This is shown schematically in (11). (11) derives the same sentence as (10) without invoking GenP, thus illustrating that the presence of a possessor NP does not necessitate a separate projection for Gender.

(11)



Another shortcoming of Picallo's analysis is that the motivation provided for GenP is entirely incidental. There is no direct relation between the syntax of possession and gender itself, making GenderP, particularly as the base generation position of the possessor, very unlikely. Therefore, this argument does not sufficiently motivate GenP.

A further issue with the GenP account is that it does not explain how each noun gets assigned its gender value. For instance, it does not provide the mechanism by which *novel* rightly merges with a [FEM] specified GenP, and not a [MASC] specified one. If the gender information of *novel* were lexically specified, it would make GenP redundant. Thus, the question of where and how gender-related information is stored remains unaddressed. As a corollary, the account also does not have any way to make a distinction between different types of gender such as biological and grammatical gender.

Overall, the GenP account is one where gender heads its own independent projection, and comes about on the noun during the derivational process. We will not be adopting this account as it does not present a complete picture of gender assignment and fails to make note of further distinctions within the category of gender. Thus, the search for an alternative that can provide a more nuanced approach to gender leads us towards the notion that gender can be represented as feature mounted onto some other projection. It is in this regard that we consider Alexiadou's proposal for gender.

2.3.2 Alexiadou's Bipartite Distinction

Alexiadou (2004) provides an account of how gender values are assigned to nouns in gendered languages. She proposes that there are two major ways by which nouns receive their gender values.

- All inanimate nouns receive a default gender value in the lexicon; this is inherent to the noun and should be understood as an intrinsic property of the noun itself.
- Nouns denoting human and other animate entities are different. These do not receive gender values from the lexicon. Their gender feature is valued when they refer to an individual referent.

Alexiadou uses examples from Greek (12) to illustrate the point. (12a) and (12b) denote inanimate nouns specified as [MASC] and [FEM] respectively. Alexiadou's account predicts that these gender values are an inherent part of the nouns, with the information presumably specified at the lexicon. These can be contrasted by (12c) and (12d) which denote human nouns. According to this account, the gender values of these nouns remain underspecified in the lexicon. The lexicon marks them as [+HUMAN+ANIMATE], and they await gender specification until they locate a human referent and take on the gender features of the referent. Essentially, these nouns do not contain gender specification as an intrinsic property. This account does not make explicit mention of the exact mechanisms of gender assignment to either types.

(12) from Alexiadou (2004)

a. *kipos*
garden.MSG
'garden'

b. *psifos*
vote.FSG
'vote'

c. *agori*
boy.MSG
'boy'

d. *koritsi*
girl.FSG
'girl'

It is important to note that Alexiadou's system differentiates gender values on the basis of whether they are fixed (inanimate objects) or variant (animate entities). While the mechanism given for inanimate nouns is sufficient, the one for animate beings falls short in some domains. For instance, nouns such as 'sister.FSG', 'mother.FSG', 'niece.FSG', etc., are incompatible with Alexiadou's idea of variable gender for all [+HUMAN] nouns. Despite not having a specific referent associated with them, their gender values are fixed as [FEM], much like the inanimate nouns. However, one cannot equate these with inanimate nouns, as placing 'sister.FSG' in the same category as 'vote.FSG' would miss out on the crucial fact that the former has semantic correlates and the latter does not. Essentially, the explanation provided for biologically fixed gender nouns can be refined further. In the following subsections, I make the case for finer-grained distinctions within the category of [+HUMAN] nouns and their gender values. In essence, I argue that biological gender on [+HUMAN] nouns comes in two flavours: fixed/ inherent and variable.

Additionally, another key feature of Alexiadou's system of gender assignment is that functional heads are not at all involved in it. Either the noun is inherently specified for a gender value (12a), or it obtains the value from a referent (12c). Gender is not something that is facilitated by functional heads.

I depart from this approach by maintaining that the differences within biological gender as well as those between biological gender and grammatical gender are central to any system of gender assignment, and that the way to implement this distinction is through the involvement of functional heads. Functional heads in the nominal domain, as will be explained in the following sections, are useful tools in capturing gender assignment processes. Using nominalization, I show that the involvement of functional heads in the assignment of gender values will be indicators of whether those values can be obtained derivationally or not.

In the following sections, we shall review accounts of gender assignment where functional heads play a crucial role. Thus, we consider adopting functional heads to explain gender assignment to nouns, and turn to Kramer's proposal of gender assignment for some insights.

2.3.3 Kramer's Distributed Morphology-Based Account

Building on the assumption that gender is a syntactic feature, Kramer (2014, 2015, 2016) begins with the assumption that gender is an inherent[2] property of nouns. However, the exact location of the features remains to be investigated. (13) is Kramer's starting point.

(13) DP

D NP

N

Gender

Kramer's account is cast within Distributed Morphology and therefore assumed that the lexicon contains roots devoid of any kind of attributive information. Upon combining with specific functional heads, these roots assume their categorial status. Combining a root with the nominalizer *n* yields a noun and combining it with the verbaliser *v*, a verb. As shown in (14), the root ADVICE merging with *n* is the noun 'advice' (14a) , and the one merging with *v* is the verb 'advise' (14b).

(14) a. *n*P

n $\sqrt{advice}$

b. *v*P

v $\sqrt{advice}$

Now let us evaluate Kramer's position with specific focus on the implications for gender assignment. In this approach, roots do not carry any gender-related information. Gender values are specified on *n*. A root, when it combines with *n* and becomes a noun, obtains the gender value of that *n*. While Kramer does not state this explicitly, this system implies that while learning lexical items, one does not learn the associated gender value of each individual item. All the learner knows is that the bare, unspecified root will be assigned a gender value once it combines with *n*. The assignment of gender to nouns, according to Kramer's account, is an

[2]Kramer (2015, p. 23) takes gender to be an inherent property of nouns in the sense that it remains consistent across different domains. Regardless of how a noun is inflected (definiteness, case, number, etc.) its gender value remains unchanged.

entirely derivational process with no learning involved. Note that this implication is not congruent with the well-established idea that gender values of (at least some) nouns are idiosyncratic and cannot be derived by a systematic rule.

Additionally, this approach treats the difference between biological gender and grammatical gender as a function of differently specified *n*. Both these types of gender are obtained when an unspecified root merges with a kind of *n*. The difference lies in the fact that the *n* which leads to biological gender is specified with an interpretable gender feature, while the *n* that leads to grammatical gender has an uninterpretable gender feature. This distribution stems from biological gender having semantic correlates, i.e. being semantically interpretable, and grammatical gender not having it, and therefore being semantically uninterpretable. In (15), i *n* represents an *n* with an interpretable gender feature and u *n* represents an *n* with an uninterpretable gender feature.

(15)
a. $\sqrt{mother}$ is interpreted as biological gender [FEM] because it merges with i *n*[+FEM]
b. $\sqrt{father}$ is interpreted as biological gender [MASC] because it merges with i *n*[-FEM]
c. $\sqrt{watch}$ is interpreted as grammatical gender [FEM] because it merges with u *n*[+FEM] (depending on the language)
d. $\sqrt{house}$ is interpreted as grammatical gender [MASC] because it merges with u *n*[-FEM] (depending on the language)

Naturally, the next question that arises is: How does the system ensure that a root merges with the appropriate *n* and not a different kind of *n*? For instance, what is it that drives $\sqrt{mother}$ to only merge with i *n*[+FEM]? To answer this, Kramer posits a set of licensing conditions, which address the issue of matching. Licensing conditions ensure that the root matches with the appropriate type of *n*. These conditions depend on the lexicon carrying some amount of gender information. For example, $\sqrt{mother}$ must carry some indication that it ought to be realised as a biological gender female noun, hence the root must be merged with i *n* and no other kind of *n*. The licensing conditions can operate with this information and ensure that the specified requirement is met, i.e., the root is appropriately licensed.

These licensing conditions go against a fundamental tenet of the DM-based approach, which is that roots are devoid of any distinguishing information. The system proposed here can be sustained only if the lexicon also carries some information. Another problem associated with such lexical specification is that it creates redundancy, as gender information is now expressed twice: once in the lexicon, on the roots, and once on *n*.

A third issue to bear in mind is that this is a system that involves minimum learning when it comes to gender. Gender is a part of the derivational component in Kramer's system, thus undercutting its idiosyncratic nature. If we take this idea further and situate it in the context of how gender fares in bilingual settings, it leads to some interesting implications. Since gender is not crucially dependent on learning or input, the context of acquisition should not be an important factor in how it gets realised. In other words, there is no reason why the acquisition of gender should suffer in contexts of reduced input, if gender was all derivational with no learning component. However, that is not what we observe empirically; most bilingual settings record some form of attrition or loss of gender, strongly suggesting that it has a lexical component that must be learnt.

Thus, we do not take up Kramer's account as is. However, as alluded to in the beginning of the current discussion, functional heads may be necessary in differentiating between different (sub)-types of gender. Therefore, we recognise the merit in having functional heads such as *n* play a key role in the gender assignment of nouns, and that is something that we will co-opt in our understanding of the structural representation of gender.

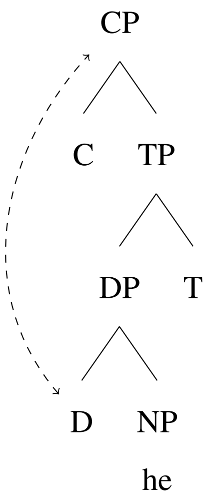
2.3.4 Sigurðsson's Edge Computation Account

Sigurðsson (2019) provides an edge-computation account for how gender is assigned to nouns, especially when discourse elements are involved. This account uses a functional head, D, to anchor gender assignment. Sigurðsson posits this mechanism to explain how pronouns, which are inherently deictic in nature, are able to encode the different gender values of the nouns they refer to. In many languages of the world, the pronouns referring to masculine (16a), feminine (16b) and neuter (16c) or non-gendered nouns are marked with distinct morphology. Tamil is one such language, with morphologically distinct terms for male and female referents, and non-gendered entities.

(16) a. 'he' b. 'she' c. 'it'

The question here is: How is the DP able to access the gender value of an individual outside of the sentential context and use it for verbal agreement? Sigurðsson answers this question by proposing the involvement of a mechanism called Context Scanning, which allows the DP to 'look up' the left periphery (the domain of discourse) and take reference from an item there. To elaborate, the DP is regarded as a phase, the edge of which is capable of probing the domain of CP for a referent with the relevant phi features. The nodal head for this mechanism is D. The information thus obtained is recycled into the *n*P below and then used for clausal or DP internal agreement. This process is illustrated in (17) below, where 'he' receives its gender value from the CP domain by virtue of the DP scanning the context to obtain this information, and passing it on to the noun via D.

(17)



This mechanism is particularly relevant for the cases highlighted in (18) where nouns such as 'Clinton' carry a gender value that corresponds to a referent (generic or particular) in the discourse. The lexical entry by itself is underspecified for gender. When 'Clinton' refers to a female referent, the lexical item is marked with the [FEM] form, and it changes accordingly when the referent is male.

(18) from Sigurðsson (2019)

a. Clinton$_i$ campaigned hard, but she$_i$ lost in the end.

b. Clinton$_j$ campaigned hard, and he$_j$ won in the end.

Sigurðsson's account of context-scanning and the subsequent recycling of discourse level features can also explain to a set of common nouns, such as 'student', 'teacher', etc. Laleko

(2018) provides a detailed sketch of animate nouns in Russian that are underspecified for gender values and depend on the sex of their referent in the discourse to endow them with a gender value. Similar cases in Spanish are discussed by Alexiadou (2004). The focus of these works is on identifying these nouns as a distinct natural class - nouns without inherently specified gender - and to account for the various ways in which the gender values obtained from the discourse are encoded on them morphologically. The novelty of the current proposal lies in the fact that we can take Sigurðsson's account of gender assignment to pronouns, and extend it to explain gender assignment to these nouns as well.

As outlined by Laleko and Alexiadou independently, languages have certain animate nouns that crucially depend on a referent in the discourse to assign them a gender value. These also come under the umbrella of biological gender, but with the distinction that these are cases of variable biological gender, as opposed to the previously discussed fixed biological gender. What we need is a mechanism to assign gender to these nouns.

Tamil has a set of nouns that fall under this category. All the common nouns given in (19) have [MASC] and [FEM] counterparts, each one corresponding to a set of referents.

(19)
a. *kadaikkaar-ar*
shopkeeper-MSG.HON
'male shopkeeper'

b. *aasiriya-r*
teacher-MSG.HON
'male teacher'

c. *paNakkaar-an*
rich person-MSG
'rich man'

d. *kadaikkaar-amma*
shopkeeper-FSG.HON
'female shopkeeper'

e. *aasiriy-ay*
teacher-FSG.HON
'female teacher'

f. *paNakkaar-i*
rich person-FSG
'rich woman'

We adopt Sigurðsson's Edge-Computation mechanism to account for all those cases where common nouns, which are inherently unspecified for gender, obtain a gender value from the discourse domain. Using the same derivational mechanism in (17), gender assignment to the nouns in (19) can also be explained. The lexical items are first merged as N. At this point in the derivation, they are not specified for gender values. The functional D liaises with the discourse-domain and obtains a gender value from there. The obtained gender value is then encoded on the noun, to be realised by the PF component with appropriate morphology.

A major advantage of this proposal is that allows for different gender specifications to exist on the same root without having to represent them multiple times in the lexicon. To elaborate, (19a) and (19d) are realised as two distinct lexical items, but the are not represented as two distinct items in the lexicon. Because it is possible to obtain the two forms derivationally, the lexicon need not make note of them as separate entries. Thus, we are able to reduce some amount of redundancy in the lexicon by positing a derivational mechanism to obtain gender values from the discourse.

Sigurðsson's account also bolsters the idea that functional heads are crucially involved in gender assignment. In this case, it is the head D that aids in the process of assigning gender to gender-variable common nouns and pronouns. For ease of exposition, we shall call this entire process and its outcome as discourse-based, or variable biological gender, whereby a functional head (D) scans the context for a referent in the discourse that can lend its gender value to items inside the DP.

2.3.5 Interim Summary I

In this section we reviewed four accounts of gender assignment, with two of them having preferences for the lexicon as the locus of gender assignment and two of them favouring the position that functional heads need to be involved. We rejected Picallo and Longobardi's GenP hypothesis because of the issues pointed out in Section 2.3.1. From Kramer's proposal we adopt the idea that *n* is involved in the assignment of one kind of gender (grammatical), and from Sigurðsson's account that D is involved in cases where the gender value of a nouns comes about from a referent in the discourse. Overall, we are still within the realm of lexicalism, and posit that gender information is stored in the lexicon. Biological gender (when it is fixed) is present on the lexical head N. Other types of gender are introduced to the derivational spine by designated functional heads.

With this setting the backdrop, let us recall the empirical facts discussed in Section 2.2: Tamil has only biological gender, and Hindi-Urdu has both, biological as well as grammatical gender. A crucial position maintained here is that the split between biological gender and grammatical gender is a clear and significant one, and one that is marked in terms of whether the feature is carried on a lexical or a functional head. In the subsequent section, we shall delve deeper into this idea and provide a comprehensive account of the gender assignment taking place in Tamil and Hindi-Urdu.

2.4 Structural Representations of Gender

The theoretical accounts of gender surveyed in the previous section are diverse in their approaches to how gender is assigned to nouns. However, there is one recurring theme that almost all of them follow; the presence or absence of semantic correlation is a productive way to classify different types of gender. The gender value of animate entities is understood to have a semantic basis, whereas that of inanimate objects is understood as not having any semantic basis; it could be assigned arbitrarily to nouns.

While this kind of characterization is informative about the semantic identity of gender features, it is not informative of the structural representation of gender. To go back to the literature, Kramer's work raises the question about where gender features are located, but takes recourse to semantics to answer the question. Kramer posits that gender features are located on *n*, but the different types of gender are distinguished solely on the basis of whether the feature is semantically interpretable or not. The idea that biological gender and grammatical gender have distinct semantic correlates finds support in other accounts as well, such as those of Alexiadou and Sigurðsson.

The question that remains unanswered is whether different types of gender have distinct representations in the syntactic component. In other words, can syntax independently differentiate between biological gender and grammatical without invoking the semantic correlates? The syntactic encoding of gender can be realized in two ways:

- In terms of location where the different types of gender are located in the derivational spine
- In terms of the different mechanisms by which different types of gender are realised on nouns

Option (i) is proposed in the Distributed Gender Hypothesis posited by Steriopolo and Wiltschko (2010). Steriopolo and Wiltschko's account is based on the premise that gender a syntactically heterogeneous category, and that it occupies more than one position in the DP spine. According to this account, biological gender is directly present on the root/ lexical item. Grammatical gender is present on the functional head *n*, which is higher up the DP. These distinct positions for biological and grammatical gender also reflect a lexical-functional divide between the two. Biological gender is located on a lexical head, whereas grammatical gender is the domain of functional heads such as *n*. Connected to Steriopolo and Wiltschko's account, we move on to option (ii): How do we explain these differences in terms of syntactic mechanisms? In order to answer this question, we assemble evidence from nominalisation.

Nominalisation is the syntactic process by which either uncategorized roots, or items of other categories such as verbs and adjectives are transformed into nouns (Alexiadou, 2010). It is well-established that the node that facilitates this process is *n*. To simplify things, merging *v*P with *n* results in a deverbal nominal, and merging *a*P with *n* results in a deadjectival nominal. In languages where all nominals have a gender value, the syntactically formed nominal is no exception. Deverbal and deadjectival nominals take on the gender value associated with the particular *n* they merge with.

With this context, let us now look at some examples of nominalisation. It is particularly relevant that not all types of gender are found on nominalised structures. Nominalised structures occur with grammatical gender (20) or by taking gender reference from the discourse (23), but never with biological gender, as shown by the Hindi-Urdu and Amharic examples given below.

(20) Deverbal Nominals: Hindi-Urdu

a. *bhaag* –> *bhaagnaa*
'run'(V) –> 'running.MSG'(N)

b. *so* –> *sona*
'sleep'(V) –> 'sleeping.MSG'(N)

c. *dho* –> *dhulaai*
'wash'(V) –> 'washing.FSG'(N)

d. *paDh* –> *paDhaai*
'study'(V) –> 'studying.FSG'(N)

(21) Deadjectival nominals: Hindi-Urdu

a. *komal* –> *komaltaa*
'soft'(A) –> 'softness.FSG'(N)

b. *aslii* –> *asliyat*
'real'(A) –> 'reality.FSG'(N)

c. *mehengaa* –> *mehengaai*
'expensive'(A) –> 'expensive-ness.FSG'(N)

The presence of gender on the nominalised structures in (20) and (21) is evidenced by the agreement relations between the deverbal nominal and its modifiers in (22) below:

(22) a. *usk-aa bhaagnaa*
3SG-MSG running.MSG
'his/her running'

b. *usk-ii paDhaai*
3SG-FSG studying.FSG
'his/her studying'

c. *usk-ii komalta*
3SG-FSG softness.FSG
'his/her/its softness'

d. *baDht-ii mehengaai*
rise.IMPF-FSG expensive-ness.FSG
'rising expenses'

The cases above showed nominalised structures taking up grammatical gender values. Nominalisation is also open to the possibility where gender values come from a referent in the discourse, as in the Amharic examples in (23), where the gender value of the nominalised 'Ethiopian' varies with respect to whom the term is referring to. If the referent in the discourse has MASC gender, the same feature is passed on to the nominalized structure, and the same happens for the FEM feature.

(23) Amharic: (Kramer, 2016, p. 190)

a. *ityop'p'iya –> ityop'p'iy-awi*
'Ethiopia' –> 'Ethiopian' (MASC referent)

b. *ityop'p'iya –> ityop'p'iy-awit*
'Ethiopia' –> 'Ethiopian' (FEM referent)

However, this option is not available for biological gender. We do not see instances where nominalised structures receive a gender value that comes from a biologically gendered entity, such as 'boy.MSG' or 'girl.FSG'. This is one type of gendered nominalised that is not obtained in grammars. Essentially, lexically specified biological gender is not an option for nominalised structures.

In other words, grammatical gender can be obtained either through inherent specification (as seen on inanimate nouns in some languages) or by the syntactic process of nominalisation. Similarly, variable biological gender is available to prototypical nouns as well as nominalised structures. However, fixed biological gender is different. It can be obtained only via inherent specification. Only when a noun is inherently specified with a gender feature can this feature refer to biological gender. Fixed biological gender cannot be attributed through derivational processes like nominalisation.

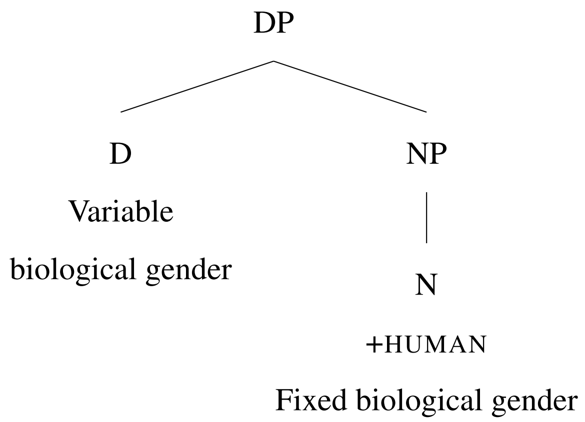
When we begin probing into the cause of the above noted disparity, we find ourselves turning back to the first question, about the location of these gender features. We then speculate that biological gender and grammatical gender must be on different locations in the derivational spine, and that the difference between them must be strong enough to allow only one of the two (grammatical gender) to participate in nominalisation. Recall from Steriopolo and Wiltschko's account that there is a lexical-functional divide between biological and grammatical gender. The empirical evidence from nominalisation corroborates this divide by allowing access to nominalisation for the type of gender located in the functional domain, and not to the one in the lexical domain. Further, there is also a difference between fixed and discourse-dependent biological gender. The former comes about through lexical specification, whereas the latter involves the functional head D in order to materialise on nouns. The fact that nominalisation can have access to one of these but not the other encourages us to pursue the idea that there is indeed a split within biological gender, with the fixed and the variable iterations having distinct structural representations.

To summarise, we now have empirical evidence to confirm the distributed gender hypothesis. The presence of gender on nominalised structures is in line with the idea that biological gender and grammatical gender are located on distinct nodes inside the DP, and that they can be accessed differently, as shown by the nominalisation examples. This answers our question that there is indeed a syntactic/ structural distinction between biological and grammatical gender. With this architecture in mind, and by taking into consideration the empirical facts in Section 2.2, we shall now provide distinct structural representations for the gender systems of Tamil and Hindi-Urdu.

2.4.1 Representation of Gender in Tamil

Recall from Section 2.2.1 that Tamil has gender only on [+HUMAN] nouns, i.e., biological gender, and a mechanism to assign gender values to pronouns and common nouns using Edge-Computing. According to the distributed gender hypothesis, biological gender originates on the lexical head N, and following Sigurðsson's account, we posit D as the locus of discourse-dependent gender. This is illustrated in (24) below:

(24)



As shown above in (2), and repeated here as (25), Tamil has gendered III person pronouns. The gender values for these and the common nouns in (26) comes when the edge of the DP scans the context for a referent that can act as an antecedent for the pronoun. To be clear, these are also instances of biological gender. They are, however, distinguished by not having an inherent specification for gender values. The domain of discourse plays a central role in the process of assigning gender to these nouns.

(25)
a. *avan*
'he' (for all human male nouns)

b. *ava*
'she' (for all human female nouns)

c. *adu*
'it' (for all non-human nouns)

(26)
a. *maana-van*
student-MSG
'male student'

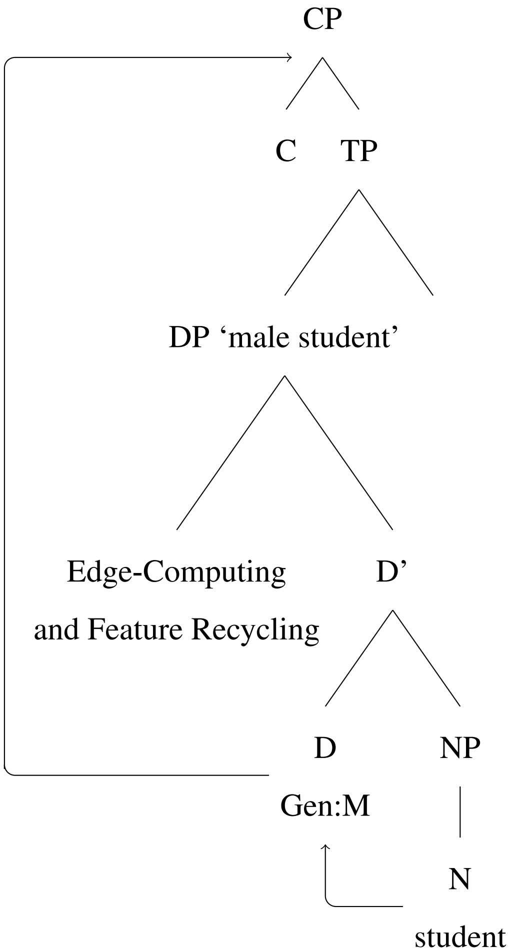
b. *aasiri-yar*
teacher-MSG.HON
'male teacher'

c. *maana-vii*
student-FSG
'female student'

d. *aasiri-yai*
teacher-FSG.HON
'female teacher'

The transfer of gender values from the discourse domain to a lexical item happens via the functional head D, which can scan the left-periphery for a potential referent and then recycle those phi features to the lexical item on N. The representation for (26a) is given in (27).

(27)



Apart from this, there is also fixed biological gender in Tamil. This type of gender is present in Tamil only on [+HUMAN] nouns , shown in (1) and repeated here as (28).

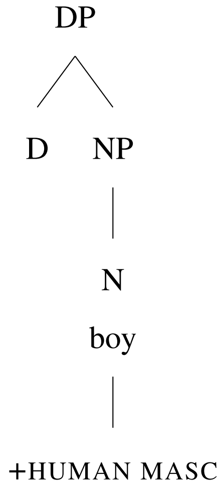
(28) a. *paiyan*
'boyMSG'

b. *poNNu*
'girl.FSG'

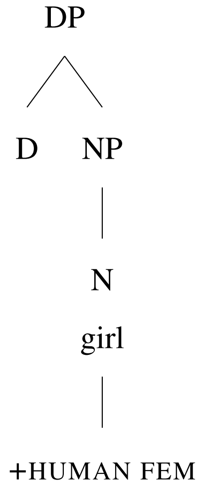
This type of gender is located at N, and will be interpreted semantically when the derivation reaches the LF component. N in Tamil has a binary specification of [MASC] or [FEM]. These specifications are inherent; present in the lexicon as part of attributive information for the lexical item. The representations of (28a) and (28b) are shown below as (29a) and (29b) respectively.

(29)

a.

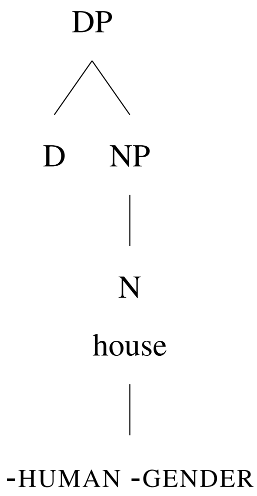


b.



When the noun is specified as [-HUMAN], there is no gender specification on N, as shown in (30). This can be captured by invoking the feature geometry approach proposed by Harley and Ritter (2002). According to this framework gender is included in the feature composition of nouns in Tamil only when the representation includes an activated [HUMAN] specification. In other words, gender features are parasitic on [+HUMAN] in Tamil.

(30)



Essentially, the two heads crucial for gender in Tamil are D and N. D is responsible for discourse-based biological gender on pronouns as well as some common nouns, which is variable depending on the gender value of the antecedent. N comes in two versions - gendered and ungendered. The gendered N is inherently specified for a fixed gender value, either MASC or FEM. There is no grammatical gender, i.e., *n* in Tamil does not carry gender values.

The discussion about gender in Tamil needs further clarification, with respect to nominalization. Kramer (2015) posits that cross-linguistically nominalizations are capable of being gendered when *n* carries gender features. Markova (2010) uses Bulgarian nominalization to demonstrate the connection between gender and nominalization. Her claim is that Bulgarian gender suffixes (or various derivational affixes marked for gender) can act as nominalizers. Thus, any nominalized structure is invariably endowed with gender features. Markovskaya (2012) uses Russian to show that the gender of derived words is determined by nominalizing affixes, which are associated with a particular gender value. Once again, the gender specification on n is central to the gendered nature of nominalizations. Soare (2019) also echoes this view. Using French nominalization data, she claims that the nominalizing affixes (*–ation, -ment, -age, -ée* in French) introduce gender specification.

There is a common theme seen among the works that discuss the connection between gender and nominalizations: all of them attribute the presence of gender on nominalized entities to the nominalizing affix. Additionally, another commonality between the languages used as empirical evidence for the same (Bulgarian, Russian and French) is that they have natural as well formal systems of gender. In other words, these are languages with grammatical gender. When a language has the option of semantically vacuous (grammatical) gender on inanimate nouns, it naturally follows that derivational/ nominalizing affixes too will be gendered. Hence, nominalizations are gendered in the languages considered here.

However, Tamil is a system that is built differently. As discussed previously, it has a gender system that is entirely based on semantics. Grammatical gender values are not present in the language. An expected correlation in such a system is that derivational/ nominalizing suffixes, too, are devoid of gender features. Thus, nominalised entities in Tamil lack a gender value. Consider (31), which has a gendered noun (31a), as evidenced by [FEM] verbal agreement. When the same noun is attached with a nominalizing affix *-mai*, the resulting structure is not gendered (31b). This is evidenced when (31b) triggers the same agreement morphemes as a non-gendered noun 'hand' does in (31c). [FEM] marking is not allowed on the verb in (31b).

(31) a. *anda peN kannadi-la teri-nj-aa*
that woman.FSG mirror-LOC visible-PST-FSG
'That woman was visible in the mirror.'

b. *avaL-oda peN-mai kannadi-la teri-nj-udu/*aa*
she-POSS woman-hood mirror-LOC visible-PST-NEUT(non-gendered) / *FSG
'Her woman-hood (femininity) was visible in the mirror.'

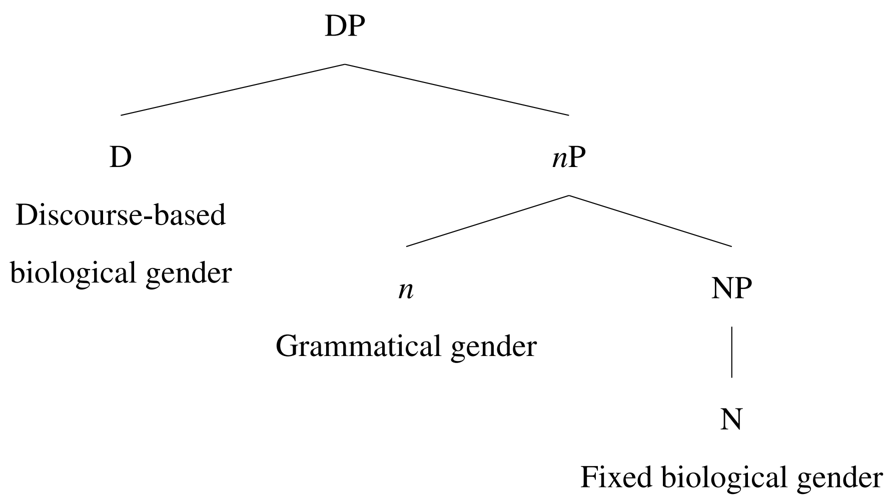
c. *avaL-oda kai kannadi-la teri-nj-udu*
she-POSS hand mirror-LOC visible-PST-NEUT (non-gendered)
'Her hand was visible in the mirror.'

Essentially, there is no way to form gendered nominalizations in Tamil. Even a gendered lexical item like 'woman' loses its gender value upon being nominalized. Nominalizers are strictly non-gendered in Tamil. With these facts, we are able to establish that gender features are not derivationally accessible in Tamil. *n* in Tamil can nominalize, but it neither carries gender features, nor accesses the gender features of the [+HUMAN] nouns it applies to. The only way a lexical item can be assigned a gender value is through inherent specification at the lexicon.

2.4.2 Representation of Gender in Hindi-Urdu

Section 2.2.2 showed that Hindi-Urdu has biological gender on animate nouns, grammatical gender on inanimate nouns as well as a mechanism to obtain the gender values of a referent in the discourse. Biological gender, a lexical property, is situated on N, grammatical gender on *n* and gender values from the discourse are captured and brought into the DP via D, as shown in (32).

(32)



As shown above in (6), repeated here as (33), Hindi-Urdu has common nouns, the forms of which vary relative to the gender of the discourse participant they refer to.

(33)

a. *chaatr*
'male student'

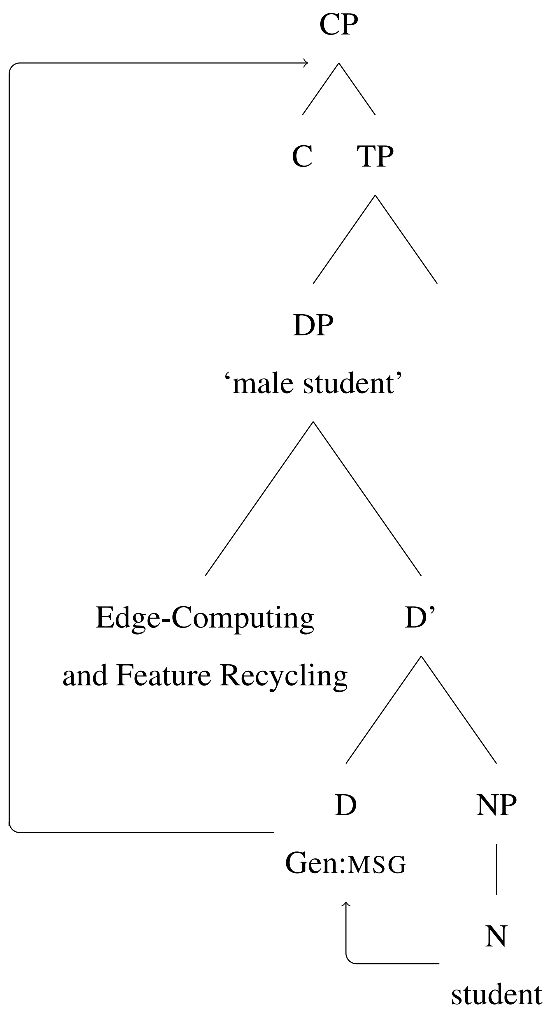
b. *adhyaapak*
'male teacher'

c. *chaatraa*
'female student'

d. *adhyaapikaa*
'female teacher'

Similar to the process laid out for the examples in Tamil, the functional head D interfaces with the discourse domain to access the gender features from there and then instantiate them inside the DP. This is illustrated for (33a) as (34) where the [MASC] features of a referent in the discourse is passed down the DP spine to the lexical item 'student'.

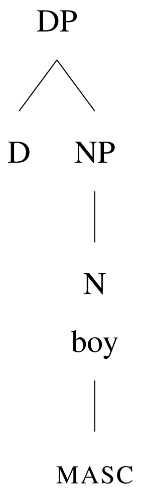
(34)

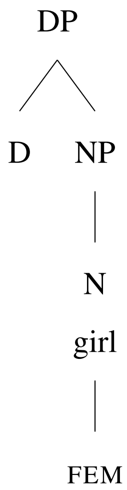


The next type of gender is fixed biological gender, represented on N. In Hindi-Urdu, all animate nouns are allotted a gender value. This is one type of inherent specification on Hindi-Urdu nouns. Gender on N can be [MASC] or [FEM], as shown in (35) below. (36a) shows the representation for [MASC] nouns and (36b) shows the representation for [FEM] nouns.

(35) a. *aadmii*
‘man.MSG’

b. *aurat*
‘woman.FSG’

(36) a.


b.


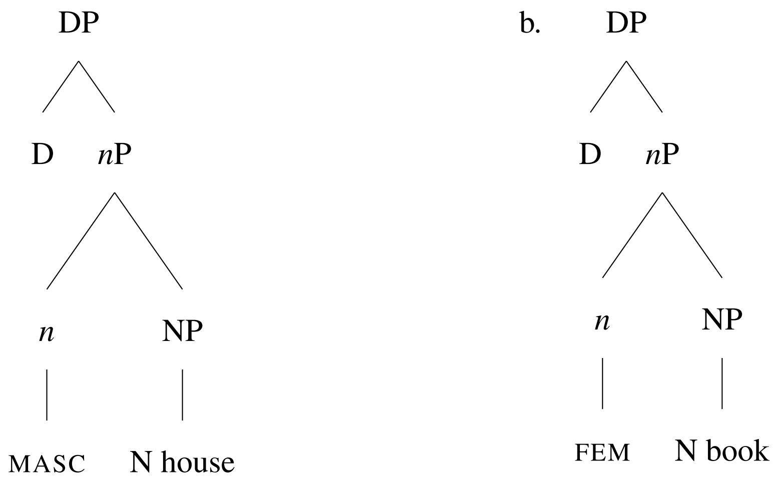
The third type of gender found in Hindi-Urdu is grammatical gender, available inanimate objects (37).

(37) a. *ghar*
‘house.MSG’

b. *kitaab*
‘book.FSG’

The locus of grammatical gender is the functional head *n*, and this is the type of gender that participates in nominalization, as discussed previously. (38a) gives the representation of [MASC] *n*, which generates nouns like (37a), and (38b) gives the representation of [FEM] *n*, which gives rise to nouns like (37b). These gender values are different from biological gender in that when the derivation reaches the LF component, these features will not be interpretable to the semantic component; they will be ignored. As noted above, *n* is the node responsible for grammatical gender and nominalisation.

(38) a. b.



To summarise, nouns in Hindi-Urdu can be assigned gender in two different ways: through inherent specification (fixed biological and grammatical gender) and using Edge-Computation mechanisms (discourse-based biological gender). Thus, the functional heads involved here are D and *n* along with the lexical head N.

2.4.3 Interim Summary II

At the outset of this chapter, we adopted the position that gender assignment is structural; the DP spine with its set of lexical and functional heads is collectively involved in the assignment of gender to nouns. In this section, we have reviewed some existing accounts of gender assignment and augmented them by providing exact mechanisms by which it can take place. Another major point was that a clear distinction must be made between the different types of gender assigned to nouns, and that these distinctions must be represented as such in the structure. We have shown that by invoking certain functional heads inside the DP. Biological gender, when it is fixed, is represented at the lexical head N, and when it is variable, at the functional head D. Grammatical gender involves the functional head *n*.

To reiterate the point, gender assignment in Tamil and Hindi-Urdu differs in terms of which heads are involved. Tamil only has the involvement of D and N, whereas Hindi-Urdu has, along with these two, the involvement of *n* to account for grammatical gender. Another major point of differences between the two languages is in terms of the connection with [+/-HUMAN] features. Gender is parasitic on [+HUMAN] in Tamil, but not so in Hindi-Urdu. Thus, while all nouns in

Hindi-Urdu are assigned gender values, only those with [+HUMAN] specification are in Tamil. With this information setting the context, we shall now begin to lay out some hypotheses and predictions for how these categories will fare when Tamil as a heritage language co-exists with Hindi-Urdu as the dominant language.

2.5 Current Proposal: Implications for Language Contact

This section will have two parts: first, we present some case studies of gender in language contact that focus particularly on different types of gender and the split between the two. After an overview of these studies, we shall derive hypotheses from these and base our predictions on them.

2.5.1 What do existing accounts tell us?

The idea that gender agreement is not entirely resilient to change in a heritage grammar has received a lot of attention within the domain of generative syntax. One of the most well known proposals is from Lohndal and Westergaard (2016). Lohndal and Westergaard take up the case of heritage Norwegian spoken in the US with English as the L2. The global claim of their paper is that gender is indeed a vulnerable category in language contact. The empirical evidence for the claim comes from a study of the Corpus of American Norwegian. The main phenomenon under question is nouns with their gender value appearing on indefinite determiners. All nouns in Norwegian are gendered, and indefinite determiners canonically agree with the gender value of the noun, as shown in (39), where the indefinite determiner agrees with the [MASC], [FEM] or [NEUT] feature specified on the noun.

(39)	a. *en* *butikk* a-MSG store.MSG 'a store'	b. *ei* *uke* a-FSG week.FSG 'a week'	c. *et* *tre* a-NSG tree.NSG 'a tree'

As can be seen from (39), the gendered noun in Norwegian comes in three variants [MASC], [FEM] or [NEUT]. Lohndal and Westergaard's analysis of Norwegian gender is centered around this. The most significant outcome of their study was to identify that heritage Norwegian contained rampant overgeneralization of the [MASC] form of the determiner; nouns which are encoded as [FEM] or [NEUT] in the baseline variety of Norwegian are also used with the [MASC] determiner in the heritage variety. Within the two affected variants, it was recorded that [NEUT] was the most vulnerable category. Lohndal and Westergaard's work treats the heightened vulnerability of the value [NEUT] as puzzling, but I propose an alternate explanation for the phenomena observed.

If one were to consider the distinction between biological and grammatical gender, these facts have a straightforward explanation. The value [NEUT] contains nouns with grammatical gender, the assignment of which tends to be opaque and requires noun by noun learning. As an extension of this explanation, the reason why [MASC] and [FEM] are relatively stable types could be that they also contain biologically masculine and feminine nouns respectively. It is plausible that the target inconsistent nouns in the MASC and FEM categories are mostly the grammatical gender nouns. Thus, heritage Norwegian presents us with an example where analysing heritage grammars in terms of the different types of gender available certainly adds depth to the explanation.

A similar proposal is carried out by Laleko (2018), who explores heritage Russian in the US by taking into consideration the different types of gender assignment that are available in the grammar. To provide a gist of the study, Laleko's study contains three types of gendered nouns:

- Lexical gender assignment
- Morphological gender assignment
- Discourse based gender assignment

Lexical assignment of gender refers to nouns such as (40) which are lexically marked for a fixed gender value. This value is predetermined and independent of the context of the noun's occurrence.

(40) a. *sestra*
sister.FSG

b. *brat*
brother.MSG

Apart from this, there are some animate nouns where the addition of derivational morphology to the [MASC] form of the noun results in the [FEM] form, as seen below in (41).

(41)

a. *student*
student.MSG
'male student'

b. *slon*
elephant.MSG
'male elephant'

c. *student-**ka***
student-**FSG**
'female student'

d. *slon-**ixa***
elephant-**FSG**
'female elephant'

The cases presented above are examples of fixed gender values. In contrast, there are nouns in Russian that display variable gender, taking cues from an antecedent in the discourse domain. When a particular referent is known to be male or female, the noun must obligatorily control masculine or feminine agreement on agreeing heads. These are called hybrid or common gender nouns.

(42) a. *sudja*
'male or female judge'

b. *kollega*
'male or female colleague'

With these systems in place, Laleko conducted a large scale acceptability judgement study with the prediction that lexical and morphological assignment of gender will be stable in heritage Russian, whereas hybrid and common gender nouns will show maximum disparity in comparison to baseline Russian. This prediction is borne out; The data confirm that domains which involve the interface between the syntactic spine and the discourse component (42) tend to be the most vulnerable category in terms of gender assignment strategies being preserved in heritage grammars.

While Laleko does not illustrate the differences between fixed (40-41) and variable (42) gender in terms of their internal structure, the point still remains valid that a distinction must be made between these two types of gender assignment: lexically endowed gender and discourse-related gender assignment.

These findings are in line with previous section that argued for a distinction between different types of gender. The distinction was made in terms of the syntactic location of the features and the derivational processes in which they participate. In this vein, we now proceed to observe how this distinction fares in contexts of language contact, particularly those with some amount of asymmetry in terms of the two languages involved.

Sorace and Filiaci (2006) provide one of the most widely accepted hypotheses regarding how different grammatical phenomena fare in situations of language contact. They hypothesise that narrow syntactic properties are completely acquirable in a bilingual context, whereas interface properties involving syntax and another cognitive domain may not be completely acquirable. This is now referred to as the Interface Hypothesis (IH). IH mainly emerged from cross-linguistic studies on null subjects. In languages that allow pro-drop, the speaker's grammar must encode the fact that a null subject is syntactically licensed, and that there may be semantico-pragmatic conditions governing the appropriated usage of null or overt subjects, depending on factors such as the context of the utterance. It has been found that grammatical phenomena like the null subject condition tend to be treated as optional in contexts of language contact: The pro-drop condition may undergo significant changes in a language variety where it coexists with other L2, L3, etc, thus leading to the generalisation that discourse related phenomena are vulnerable in bilingual contexts.

While IH has predominantly aided the study of pro-drop in bilingual contexts, there are other venues where it can be applied fruitfully. In this instance, we use IH to help characterise how one type of gender assignment mechanism identified in this chapter, Discourse-based biological gender in particular, fares in heritage Tamil. As illustrated in Section 2.4, the gender value found on pronouns and gender-unspecified common nouns originates in the domain of discourse. Through the mechanism of Edge-computing, the functional head D scans the layer of context in pursuit of a discourse referent that can pass on its gender values to the DP.

This is a clear example of a grammatical phenomenon that involves an interface between the syntactic configuration of the DP and the context layer to extract gender values from there. Thus, we understand Discourse-based gender assignment as an interface phenomenon. Extending IH to this phenomenon, we expect it to undergo significant change in the variety of Tamil that exists as a heritage language in New Delhi. Based on this, we then predict that assignment of Discourse-based biological gender will not be at ceiling in the grammar of heritage Tamil. Empirically we expect to see mismatches in the assignment of gender to pronouns and common nouns. Considering the fact that there are two gender values in Tamil, [MASC] and [FEM], we expect to see considerable overgeneralisation of the [MASC] form of the pronouns as well as common nouns, with the [FEM] counterparts not being assigned systematically.

Another important or well-attested hallmark of heritage grammars cross-linguistically is that some aspects of the grammar are consistently resilient to structural change, whereas others are invariably vulnerable (Westergaard and Kupisch, 2021). Among the phenomena identified to be resilient are syntactic features on lexical heads. An ancillary benefit comes in the form of these features being also interpretable; i.e. having a semantic correspondence.

The second prong of our inquiry into gender assignment deals with biological gender, which is represented on N and considered to be a lexical property. This type of gender is deemed to be interpreted when the derivation reaches the LF component. With this as the backdrop, let us now take a step back and return to the idea that there are three types of gender, a distinction that merits making. Now our question is: How can we implement the gradation identified in gender so far within the context of a heritage grammar? With this question framing our inquiry, we constructed a study that will let us test our hypotheses empirically.

2.5.2 The Current Study

The current study aims to provide empirical validation for the structural distinctions carved out in gender in the previous sections. In essence, Section 2.4 chalked out three distinct structural positions for gender within the nominal spine. Two of them - fixed biological gender and discourse based biological gender - are posited to be active in Tamil while Hindi-Urdu has grammatical gender in addition to these two. The context here is then that the grammar of the heritage language (Tamil) has a subset of the gender representations available to that of the dominant language.

Given such an alignment of features, the current study proceeds to investigate how the gender representations available to baseline Tamil fare in Tamil as a heritage language. Zooming in on the representation of gender in Tamil, it was proposed that there are two distinct mechanisms of gender assignment to nouns. In Tamil only [+HUMAN] nouns carry gender values, and so the two mechanisms have it in common that the gender value that ultimately appears on nouns is semantically interpretable to the LF component. The core argument, then, is that despite both of them being semantically identical, the structural differences between them are salient enough to direct the two to different paths in a heritage grammar.

This study is an investigation into how the two types of gender assignment cases fare in a heritage grammar, and this is done by eliciting data from heritage speakers in a picture description task that is explained below.

2.5.2.1 Questionnaire and Methodology

Recall that that aim of the study was to test for the presence of fixed and variable biological gender values in heritage Tamil. Methodologically, this is done by eliciting the production of gendered lexical items and pronouns. A picture description task was set up to carry this out. Speakers of heritage Tamil (n=23) were presented with a series of 2-dimensional images and they were asked specifically to identify the people in the image along with the activity they were performing. The images for this task were chosen carefully to have only one human character in each frame, to reduce any ambiguity.

Data for baseline Tamil was provided and corroborated by 10 native speakers of Tamil (age range=20-29). They were exposed to the same questionnaire, and asked to produce sentences that would elicit gendered lexical items and pronouns.

For instance Fig. 2.1a and Fig. 2.1b were presented to prompt the speakers to say sentences such as (43a) and (43b), respectively.

(a) Boy

(b) Girl

Figure 2.1: Prompts for Lexically Specified Gender

(43)

a. *oru* ***paiyan*** *padi-ki-raan*
a **boy.MSG** read-PRS-MSG
'A boy is reading'

b. *oru* ***poNNu*** *school-ku po-r-aa*
a **girl.FSG** school-DAT go-PRS-FSG
'A girl is going to schoool'

Given that this task focused solely on gender assignment on nouns, we do not consider the verbal agreement markers in (43) as an indicator of how gender fares in heritage Tamil. The only variable in question here is whether the right lexical items were used for 'boy' and 'girl' by the speakers. The instruction given to the participants was that they had to look at the images shown to them on a screen and use an appropriate Tamil lexical item to describe them.

The second part of the questionnaire was concerned with gendered pronouns. In order to elicit the usage of pronouns, participants were shown a series of images with the same character performing different activities. This setup, illustrated in Fig. 2.2 is one that evokes the usage of pronouns, given that the referent has been named at the outset, and has to be mentioned repeatedly.

Figure 2.2: Prompts for Pronouns

Here, the expected set of utterances include the usage of a lexical item to anchor the utterances with the subsequent sentences containing a pronoun that refers back to the anchoring lexical item, as given below in (44).

(44) a. ***oru paiyan*** *kaalay-la ezhu-nd-aan*
one boy morning-LOC wake-PST-MSG

'A boy woke up in the morning.'

b. ***avan*** *saapi-tt-aan*
he eat-PST-MSG
'He ate.'

c. *piragu* ***avan*** *school-ku po-n-aan*
then **he** school-DAT go-PST-MSG
'Then he went to school.'

d. *school-la* ***avan*** *vilayad-n-aan*
school-LOC **he** play-PST-MSG
'He played at school.'

e. *adhuku apram* ***avan*** *padi-c-aan*
that after **he** study-PST-MSG
'After that, he studied.'

f. *viitu-ku vandu* ***avan*** *toongi-tt-aan*
house-DAT come.CPM **he** sleep-PST-MSG
'He slept after coming home.'

Finally, for common nouns which obtain their gender values from a referent in the discourse, the speakers were shown images such as Fig. 2.3a and Fig. 2.3b in order to prompt the elicitation of lexical items such as (45a) and (45b), respectively.

(a) Male teacher

(b) Female teacher

Figure 2.3: Prompts for Common Nouns

(45)

a. *aasiriyar*
teacher.MSG.HON
'male teacher'

b. *aasiriyai*
teacher.FSG.HON
'female teacher'

To summarise, the questionnaire was designed in such a way that it could elicit lexical items with inherently specified biological gender, pronouns, as well as lexical items with discourse-dependent biological gender.

2.5.2.2 Hypotheses and Research Questions

The studies presented in Section 2.5.1 outlined the importance of making a structural distinction between different types of gender assignment within the nominal spine. In that vein, it was presented in this chapter that Tamil makes a distinction between fixed or inherent biological gender and discourse-dependent or variable biological gender. Recall that the two arise from two distinct operations. The former comes about on nouns through lexical specification, whereas the latter is endowed on nouns through context scanning by D and subsequent recycling of the gender values found in the domain of discourse.

Another point of discussion in this chapter was with regard to the Interface Hypothesis, which posits that syntactic phenomena which interface with the discourse component will be susceptible to attrition, or structural change at the least, in heritage grammars. When we extend the Interface Hypothesis to gender assignment in Tamil in a heritage context, one prediction that immediately emerges is that fixed biological gender and variable biological gender will fare differently in heritage Tamil. In other words, variable, or discourse-dependent biological gender will undergo attrition but fixed biological gender will remain stable.

2.5.2.3 Results of the Study

The 10 baseline Tamil speakers recruited for the study produced gendered lexical items and pronouns in a target-consistent manner. This section elaborates on the data elicited from heritage Tamil speakers. The first task was to use gendered lexical items to describe images depicting [+HUMAN] nouns such as 'boy', 'girl', 'man', 'woman', etc. The speakers performed at ceiling with regard to this task. They produced utterances such as (46) given below. In this set of tasks, our focus is only on the usage of target consistent lexical items, so we shall set aside the issue of gender agreement morphology for the purposes of this chapter. The responses presented here contain full sentences replete with gender marking on verbs as well, but those are not considered for the current analysis.

Lexical items such as 'boy' in (46a), 'girl' in (46b), 'man' in (46c) and 'grandma' in (46d) are examples of nouns that have intrinsic, fixed gender values. These values are specified in the lexicon, so the nouns enter the derivation with their gender value pre-determined.

(46) a. *oru* ***paiyan*** *breakfast saapida-r-aan*
one **boy** breakfast eat-PRS-MSG
'A boy is eating breakfast.'

b. *anda* ***ponnu-kku*** *inniki porandaNaaL*
that **girl-DAT** today birthday
'It is that girl's birthday today.'

c. *oru* ***aaL*** *anga nada-kar-aaru*
one **man** there walk-PRS-MSG.HON
'A man is walking there.'

d. *oru* ***paati*** *poo vik-ar-anga*
one **grandma** flower sell-PRS-FSG.HON
'An old lady is selling flowers.'

As expected, these nouns continue to populate heritage Tamil. They do not pose any challenges for heritage speakers, presumably owing to their lexical specification and semantic interpretability; all these nouns come with inherent gender specification, and they will be interpretable to LF, as they denote [+HUMAN] nouns that have semantic correlates for gender. The factors mentioned above make lexically specified biological gender conducive to persist even in conditions of restricted input, such as heritage contexts. Essentially, there is lexical learning involved with biological gender too. However, it is plausible that their semantic interpretability makes them easier to learn. This provides an explanation for why heritage speakers of Tamil performed at ceiling with regard to this task. There was no divergence observed in this task. The speakers produced 105 tokens of [FEM] and 113 tokens of [MASC] nouns in total, and all 218 tokens were target-consistent.

The next part of the study focused on heritage speakers' grasp of pronouns. In order to simulate contexts that typically evoke pronouns, the speakers were asked to build a narrative using a sequence of pictures, and narrate the events as a story. As explained in the preceding section, the main character was kept constant throughout the pictures, as that could prompt the narrator to use pronouns to recall the character. Gender values on pronouns are not intrinsic to the lexical items. These are contingent upon locating a referent in the discourse layer. Because of the role of discourse in computing the gender values of these nouns, the prediction was that there will be some attrition in the domain of gendered pronouns.

The opening line of the story (47) and the subsequent utterances are given below.

(47) *oru paiyan kaalai-la ezhu-nd-aan*
one boy morning-LOC arise-PRS-MSG
'A boy woke up in the morning.'

The elicited utterances in this task can be grouped into four sub-types: (i) usage of the target consistent gendered pronoun as in (48a), (ii) pro-drop accompanied by verbal gender agreement as in (48b), (iii) repeated usage of the lexical item itself as in (48c), and (iv) absence of pronouns and any other indicators of gender as in (48d).

(48) a. *avan saapi-tt-aan*
he eat-PST-MSG
'He ate.'

b. *piragu saapi-tt-aan*
then eat-PST-MSG
'Then (he) ate.'

c. *oru paiyan saapi-tt-aan*
one boy eat-PST-MSG
'A boy ate.'

d. *saapad-ra maari*
eat-INF like
'like (someone is) eating'

We consider responses such as (48a) and (48b) to be target consistent, as both are appropriate in a context where an antecedent is specified. Responses such as (48c), while not incorrect, are not entirely felicitous and are coded as semi-consistent. On the other hand, responses such as (48d) are coded as target-inconsistent, as they do not reflect any feature of the given antecedent. Responses for a [FEM] antecedent were recorded in an identical manner, and their distribution is given below in Fig. 2.4. The distribution in Fig. 2.4 shows that target consistent pronouns are present in approximately 75% of the utterances (n=136), with semi-consistent (n=38) and inconsistent (n=12) responses being present too. The total number of responses considered for this task was 186.

It is important to focus on the semi-consistent and inconsistent responses, which together form approximately 35% of the total responses. This is particularly relevant in comparison with

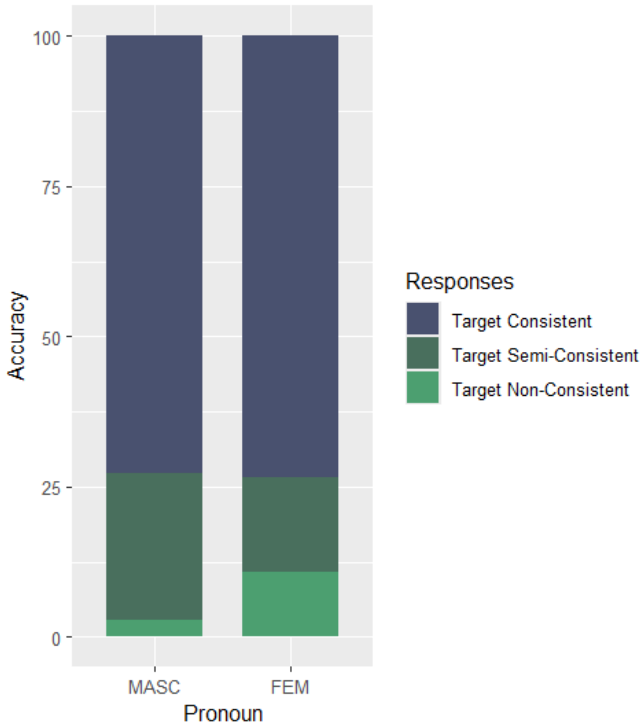


Figure 2.4: Distribution of MASC and FEM Pronouns

fixed biological gender nouns, where all the utterances were target consistent. What it tells us is that heritage speakers of Tamil do find it difficult to locate a referent and use the features of the referent in the form of a pronoun. They either choose to repeat the same lexical item that they began the narrative with, or only describe the action but without any agreement markers on the verb that could potentially establish a connection with the referent.

The final domain that we explore in this study is that of common nouns that appear in two different shapes depending on the gender value of the discourse referent they derive their features from. As explained in the previous section, the speakers were shown images of male and female humans in different professions. These terms continue to be used in common parlance in baseline Tamil. In earlier parts of the chapter, it was argued that the gender values on these nouns are also discourse-dependent and assigned through mechanisms such as edge-computation. Therefore, the current prediction is that heritage speakers will not perform at ceiling with regard to this task. We expect overgeneralisation of the [MASC] term to refer to both groups of individuals.

The elicited utterances in this task can be grouped as follows: (i) Usage of the target consistent gendered term, as in (49a), (ii) Using a [FEM] term for a [MASC] noun, or vice versa as in (49b), (iii) Using an English lexical item instead of a Tamil one as in (49c) and (iv) Not using any lexical items at all as in (49d). The context given to the speakers in this particular instance was the picture of a male milk seller selling milk.

(49) a. *paalkaar-ar paal vik-ir-aar*
milkman-MSG milk sell-PRS-MSG
'The (male) milk seller is selling milk.'

b. *paalkaar-anga paal vik-ar-aanga*
milkwoman-FSG milk sell-PRS-FSG
'The (female) milk seller is selling milk'.

c. *oru person paal vik-ar-aanga*
one person milk sell-PRS-FSG
'A person is selling milk.'

d. *paal vik-ara maari*
milk sell-INF like
'like (someone) selling milk'

For the purpose of this analysis we code (49a) as the only target consistent response, and all others - using a different gender value, using English lexical items, and not using any lexical items - as target inconsistent responses. The distribution of these responses is given below in Fig. 2.5. The total number of responses considered for this task was 178.

In Fig. 2.5, we see that the target consistent responses constitute approximately 60% (n=112) of the responses produced by the speakers. As predicted, using the edge-computation operation to recycle gender values from the discourse seems to be a challenge for this grammar. Once again, this is in stark comparison with intrinsic biological gender lexical nouns, where the accuracy of these speakers was at ceiling. Additionally, Fisher's exact test helps establish that there is a statistically significant association between the variables ($p < 0.001$). Further, the size of the sample (n=23) made it possible to verify that the target-inconsistent structures were distributed across all the speakers, thus ruling out the possibility of these structures clustering around a few speakers.

To summarise, Table 2.2 illustrates that while heritage speakers perform at ceiling when it comes to intrinsically gendered nouns, the same cannot be extended to those lexical items whose

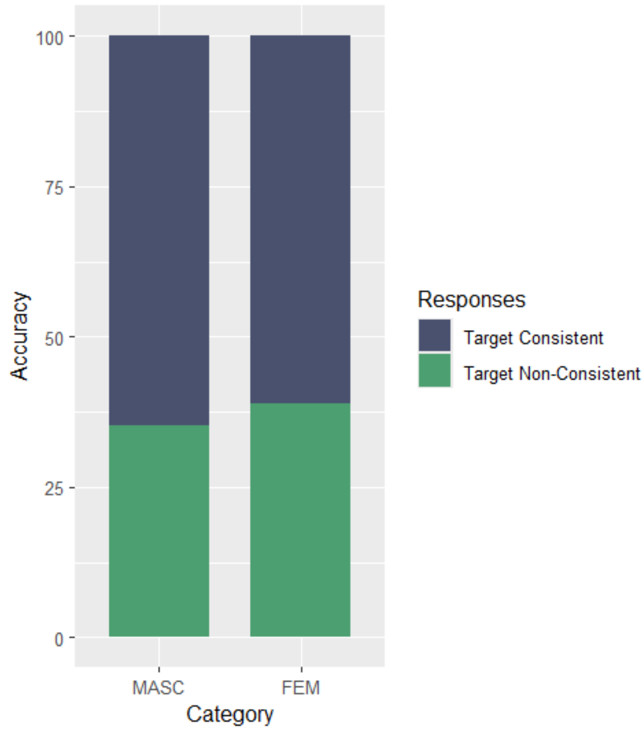


Figure 2.5: Distribution of MASC and FEM Common Nouns

Construction Type	No. of Tokens	Accuracy Rate
Fixed Biological Gender	200	100%
Gendered Pronouns	186	75%
Discourse based lexical items	178	63%

Table 2.2: Results of the study

gender assignment happens with the involvement of discourse elements. There is indeed a stark difference between the gender assignment strategies with only one of them being resistant in heritage Tamil.

2.5.2.4 Discussion

An analysis of the utterances produced by the speakers of heritage Tamil validates the predictions made above. Section 2.5.2.3 presented two entirely distinct trends in the data. As expected, lexically endowed biological gender, with its connection to [+HUMAN], was completely unaffected, with heritage Tamil speakers performing at ceiling in tasks that involved their usage. In contrast, there was noticeable attrition in the gender assignment of pronouns as well as discourse-dependent common nouns. There were several target-inconsistent structures in both tasks, with speakers not using pronouns, repeating lexical items instead of pronouns, completely omitting the gender information of the referent and so on. The difference between the two can be explained sufficiently by the stability of core syntax in contrast to the vulnerability of interface phenomena.

The first point of discussion is the semantic interpretability of biological gender; there is a strong correlation between gender and [+HUMAN]. Crucially, both the types of gender assignment considered in this chapter belong to this category. They are alike in that regard. The difference between them arises from the strategy by which they are assigned to nouns: Lexical specification or context-scanning and edge computation. Data from heritage Tamil clearly shows that the involvement of additional discourse-related operations in the latter is salient enough to induce variation in pronouns and common-gender nouns, while lexically specified gender remains stable. Based on these facts, we can further fortify the Interface Hypothesis: The domain of gender assignment is another venue where we can clearly demarcate the effects of core-syntactic operations interfacing with discourse components and how such an interface between the two makes grammatical phenomena vulnerable in the context of heritage grammars. In other words, studying gender in heritage Tamil has offered us a novel vantage point to observe the role of discourse-level operations in language contact, while making it clear that core syntax is impervious to these effects.

The second thread of discussion is related to the first. Despite attrition in discourse-related domains, there are certain grammatical phenomena which do not display any divergence in heritage Tamil. For instance, in the domain of gendered pronouns, the inconsistencies in the data obtained resulted from the absence of pronouns, or the complete absence of gender information. These were seen when the heritage speakers refrained from using pronouns at all, and instead chose to keep repeating the lexical items presented to them. Interestingly, there were no instances where speakers used [FEM] pronouns for a [MASC] referent, or [MASC] pronouns for a [FEM] referent. Those constructions could be regarded as mismatches, as they indicate the speakers were able to execute operations like Edge-Computation and Feature Scanning, but failed to map the exact gender values of the antecedent onto the pronoun. Such mismatches were nowhere to be found in these utterances. This leads us to connect to the earlier discussion about the semantic interpretability of gender in Tamil. Semantic interpretability may not make gender assignment impervious to discourse-effects, as predicted by the Interface Hypothesis, but it certainly restricts the kind of inconsistencies found in heritage Tamil.

At this point, the connection between gender features and the pronoun task in this study is worth mentioning. The findings about inconsistencies in pronoun usage in heritage Tamil is included as part of the current discussion in order to create a foil for, and highlight the stability of, gender features. The instability of edge-computation mechanisms for pronouns in heritage Tamil is in line with the predictions of the interface hypothesis. However, the fact that there are no gender mismatches even while occurring in unstable domains such as pronouns makes it clear that gender features are stable despite the instability in the surrounding domains.

The third strand of discussion is related to cross-linguistic effects of transfer from the L2. Recall that Tamil has only semantic gender, whereas Hindi-Urdu has semantic as well as formal gender. Our current study reveals that heritage Tamil is identical to the baseline in retaining the connection between gender and [+HUMAN]. None of the [-HUMAN] nouns produced in the elicitation tasks such as 'flower' in (46d) and 'milk' in (49a), for instance, appear with gender marking. In terms of the structural representations posited in this chapter, grammatical gender is a feature on the functional head *n*. Thus, *n* in Hindi-Urdu is specified for gender features, whereas *n* in Tamil is not. The non-appearance of grammatical gender in heritage Tamil can then be understood as *n* resisting transfer-induced change. This is in line with a well-accepted idea that functional features are difficult to acquire or modify in bilingual grammars.

To summarise, the study conducted in this chapter is useful in informing us about which domains may be particularly vulnerable or resilient in heritage grammars, contributing to the broad discourse about the principled ways in which heritage grammars undergo restructuring.

2.6 Conclusion

This chapter focused on understanding the gender system of heritage Tamil at its first order of representation: gender assignment to nouns. To that end, we first presented a sketch of the kinds of gender systems present in Tamil and Hindi-Urdu. The focus of the discussion was on the structural mechanisms by which different nouns are assigned their gender values. At the outset we made a distinction between biological and grammatical gender, with the former being semantically interpretable and the latter being uninterpretable. Further, within biological gender we made another split on the basis of the mechanism by which biological gender can be assigned to nouns: lexical specification, or context scanning followed by feature recycling. An elicitation study conducted on the basis of these concepts revealed empirical patterns that confirmed two broad ideas concerning heritage Tamil: a. Discourse related phenomena are vulnerable, and b. structural aspects of gender tend to be more resilient in heritage grammars.

Chapter 3

Gender in Adjectives

3.1 Introduction

Chapter 3 takes the discussion on gender forward by introducing a domain where gender agreement is observed: adjectives. The previous chapter dealt with the syntactic location of gender features, and the various mechanisms by which gender is assigned to different nouns. Subsequently, we recorded the implications of such gender assignment systems for heritage grammars in terms of stability and change. Essentially, Chapter 2 was about how gender is encoded representationally in grammars. Now we move into the domain of how gender participates in derivational processes: How does gender on nouns interact with other categories? The questions are about what happens when other categories, such as adjectives and verbs, have to co-vary with the noun with regard to the gender feature. Thus, this chapter is an exploration of gender as a part of the syntactic operation of agreement. As noted in Chapter 1, predicative adjectives are the closest agreeing heads to a noun, thus forming the locus of this chapter. We shall first present an account of adjectives in Tamil and then focus on the domains where they participate in agreement, which will then be studied from the perspective of a heritage variety of Tamil.

As with the preceding chapter, Chapter 3 continues to raise questions about which grammatical domains in heritage Tamil are expected to be more resistant to change, and which ones more susceptible to change. In this context, turning the focus towards adjectives in heritage Tamil is particularly advantageous: as we shall uncover in this chapter, there are two facets to the domain of adjectives in Tamil: (i) the derivational formation of adjectives, and (ii) the agreement relations seen on (some kinds of) adjectives. The former requires case-by-case learning on the part of the language acquirer, whereas the latter is part of core-syntax, and does not need to be learnt. By focusing on these two sub-domains of adjectives in the context of heritage Tamil, we will be able to establish a correlation between the reduced nature of Primary Linguistic Data (PLD) available, and the stability or vulnerability of grammatical domains in heritage grammars. Studying adjectives and gender agreement helps conclusively establish the duality of heritage grammars: learning-intensive phenomena undergo variation, whereas core syntax remains stable.

The chapter is organised as follows. Section 3.2 presents an understanding of what constitutes the category of adjectives in Tamil. Sections 3.3 and 3.4 capture the structural representation of adjectives in Tamil and Hindi-Urdu with equal focus on attributive and predicative variants. Section 3.5 presents the current study on how heritage Tamil represents adjectives and their agreement relations. The results of the study reveal that both domains fare differently despite sharing the same acquisition conditions. The difference is explained in terms of how much lexical learning is needed, and the extent to which there is support from the structural configuration of a grammatical phenomenon. Data from heritage Tamil will be used to establish that despite adjective formation undergoing deviation, gender features which are part of the structural component, come about even with reduced linguistic input. Finally, Section 3.6 concludes.

3.2 Adjectives in Tamil

There is a general consensus within generative theory that Tamil (among other Dravidian languages) does not feature adjectives as a distinct category. In this section, we shall review certain accounts which adopt such a position, and then provide an alternate analysis.

3.2.1 Existing Accounts: There are no adjectives in Tamil

Menon (2013, 2014) adopts the stance that adjectives do not constitute a separate lexical category in Dravidian languages. While the primary empirical support for this claim comes from Malayalam, (a language with close genealogical ties to Tamil), the argument made here spans all major Dravidian languages. Because of this, we consider Menon's account an important one while studying adjectives in Tamil. Menon's analysis of adjectives is framed by the strong claim that adjectives do not form an independent class in Dravidian languages– neither in the lexicon nor in the syntactic component. Such a lack of adjectives forces these languages to create ad hoc adjective-like structures for the purposes of modification and predication. It is important to note that Menon's account is built on the premise of Distributed Morphology (DM) according to which the lexicon is simply a container of unlabeled roots devoid of any identifying information. Within this architecture, adjectives (nouns and verbs too) are formed when roots take up certain position in the derivational spine[1]. For instance, merging with *n* results in a root being interpreted as a noun, and merging with *v* leads to the creation of a verb. So, the primary question for Menon is: How are adjectival modifiers formed?

To answer this question, Menon posits that roots come in two kinds: native (to Dravidian), and borrowed (from Indo-Aryan languages at an earlier point in the diachrony). This bifurcation is central to Menon's analysis, as the derivational process by which adjectives are formed out of roots differs significantly for these two types of roots.

Using examples from Malayalam, Menon shows that "native" roots, when adjectivized, contain a relativiser, as shown below in (1). The roots in (1a), (1b) and (1c) are all understood to be native to Dravidian, and adjectives can be formed out of them using a relativiser.

(1) Malayalam

a. *ceriy-a*
$\sqrt{small}$-REL
'small'

b. *pudiy-a*
$\sqrt{new}$-REL
'new'

c. *pacc-a*
$\sqrt{green}$-REL
'green'

On the other hand, "borrowed" roots fare differently. Here, the roots are first nominalised. They are then attached to a copula, and finally relativised to form adjectives. This strategy is exemplified in (2) where the nominalised 'happiness' and 'height' have to be attached to a verbal element in order to become adjectives.

(2) Malayalam

[1] For the purposes of this discussion we shall set aside the formation of nouns and verbs. Our focus will be solely on adjectives.

a. *sandosham-uLL-a*
happiness-COP-REL
'happy'

b. *pokkam-uLL-a*
height-COP-REL
'tall'

Essentially, according to Menon, the idea that adjectives are not an independent category in Dravidian languages is quite appealing, given how there are clearly nominal and verbal elements involved in their formation. Thus, there seems to be no incentive to prop them as a separate category.

With regard to Menon's theory of adjective formation, two issues need reconsideration: a conceptual and an empirical one. Conceptually, a system such as Distributed Morphology assumes that roots in the lexicon lack any identifying information; they are not labeled in any way whatsoever. However, Menon's theory crucially hinges on roots being labeled as "native" or "borrowed", as only when they are categorized differently can they adopt different adjective-formation strategies. Such labeling is nonetheless incongruent with the principle of Distributed Morphology. The empirical issue is that this analysis cannot be extended to Tamil, another Dravidian language. In Tamil, as we shall see in forthcoming sections, there is no distinction between native and borrowed adjectives. There are indeed two processes of adjective formation, but these are not sensitive to the native or non-native status of the root. Because of these two reasons we cannot extend Menon's account for adjectives in Tamil. As we shall see in the forthcoming sections, the empirical patterns found in Tamil require a more nuanced approach to adjectives.

Before going deeper into adjective formation in Tamil, let us review a few other existing accounts of adjectives in Dravidian languages.

Another set of arguments against positing adjectives as a distinct category comes from Jayaseelan and Amritavalli (2017). The position adopted here, however, is not as strong as that of Menon's. Jayaseelan and Amritavalli claim that adjectives are not lexically distinct categories in Dravidian languages; applying some derivational processes to nouns is what results in modificational structures in syntax. However, there is still some reluctance to label these structures definitively as adjectives. Jayaseelan and Amritavalli base their position on the fact that most putative adjectives in Dravidian can be traced back to nominal roots, and that the number of indisputable adjectives in these languages is a very low number, approximately 30.

This line of argumentation is further developed by Amritavalli (2019). Amritavalli states that given the paucity of adjectives in Dravidian languages, nouns often take on the role of these modifiers. It is a very productive strategy to "adjectivalise" a noun by adding the suffix *-aa* to it. This process is illustrated in (3) below, where the noun in (3a) and (3b) becomes a modifier (3c) with the addition of a suffix *-aa*.

(3) Tamil (Amritavalli, 2019)

a. *kastam*
difficulty
'difficulty'

b. *id-oda kastam*
it-GEN difficulty
'its difficulty'

c. *idu kastam-aa irukku*
this difficulty-aa be.PRS.3SG
'This is difficult.'

Amritavalli's position, too, is that modifiers can be productively derived from nouns and other elements, thus obviating the need for having a separate category for adjectives in this set of languages. To summarize, we see that a conventional understanding of adjectives is rooted in them not being a standalone category. A more moderate version of this claim is given by Krishnamurti (2003). Krishnamurti's account of Dravidian languages posits that there are some parts of speech that are mainly identified only in the syntax component in Dravidian languages: these include adjectives, adverbs and clitics. The central idea is that these categories might not be lexically distinct, but they are certainly identified as such by the derivational component. Krishnamurti's rationale for considering these as distinct categories stems from the fact that these are purely adjectival roots. They are distinct from elements with nominal or verbal flavours, and must be regarded as such. To illustrate his point, Krishnamurti provides the following examples:

(4)
a. *karu* 'black'
b. *cem* 'red'
c. *veL* 'white'
d. *ten* 'southern'
e. *vata* 'northern'
f. *pin* 'back/behind'
g. *mun* 'front/ forward'
h. *mutu* 'old'

i. *pudu* 'new'

j. *iLa* 'young'

k. *in* 'sweet'

l. *puLi* 'sour'

The examples above are clear instances of modifiers without any nominal or verbal origins. While acknowledging that these roots cannot be used as modifiers by themselves, and that they require the application of some derivational processes to turn them into viable adjectives, Krishnamurti argues for adjectives to be recognised as a distinct category in Dravidian languages. What we draw from the accounts surveyed here is that adjectives are a special category in Tamil. Nouns and Verbs are lexically defined as such, and used in the syntactic domain as arguments and predicates, respectively. Adjectives, on the other hand, are not a clearly marked lexical category. But we maintain that they are recognised as a distinct category in the derivational component. The current proposal draws from this approach, and is explicated in the following section.

3.2.2 The Current Proposal: Distinct Adjectives in the Syntax

As we can see from the existing accounts presented in the previous section, adjectives are not taken to be an autonomous lexical category in Tamil. However, the possibility still remains that they will be regarded as distinct syntactically. Here, we present some empirical data to evidence the claim that Tamil derives adjectives in the syntax in a principled and productive manner, and that there are some clear diagnostic tests that can confirm that these derived modifiers are indeed adjectives. However, before presenting the diagnostic tests, some clarification about the rationale behind such a theoretical position is in order.

The approach adopted here is that adjectives are marked in the lexicon as property-denoting concepts, and are realised derivationally. The reason for saying so is that even though adjectives and other modifiers like relative clauses are derived using the same syntactic processes (relativizers), there are some key differences between them. Adjectives are gradable modifiers, they are comparable, they have attributive and predicative counterparts. These properties do not apply to other relative clause modifiers. The fact that there is such a distinction between adjectives and other modifiers compels us to posit that the adjectival nature of certain structures has to be indicated in the lexicon itself. Not marking adjectival structures as such in the pre-derivational stage would lead to overgeneralization as illustrated below.

To elaborate, both adjectives and relative clauses are formed with the same relativizing morpheme *-a*. Thus, they look identical on the surface (5a) and (5b).

(5) a. *nall-**a** paiyan*
$\sqrt{\textit{good}}$-**REL** boy
'good boy'

b. *nadakar-**a** paiyan*
walk.PRS-**REL** boy
'boy who is walking' or 'walking boy'

However, only (5a) has a predicative counterpart (6a), as it is an adjective. (5b) cannot have a predicative counterpart (6b).

(6) a. *anda paiyan nall-**a**-van*
that boy $\sqrt{\textit{good}}$-**REL**-MSG
'That boy is good.'

b. **anda paiyan nadakar-**a**-van*
that boy walk.PRS-**REL**-MSG
'That boy is who is walking.'

The distinction between (6a) and (6b) evidence that adjectives are treated differently in the syntactic domain.

Having established the rationale behind the position adopted, we can review the set of diagnostic tests which help conclusively establish modifiers formed using relative clauses as adjectives in Tamil.

The first test appeals to what has been claimed (Dixon, 2004, p. 10) as the most central property of adjectives: Adjectives are expressions that "alter, clarify or adjust the meaning contributions of nouns". In this regard, they are attributive modifiers. It is an intrinsic property of such attributive modifiers that they can be used as predicates of a sentence. The items in (7) exemplify these traits[2]. (7a) and (7b) attributively modify the noun 'boy' with the same lexical items appearing as the predicate of the sentences in (7c) and (7d), respectively.

(7) a. *anda nall-a paiyan*
that $\sqrt{\textit{good}}$-REL boy
'that good boy'

b. *anda uyaram-aana paiyan*
that height-ADJR boy
'that tall boy'

c. *anda paiyan nall-a-van*
that boy $\sqrt{\textit{good}}$-REL-MSG
'That boy is good.'

d. *anda paiyan uyaram-aaga irukaan*
that boy height-ADJR be.MSG
'That boy is tall.'

[2] For the moment, I use 'adjectivizer' ADJR as a single morpheme. Forthcoming sections will break it down further and elaborate on its structure.

(7) demonstrates that a fundamental property ascribed to adjectives cross-linguistically can be observed to modifiers in Tamil. Based on this, we continue to entertain the idea that adjectives exist in Tamil in the syntactic domain, if not as an independent category in the lexicon.

The second test checks whether these modifiers can co-occur with other modifiers that in parallel qualify the noun that they modify. This can be executed using numerals. We can see that these modifiers indeed occur with numerals (8a) and (8b), further strengthening the proposal.

(8) a. *anda naalu nall-a pasanga*
that four $\sqrt{good}$-REL boys
'those four good boys'

b. *anda anju uyaram-aana kolandainga*
that five height-ADJR children
'those five tall children'

The third test is to with the application of intensifiers. It is an intrinsic property of adjectives that they can be further qualified by intensifiers such as 'very', a diagnostic used in Kennedy (2013) and Beck (1999). It is also well attested that these intensifiers do not modify any other parts of speech, such as nouns. In Tamil, we see that the intensifier can qualify these derived modifiers (9a) and (9c), but not regular nouns (9b) and (9d).

(9) a. *anda romba nall-a paiyan*
that INTF $\sqrt{good}$-REL boy
'that very good boy'

b. **anda romba paiyan*
that INTF boy

c. *anda romba uyaram-aana ponnu*
that INTF height-ADJR girl
'that very tall girl'

d. **anda romba uyaram*
that INTF height

The final diagnostic test comes from the availability of degrees of comparison (Kennedy, 2013). By their nature, adjectives allow for comparability; there is the possibility of having comparative (10a) and (10b), and superlative (10c) and (10d) forms of adjectives.

(10) a. *john peter-a vida nall-a-van*
John Peter-ACC than $\sqrt{good}$-REL-3MSG
'John is better than Peter'

b. *mary lisa-va vida uyaram-aa iru-k-aa*
Mary Lisa-ACC than height-ADJR be-PRS-3FSG
'Mary is taller than Lisa.'

c. *john ellarayum vida nall-a-van*
John everyone.ACC than $\sqrt{good}$-REL-3MSG
'John is better than everyone. = John is the best.'

d. *mary ellarayum vida uyaram-aa iru-k-aa*
Mary everyone.ACC than height-ADJR be-PRS-3FSG
'Mary is taller than everyone. = Mary is the tallest.'

To summarise, these diagnostic tests reveal that the generally attested key hallmarks of adjectives cross-linguistically are commensurate with the derived modifiers in Tamil too. Based on this, we infer that despite adjectives not being a primitive lexical category in Tamil, they are recognised as a distinct class syntactically.

A cursory glance at the adjectival structures provided so far suggests that there must be principled ways of deriving them, and we shall explore these in the following section.

3.3 Deriving Adjectives in Tamil

The previous section established conclusively that adjectives are recognised as a distinct category syntactically. In other words, adjectives are derived from other elements, such as nouns or uncategorised roots. In what follows, we will illustrate the processes by which adjectives are derived in Tamil.

3.3.1 A Two-way Distinction

By looking at the morphological shape of the adjectives presented so far, we can see that they come in two types. From this I infer that there must be at least two routes to deriving adjectives in Tamil. In this subsection, we shall delve into the two types and understand how they are derived. The two types that will be discussed in this section are presented in (11) and (13).

In (11), we present an example of the first type. Here, an uncategorised root such as 'good' is relativised to form an adjective (11a). The derived adjective can be used in the attributive position (11b), as well as predicatively (11c). In the predicative position, the adjective exhibits agreement with the features of the subject.

(11) a. *nall-a*
$\sqrt{good}$-REL
'good'

b. *nall-a paiyan*
$\sqrt{good}$-REL boy
'good boy'

c. *anda paiyan nall-a-van*
that boy.MSG $\sqrt{good}$-REL-MSG
'That boy is a good person.'

For the predicative adjectives in (11), it is possible to add tense information. This can be done with the help of time adverbials, such as 'back then', 'earlier', etc (12).

(12) a. *appo avan romba nall-a-van*
back then he INTF $\sqrt{good}$-REL-MSG
'Back then he was a very good person.'

b. *munnadi avan romba nall-a-van*
earlier he INTF $\sqrt{good}$-REL-MSG
'Earlier, he was a very good person.'

Since there are no overt verbal elements in (12), it is not possible to affix tense morphology to the predicative adjective.

The second sub-category of adjectives are formed by verbalizing a noun, and then relativising it (13a). In (13b), the lexical noun 'happiness' is attached to a verbal element, and then relativised to form the attributive adjective. The predicative form of these adjectives is derived when the noun+verbal element complex is used with a copula (13c) . Once again, there is agreement in the predicative domain, with the copula agreeing with the subject, whereas there is no agreement in the attributive domain.

(13) a. *sandosham-aa-na*
happiness-V-REL
'happy'

b. *sandosham-aa-na paiyan*
happiness-V-REL boy
'happy boy'

c. *anda paiyan sandosham-aa iru-k-aan*
that boy.MSG happiness-V be-PRS-3MSG
'That boy is happy.'

The adjectival predicate in (13c) has an overt verbal element present: the copula BE. Therefore, it is possible to directly affix tense morphology onto them in the case of predicative adjectives formed this way (14).

(14) a. *anda paiyan sandosham-aa iru-nd-aan*
that boy.MSG happiness-V be-PST-3MSG
'That boy was happy.'

b. *anda paiyan vaaɹkai-la sandosham-aa iru-pp-aan*
that boy.MSG life-LOC happiness-V be-FUT-3MSG
'That boy will be happy in life.'

When we look at more examples of adjectives in Tamil, we see that they fall into either of these two patterns. (15) shows the first type, where an uncategorised root is relativised to form adjectives. The root in question can never exist by itself; it minimally needs a relativiser in order to be present in a derivation. (16) presents the other type, where an existing noun is attached to a verbal element, and then relativised. This strategy also forms adjectives. It is interesting to note at this point that the native vs borrowed distinction does not hold good in these cases. (16) contains native (16a)-(16d), as well as borrowed (16e)-(16f) lexical items. It is true that we do not find any loanword cases in (15), but that could be attributed to the fact that it is not as productive a strategy in Tamil as (16).

(15) Root + Relativiser

a. *nall-a*
$\sqrt{good}$-REL
'good'

b. *kett-a*
$\sqrt{bad}$-REL
'bad'

c. *periy-a*
$\sqrt{big}$-REL
'big'

d. *cinn-a*
$\sqrt{small}$-REL
'small'

e. *pudiy-a*
$\sqrt{new}$-REL
'new'

f. *paɹay-a*
$\sqrt{old}$-REL
'old'

(16) Noun + V + Relativiser

a. *amaidi-aa-na*
silence-V-REL
'silent'

b. *porupp-aa-na*
responsibility-V-REL
'responsible'

c. *sood-aa-na*
heat-V-REL
'hot'

d. *porumai-aa-na*
patience-V-REL
'patient'

e. *sogam-aa-na*
sadness-V-REL
'sad'

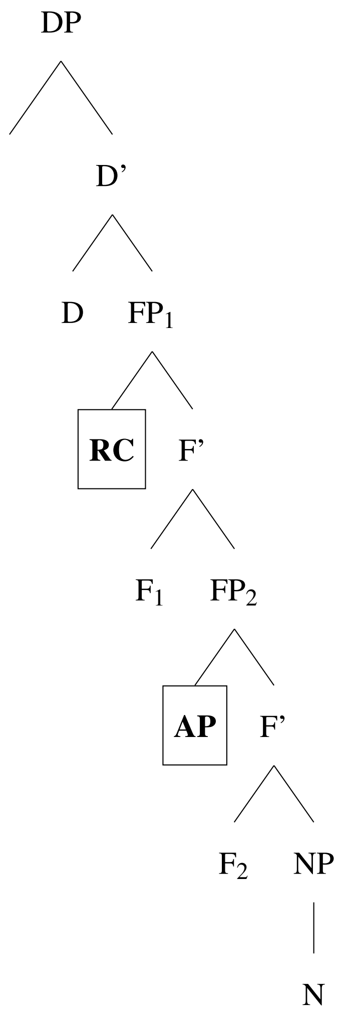
f. *dairyam-aa-na*
courage-V-REL
'courageous'

Given such a dichotomous divide in the final morphological shape of adjectives in Tamil, we then ask if the underlying process that led to such structures also has a strict bifurcation. Essentially, the question we now ask is: Are there two different ways of deriving adjectives in Tamil? The following sub-section takes up this question, and provides a possible illustration of these derivational processes.

3.3.2 Deriving the Two-way Distinction

Cinque (2010) and Alexiadou (2014) provide a comprehensive account of how adjectives are derived across languages. Cinque posits that languages can have two strategies by which adjectives are represented in the derivation. They can either be phrasal specifiers of dedicated functional projections, or they can be introduced as predicates of reduced relative clauses. This conceptualisation is represented as (17) below, where APs can be either directly merged as specifiers of functional projections, or they can be part of a (reduced) relative clauses, in which cases they would be considered as indirect modifiers of the noun.

(17)



It is imperative to note that this proposal does not stipulate that all languages must have these two means of deriving adjectives. Rather, the implication is that, when languages use adjectives while building structures, they do so via one of these two mechanisms. From Cinque's proposal, we can infer that the availability of these strategies is subject to parametric variation across languages.

Taking Cinque's account further, we propose that in languages in which adjectives form a primitive and independent lexical category, both strategies are available. Essentially, Germanic and Romance languages can introduce adjectives into the derivation either by merging them directly as APs or by introducing them through relative clauses. However, for Dravidian languages such as Tamil, only the latter is available. The morphological shape of adjectives indicates that they are not unary items; they are always composed of at least two elements. This is a clear indicator of adjectives not being a primitive category in the lexicon, and thus not heading their own projections as APs. Therefore, we presume the option of directly merging APs to be closed off to Dravidian languages.

Based on the discussion so far, I propose that adjectives in Tamil are derived invariably through relativisation. The unavailability of adjectives as a primitive category causes the derivational space to look for alternate ways to introduce modifiers into structures, and this is done majorly by relative clauses. The previous section demonstrated that there are two ways to derive adjectives with the help of relativisers. One strategy is to relativise an unlabeled root in the lexicon to form an adjective, and the second route is when a noun takes on a verbal marker and then gets relativised.

In terms of the structural representations underlying them, in Tamil, CPs formed by one of the two strategies listed above are interpreted as adjectives. Structurally, there is no A projecting to AP in the grammar; certain CPs are indicative of adjectives.

In the forthcoming sections, we shall explore how these adjectives occur in different structural positions, and how those structures can be derived using the mechanisms posited so far.

3.3.3 Attributive Adjectives

Attributive adjectives are defined in terms of their positional distribution. These occur within the nominal domain either pre-nominally or post-nominally. These are generally considered to be modifiers of the noun, and are understood to not have an argument-predicate relationship with the noun that they modify. In this section, we shall explore the structure of attributive adjectives in Tamil, discuss how they are derived in the structure.

As noted previously, there is a bifurcation in terms of the morphological shapes of adjectives in Tamil. We continue to see this division in the domain of attributive adjectives. One strategy of deriving attributive adjectives is by relativising an uncategorised root such as ‘good’, ‘bad’ etc. The resultant adjective modifies the nouns in (18), structurally represented as (19).

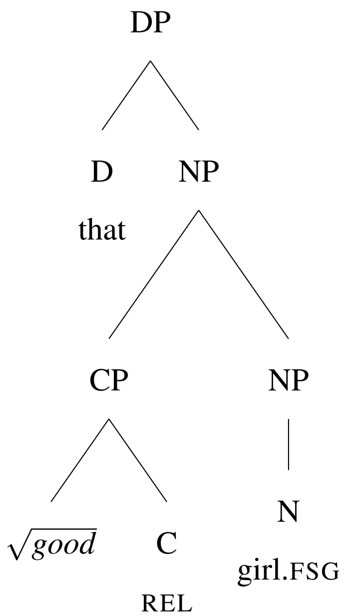
(18) a. *anda nall-a poNNu*
that $\sqrt{good}$-REL girl
‘that good girl’

b. *anda nall-a paiyan*
that $\sqrt{good}$-REL boy
‘that good boy’

c. *anda nall-a kolandainga*
that $\sqrt{good}$-REL children
‘those good children’

In (19), we see that the adjective is formed when the root ‘good’ attaches to a relativser, forming a CP. The relative clause is merged as an adjunct modifier to the NP containing the head noun ‘boy’. This creates a head-modifier relationship between ‘good-REL’ and ‘boy’. It is essential to note that the root ‘good’ cannot exist independently; in order to exist in a derivation it needs to be hosted by other elements, such as the relativiser here. Essentially, the root ‘good’ can be recognised as a morphological word only upon merging with relativisers. Without relativisers, these adjectives cannot be formed in Tamil.

(19)



The other way to obtain attributive adjectives is by verbalizing a noun and then attaching a relativiser to it. Here, this strategy is illustrated using the noun ‘beauty’, which functions in Tamil as any prototypical noun would: it can be contained by a possessor (20a), it can carry case markers (20b), and serve as the subject of a sentence (20c).

(20) a. *avaL-oda aɹagu*
her-GEN beauty
‘her beauty’

b. *avaL-oda aɹag-a patti ena-kku teriyum*
her-GEN beauty-ACC about I-DAT know.INF
‘I know about her beauty’

c. *aɹagu nirandaram illa*
beauty permanent NEG
'Beauty is not permanent.'

Using nouns such as (20), we can form adjectives by first adding a verbal marker to them, and then relativising the N+V complex (21).

(21) a. *inda aɹag-aa-na poNNu*
this beauty-V-REL girl
'this beautiful girl'

b. *inda aɹag-aa-na paiyan*
this beauty-V-REL boy
'this beautiful boy'

c. *inga aɹag-aa-na kolandainga*
this beauty-V-REL children
'these beautiful children'

The next objective is to understand the exact nature of the verbal element in (21). The verb in (21) crucially enables a nominal element 'beauty' to be relativised. Thus, I infer that the verb must contain semantics that can indicate 'occurence' or add some eventive meaning to the noun before it can be relativised. We can see that the verb *aa* consistently executes this in Tamil. In (22) below, it gives the meaning of 'happen' or 'occur', which is how we understand the function of the verbal element in (21).

(22) a. *neram aa-chu*
time happen-PST
'It's time. (lit: Time happened.)'

b. *ena-kku vayasu aa-chu*
1Sg-DAT age happen-PST
'I am old. (lit: To me age happened.)'

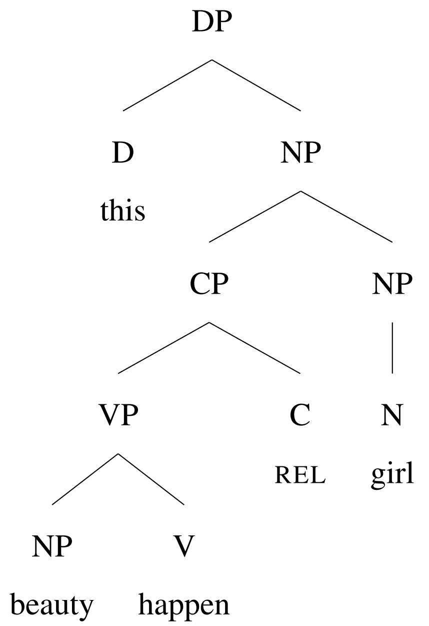
c. *enna aa-chu*
what happen-PST
'What happened?'

d. *paisa selavu aa-gum*
money spending happen-FUT
'(something) will cost money. (lit: Money spending will happen.)'

e. *onnum aa-la*
nothing happen-NEG.PST
'Nothing happened.'

Based on these facts, we can now understand the verbal element in (21) as ‘happen’. Once again, a relativiser is crucially needed to form an adjective out of the noun+happen complex. This process is structurally represented as (23) below, where the CP containing a relativiser and the noun+verb complex is merged as an adjunct to the NP containing the head noun ‘girl’. The resultant construction is presented as (24) below.

(23)



(24) *inda aɹag-aa-na poNNu*
this beauty-happen-REL girl.FSG
‘this beautiful girl’

It is important to note that the relativiser in (24) is the same as the one found in (18). The overt realisation is *-na* in the former and *-a* in the latter, because of a morpho-phonological constraint operating in the former. In (24), the verb *aa-* and the relativiser *-a* would form a cluster of vowels, which is now broken by the epenthetic *-n-*. Structurally, they denote the same morpheme. To summarise, this section discussed two strategies to form attributive adjectives in Tamil. We now proceed to predicative adjectives.

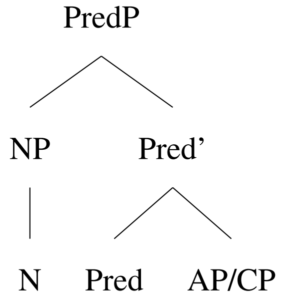
3.3.4 Predicative Adjectives and Gender Agreement

Predicative adjectives occur in the verbal domain. These adjectives modify nouns as predicates. Thus, the nouns are arguments of the predicative adjective. In order to capture the argument-predicate relation that differentiates predicative adjectives from their attributive counterparts, it is generally understood that the predicative domain is headed by a functional projection - PredP - that fosters this relation. Heycock (2013) presents a comprehensive overview of the notion of predication, and upholds the view that predication should be understood as a structural relation between the subject of a clause and the verbal expression at the core of the clause. It is essentially a relation that holds in a particular structural configuration - mutual C-Command - between the subject and the verbal expression. Baker (2008, 2011) provides one way of implementing this structural configuration to the idea of predicative adjectives by positing a structure such as (25) below.

In (25), Pred is a special copular category that can facilitate a relation of predication originating from the adjective directed at the noun. It is understood that adjectives have an implicit thematic relation as part of their internal structure. The semantic function of Pred is to retrieve the thematic role in the adjective and then use it to establish a connection to the noun. With a thematic role from the adjective assigned to it, the noun can then be structurally identified as the subject of the predicate. It is crucial to note that the subject of the adjectival predicate is not base-generated inside AP. The specifier position of PredP is the host for the subject, signaling the importance of Pred in facilitating this relation between the noun (argument) and the adjective (predicate).

As we shall soon see, not all predicative adjectives contain an overt copula. However, despite the lack of a copula in these, my analysis of these constructions posits a PredP to mediate the structural relation between the subject and the adjectival predicate. This is done following Baker's (2008) characterization of predicates. Baker differentiates verbal predicates from nominal and adjectival ones on the basis of their ability to merge with the agreement-bearing tense node (Baker, 2008, p. 30). Verbs can directly merge with T; adjectives and nouns cannot. They require a (null) PredP to be realised as predicates. Pred can be null; it is posited here only for establishing an argument-predicate relationship between the noun in the subject position, and the adjective in the predicate.

(25)



Following this account of predicative adjectives, we propose that PredP generates predicative adjectives in Tamil. As observed previously, there are two distinct realisations of the predicative adjective in Tamil. One strategy involves relativising an uncategorised root and then merging it as the complement of the head Pred. Here, Pred acts as a covert copular structure that supports the entire relation of predication. The other strategy begins from a lexical noun that is verbalized, and then acts as a predicate with the help of an overt copula. Recall that such a bifurcation finds coordinates in the attributive domain as well, where the two strategies to realise adjectives stemmed from the relativisation of a root, and from relativising a verbalized noun.

The first strategy is exemplified in (26). Additionally, we also find an agreement relation in these structure, a feature that distinguishes predicative adjectives from attributive adjectives in Tamil. Recall that attributive adjectives do not exhibit any kind of agreement with the noun they modify. However, predicative adjectives do. In (26a) we see the [FEM] features of the subject on the adjective, [MASC] on the adjective in (26b) and [PL] on the adjective in (26c). The predicative adjective inflects for the number and gender specifications of the subject of the predicate.

Based on this I infer that the presence of Pred in the predicative domain is what facilitates agreement relations between the noun and the adjective in the predicative domain. In the attributive domain, there were no functional heads that could act as probes for the phi features of the head noun.

(26) a. *anda poNNu nall-a-va*
that girl.FSG $\sqrt{good}$-REL-FSG
‘That girl is good.’

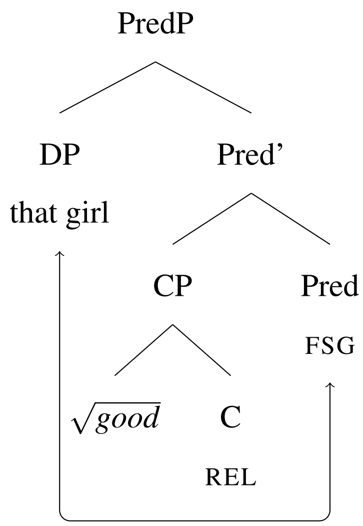
b. *anda paiyan nall-a-van*
that boy.MSG $\sqrt{good}$-REL-MSG
‘That boy is good.’

c. *anda kolandainge nall-a-vanga*
that children.PL $\sqrt{good}$-REL-PL

'Those children are good.'

The formation of predicative adjectives from uncategorised roots is illustrated in (27) where the root 'good' undergoes relativisation as the first step. The resultant reduced relative clause can now act as an adjectival modifier. As explained above, PredP is essential to the formation of predicative adjectives. Therefore, in order to derive a predicative adjective out of 'good', it is then merged with the head Pred. The maximal projection of Pred hosts the subject of the predicate 'that girl' in its specifier position. Pred mediates a relation of argument-predicate between the DP and the CP.

(27)



Recall that there is no agreement with attributive adjectives. This fact can be explained if the agreeing head in (26) is Pred and not the adjective itself. Essentially, agreement morphemes are seen on the adjective only if the root+REL complex is directly merged to Pred, as in the case of (27). The attributive domain, in contrast, lacks a Pred projection, resulting in there being no agreement relations in attributive adjectives.

In the predicative domain, agreement can be explained as a result of the syntactic operation Agree (Chomsky, 2001, 2000). The functional head Pred probes for phi features, and finds a suitable goal in the subject DP. This results in agreement between the two[3].

Additionally, we see number and gender agreement realised on the adjective. In order to confirm that it is only number and gender agreement with the exclusion of person agreement, we

[3]Within the generative tradition, there are alternative proposals for agreement, for instance Chandra (2011) presents agreement as a result of movement and Bobaljik (2008) considers agreement to be post-syntactic phenomenon. I do not take up these accounts here.

produce the following examples in (28). Examples (28a) to (28c) show the adjective inflecting for the number and gender specifications of the third person subject. (28d) and (28e) show the adjective inflecting for the gender specification of the first person subject. (28f) and (28g) do so for the gender features of the second person subject. These examples show that agreement on predicative adjectives does not indicate person features. Essentially, we only see gender and number agreement on the predicative adjective. This is in line with Baker's characterisation of phi agreement, where person agreement is observed only when the head T is involved.

(28) a. *avan nall-a-van*
he $\sqrt{good}$-REL-MSG
'He is a good person.'

b. *avaa nall-a-va*
she $\sqrt{good}$-REL-FSG
'She is a good person.'

c. *avanga nall-a-vanga*
they $\sqrt{good}$-REL-PL
'They are good people.'

d. *naa nall-a-van*
I $\sqrt{good}$-REL-MSG
'I am a good person.(Male speaker)'

e. *naa nall-a-va*
I $\sqrt{good}$-REL-FSG
'I am a good person. (Female speaker)'

f. *nee nall-a-van*
you $\sqrt{good}$-REL-MSG
'You are a good person.' (Male addressee)

g. *nee nall-a-va*
you $\sqrt{good}$-REL-FSG
'You are a good person.'(Female addressee)

Based on the facts presented in (28), we infer that in the case when predicative adjectives are formed out of relativised roots, the functional head that facilitates agreement is Pred, and not T.

The agreement relations noted here can be explained as the consequence of a Probe-Goal relationship between Pred and the DP. Baker (2008) proposes that agreement can happen under configurations of bidirectional Agree, where either the Probe C-Commands the Goal, or the Goal C-Commands the Probe. As illustrated in (27), the goal DP C-Commands the Probe in

Pred, forming the precursor for the operation Agree to be established between the two. As a result, there is agreement between the DP and Pred. It is important to note that Pred is simply the functional projection that facilitates agreement. The ultimate exponence of the agreement morphemes is performed by suffixing the relevant agreement morphemes to the relativiser.

The other strategy to obtain adjectives begins from a lexical noun. As recorded in the previous section, nouns are merged with a 'happen' V in order to form a quasi-verbal structure. This strategy crucially diverges from the one outlined previously by having an overt copula. Here too, the head Pred supports the noun+verbal+C complex[4], and mediates an argument-predicate relation between the noun and the created verbal complex. In (29), it is demonstrated that the copula agrees with the feature specifications of the subject.

(29) a. *anda poNNu uyaram-aa(-ga) iru-k-aa*
that girl.3FSG height-happen(-REL) be-PRS-3FSG
'That girl is tall'

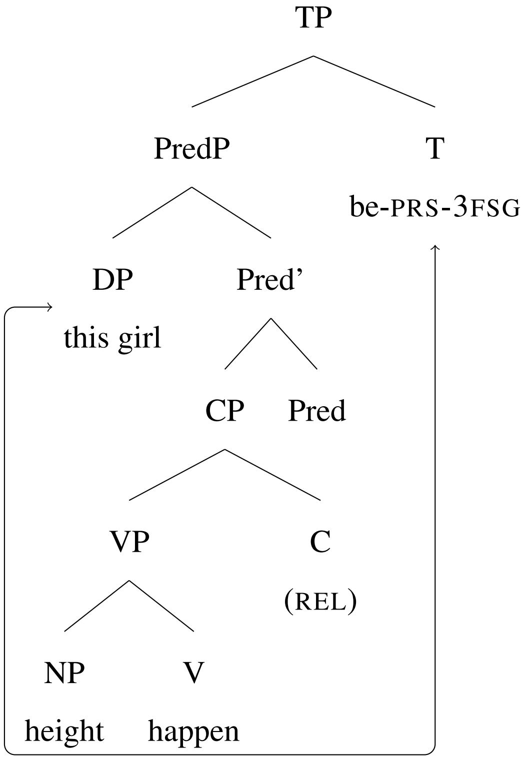
b. *anda paiyan uyaram-aa(-ga) iru-k-aan*
that boy.3MSG height-happen(-REL) be-PRS-3MSG
'That boy is tall.'

c. *anda kolandainga uyaram-aa(-ga) iru-k-aanga*
that children.PL height-happen(-REL) be-PRS-PL
'Those children are tall.'

The formation of these predicative adjectives is represented in (30) where the noun 'height' is first verbalized with the verb 'happen'. The composite VP is the complement of Pred, signifying that it is the predicate in this construction. As proposed earlier, Pred enables the assignment of a thematic role to the subject from the verbal complex. The presence of an overt copula along with a tense marker indicates that PredP is nested inside TP. Agreement with the subject takes place with the functional head T.

[4]The relativizer can be optionally dropped in these constructions in spoken Tamil. Diachronic evidence from literary Tamil provides evidence for the relativizer morpheme.

(30)



When we delve further into the notion of agreement in these predicative adjectives, we find that there is number, gender, as well as person agreement exhibited by the overt copula. We illustrate this by placing different pronouns as the subject of these predicates in (31). (31a) to (31c) show number and gender agreement with the third person subject. (31d) and (31e) show a differential marking for the first person subject, and (31f) and (31g) show that the copula also marks second person agreement morphemes distinctly.

(31) a. *ava uyaram-aa(-ga) iru-k-aa*
she height-happen(-REL) be-PRS-3FSG
'She is tall.'

b. *avan uyaram-aa(-ga) iru-k-aan*
he height-happen(-REL) be-PRS-3MSG
'He is tall.'

c. *avanga uyaram-aa(-ga) iru-k-aanga*
they height-happen(-REL) be-PRS-3PL
'They are tall.'

d. *naa uyaram-aa(-ga) iru-k-en*
I height-happen(-REL) be-PRS-1SG
'I am tall.' (Male or female speaker)

e. *naanga uyaram-aa(-ga) iru-k-om*
we height-happen(-REL) be-PRS-1PL
'We are tall.' (Male or female speakers)

f. *nee uyaram-aa(-ga) iru-k-a*
you height-happen(-REL) be-PRS-2SG
'You are tall.' (Male or female addressee)

g. *neenga uyaram-aa(-ga) iru-k-eenga*
you.PL height-happen(-REL) be-PRS-2PL
'You are all tall.' (Male or female addressees)

Essentially, we see person, number and gender agreement on the overt copula present in these types of predicative adjectives. This instance of full phi agreement leads us to infer that the functional head T may be involved in generating these adjectival structures, and executing the observed agreement relations.

3.3.4.1 Two Different Agreeing Heads

This section covered two ways of deriving predicative adjectives in Tamil. Both of them crucially involve the functional projection PredP, that can mediate a relation of argument-predicate between the noun and the derived adjective. One important difference between the two adjective formation strategies is that Strategy A adjectives directly take on agreement markers (26), as represented in (27). Strategy B adjectives, on the other hand, require the presence of another lexical verb which acts as an overt copula. Agreement morphemes are mounted on the copula (29), not on the adjective itself, as represented in (30). Along with these, there are a few other salient differences between the two strategies in the predicative domain that are worth mentioning:

(i) Predicative adjectives in (32a) do not mark tense overtly. Tense can be indicated by time adverbials, but not morphologically on the adjective. In contrast, (32b) marks tense overtly on the copula.

(32) a. Strategy A

appo avan nall-a-van
back then he $\sqrt{good}$-REL-MSG
'**Back then** he was a good person.'

b. Strategy B

avan sandosham-aa(-ga) iru-nd-aan
he happiness-V(-REL) be-PST-3MSG
'He **was** happy.'

(ii) In terms of the array of features participating in agreement, (33a) is a subset of (33b). A full paradigm of PNG agreement data, such as the one presented in (31) shows that agreement in (33a) is only with (number and) gender features, whereas in (33b), there is person, (number) and gender agreement. This supports the idea that the probes of agreement are different in the two. T is the probe in (33b), whereas (33a) has a lower probe such as Pred.

(33) a. Strategy A

avan nall-a-van
he $\sqrt{good}$-REL-MSG
'He is good.'

b. Strategy B

avan sandosham-aa(-ga) iru-k-aan
he happiness-V(-REL) be-PRS-3MSG
'He is happy.'

These patterns lead us to the question: Why have two different structures for predicative adjectives in Tamil? What prompts Strategy B to behave differently from Strategy A? A preliminary answer to this question emerges when we turn to the descriptive account of Tamil provided by Lehmann (1989). Lehmann presents that Tamil has two types of clauses: clauses with verbs and verbless clauses. (34), for example, has two NPs: one in the subject (education) and one in the predicate (importance) position. There is no verbal element in the sentence; it is an equative copular sentence with no verb in it.

(34) (Lehmann, 1989 p.172)

padippu mukkiyam
education importance

'Education is important.'

However, there are clauses which require the support of a verbal element, particularly when the sentence expresses an ascriptive proposition. For these instances, the grammar enlists the help of *iru* 'to be in a state of' or 'exist'. In his account of Tamil grammar, Lehmann states that "when a nominative NP_1 and an NP_2+*aa* combine as subject and predicate", the copular element *iru* at T is necessary. Essentially, *iru* predicates the state of NP_1 as expressed by NP_2+*aa*.

Further elaboration of -*aa* is not provided in Lehmann's account. However, we know from empirical evidence (22) that -*aa* expresses the meaning of 'happen'. Once -*aa* attaches to NPs such as 'height', 'happiness' or any other examples given in Strategy 2, the predicate is no longer purely nominal. It is half-way in the process of becoming a modifier. To fully express that NP_1 (the subject) is predicated of NP_2+happen, the structure requires a copula. The same structural requirement is not present in predicative adjectives derived through Strategy 2, therefore copulas are not required in those adjectives.

Additionally, once *iru* is present at T, it is the highest functional head in the structure. As such, it probes for phi features and ultimately realizes the features as agreement morphemes. Pred no longer displays agreement. In Strategy A adjectives, this is not the case. Pred is the highest (and only) functional head. Therefore, it exhibits agreement in these adjectives. These patterns can be explained by the simple empirical fact that in Tamil, each sentence displays maximally one agreement relation. As explained in Chapter 1, it is the highest functional head that carries morphological exponence of agreement in Tamil. When the higher functional projection, T, participates in agreement, the lower head, Pred, automatically stops displaying agreement. Therefore, we see the subject's features on T instead of on Pred whenever the former is present.

3.3.5 Interim Summary

The discussion in the current section is centred around how adjectives are derived in Tamil. Having established previously that adjectives are not a distinct category in the lexicon, but that the syntactic component certainly recognises them as a category, it was imperative to show this process structurally. Thus in this section, we provided the derivational mechanisms by which attributive as well as predicative adjectives are derived in Tamil. Additionally, we also demonstrated that there is a strict bifurcation that conditions how adjectives are derived regardless of the position of occurrence; there are two different strategies to derive adjectives in Tamil, and these differences are observable in the final morphological shape of the adjective. One way to derive adjectives is from uncategorised roots in the lexicon. Upon being relativised, these can serve as adjectival modifiers in the attributive position, and within PredP in the predicative position. The other strategy takes a lexical noun, verbalizes it and then relativises the verbal complex to form adjectival modifiers. This verbal complex can also be used in the predicative position along with an overt copula. These generalisations are presented below in Table 3.1. Additionally, another crucial difference between the attributive and the predicative versions of

	Attributive Position	Predicative Position
Strategy A	Root + Relativiser (C)	Root + Relativiser + Pred
Strategy B	Noun + V + Relativiser (C)	Noun + V + Pred + Overt Copula

Table 3.1: Adjective Formation in Tamil

the adjective arises from agreement with the subject. There is no agreement at all in the former domain, whereas there is partial phi feature agreement in Strategy A Predicative adjectives and full phi feature agreement in Strategy B Predicative adjectives. This difference between the attributive and the predicative domains is chalked up to the lack of probing functional heads in the former and the presence of Pred in the latter. Within agreement, my focus for this study is on gender agreement in both types of predicative adjectives. As established in Chapter 2, Tamil has only biological gender, assigned to [+HUMAN] nouns. Consequently, the agreement relations observed with predicative adjectives are also biological gender agreement.

Thus, I have presented the entire paradigm of adjectives in Tamil in this section. Moving forward, we shall investigate the implications of such a system for a heritage variety of Tamil, which exists under conditions of restricted input, and potential transfer effects from the dominant language Hindi-Urdu.

3.4 Adjectives in Hindi-Urdu

In this section we paint a picture of the adjective system in Hindi-Urdu, the L2 in this bilingual context. Recall that according to the theory of adjectives posited by Cinque, there are two possible sources for adjectives in languages. Adjectives are either directly merged to the DP spine as APs, or they are introduced by relative clauses. The previous section explained that the adjectives in Tamil are obtained through the latter. However, Hindi-Urdu is distinct from Tamil. Here, adjectives can be traced back to an independent lexical category (Agnihotri, 2022; Kachru, 2006) and can be represented directly as APs.

A prototypical characteristic of adjectival modifiers, these adjectives also occur in both, the attributive (35a), (35b) and (35c), and the predicative (35d), (35e) and (35f) positions. Adjectives agree with the number and gender features of the noun they modify, regardless of whether they occur in the attributive or the predicative position.

(35) a. *achch-ii ladki*
good-FSG girl.FSG
'good girl'

b. *achch-aa ladka*
good-MSG boy.MSG
'good boy'

c. *achch-e bacce*
good-PL children.PL
'good children'

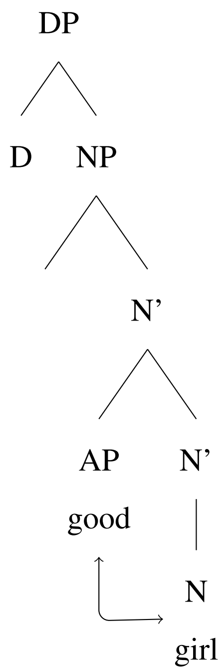
d. *wah ladki achch-ii hai*
that girl.FSG good-FSG be.PRS
'That girl is good.'

e. *wah ladka achch-aa hai*
that boy.MSG good-MSG be.PRS
'That boy is good.'

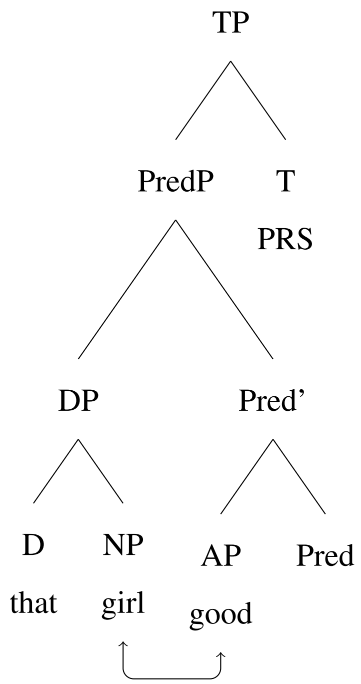
f. *we bacce achch-e hai-n*
those children.PL good-PL be.PRS.PL
'Those children are good.'

Referring back to the theoretical apparatus provided by Cinque, we can understand adjectives in Hindi-Urdu as APs directly merged to the specifier of NPs in the attributive position (36), and as AP complements of PredP in the predicative (37) domain. The argument-predicate relation in (37) is facilitated by the functional projection PredP.

(36)



(37)



Adjectives in Hindi-Urdu agree with the noun they modify, regardless of whether they occur attributively or predicatively. In (35), we observe both the attributive and predicative adjectives agreeing with the number and gender features of the noun. These instances of agreement can be explained as falling out from Agree under C-Command. In the attributive domain, (36), the probe AP finds the NP 'girl' within its C-Command domain. Thus the FSG features of the noun are affixed onto the adjective. Similarly, in the predicative domain, (37), the probe Pred extends a search for a suitable goal to agree with. Agreement in this instance can be explained as a result of Upward Agree. The subject DP C-Commands AP, thus satisfying the requirements for agreement.

To summarize, there are two main differences between the system of adjectives in Tamil and Hindi-Urdu. Firstly, adjectives constitute an independent category in Hindi-Urdu, but do not do so in Tamil, making the system reliant on relative clauses to introduce adjectives. The second major difference is in the domain of agreement. In Hindi-Urdu both attributive as well as predicative adjectives agree. In Tamil, there is no agreement in the attributive domain. In the predicative domain, there is agreement facilitated by the highest functional head at the clausal level: either Pred or T. These two points of divergence between the L1 and L2 will form the focus of the current investigation into heritage Tamil developing in close proximity with Hindi-Urdu.

3.5 Current Proposal: Implications for Language Contact

The previous section presented an illustration of how adjectives are derived and represented in Tamil and Hindi-Urdu. Now, we proceed to uncover the implications for such a configuration with Tamil as the heritage language and Hindi-Urdu as L2. Firstly we shall review existing accounts that deal with the issue of adjectives and adjectival agreement in heritage grammars. Based on the generalisations obtained from these studies, we shall then hypothesise about how the adjective formation strategies in Tamil might undergo change when Tamil exists as a heritage language in New Delhi.

3.5.1 Existing Accounts of Adjectives in Heritage Grammars

Polinsky (2015) discusses the topic of adjectives and their representation in heritage languages, and reports that heritage speakers do maintain the distinction between different lexical categories, such as nouns, verbs and adjectives. They do not conflate these in the heritage grammars if the language in its baseline form makes that distinction. This point is of particular relevance to our investigation into the state of adjectives in heritage Tamil, as it provides grounding for the assumption that adjectives as a category find continued support in heritage grammars. In this section, we shall discuss case studies which take up specific aspects of how adjectives are organised and represented across languages.

Camacho (2018) presents an account of adjectives in heritage Spanish that offers insights into how adjectives are represented in heritage grammars. The empirical domain of this study is provided by heritage speakers of Spanish living in the US, with English as their dominant language. Camacho considers one major difference between adjectives in the L1 (Spanish) and the L2 (English): the position of the adjective relative to the noun. Both the languages allow both, prenominal as well as post nominal adjectives, but with some interpretive differences. In Spanish, the Noun-Adjective order is the default one, with Adjective-Noun having specific meanings. Conversely, in English, Adjective-Noun is the unmarked ordering, with Noun-Adjective being reserved for restricted interpretations. These are well defined conditions present in both the languages. Camacho's line of investigation is about whether heritage speakers of Spanish tend to change the representation of Spanish adjectives to align more towards the system in English.

Using an acceptability judgement task, this study aimed to ascertain if the correspondence between Noun and Adjective ordering, and the resultant interpretation of the adjective that holds in baseline Spanish continues to be followed in the heritage variety.

The results of this study did not show much variation between the responses of monolingual speakers of Spanish, who formed the baseline, and the heritage speakers. The inference drawn from this study was that heritage speakers were indeed sensitive to the existing correspondence between the ordering of adjectives and the interpretive restrictions associated with them. The overall conclusion of the study was that heritage speakers have a good understanding of the underlying structure of adjectives and the repercussions in terms of their interpretation. Essentially, effects of transfer from the L2, English, into the L1 were minimal.

Sánchez and Camacho (2021) present another study on adjectives in heritage Spanish in the US. The focal point of this study was the difference between English and Spanish in terms of the ordering of adjectives: In English, adjectives may appear only in a fixed order with adjectives denoting size occurring before those denoting colour and those denoting nationality. Spanish, on the other hand, allows considerable flexibility in the ordering of adjectives. This difference between the two arises from certain DP internal movements being allowed in Romance languages, but not in Germanic. Given this discrepancy between the representation of adjectives in this group's heritage language (Spanish) and L2 (English), Sánchez and Camacho set out to test the extent to which the nature of adjectives in the L2 could influence the corresponding structures in L1. The two major hypotheses of the study were centred around the idea of cross-linguistic influence from English to Spanish:

(i) Owing to transfer effects from English, heritage speakers would prefer a more rigid system of adjective ordering in their Spanish

(ii) If there are strong cross-linguistic effects of English, the heritage speakers should not be able to access certain interpretations of adjectives, particularly those that are a result of covert movement within the DP

In order to test these two hypotheses, Sánchez and Camacho conducted two acceptability judgment tasks with heritage speakers of Spanish. The results of their study showed no evidence for transfer effects from English. For the first task, they found that the heritage speakers did indeed allow flexible ordering of adjectives in Spanish. They did not impose the rigidity found in English onto Spanish adjectives. As for the second task, the results once again showed that multiple interpretations that may be inaccessible in English, but are available in Spanish were chosen by heritage speakers. Here too, there seemed to be minimal transfer effects from English.

Based on this experimental evidence, Sánchez and Camacho concluded that the representation of adjectives in heritage Spanish was not susceptible to cross-linguistic transfer effects. Rather, these continued to remain strongly in heritage grammars.

The main inferences that we can gather from these two studies is that the development of adjectives in heritage grammars is not susceptible to cross-linguistic transfer effects. Despite a significant reduction in the amount of primary linguistic data that heritage speakers of Spanish are exposed to, they do not compensate for it by using adjectival structures in the L2 to guide their acquisition of adjectives in L1.

Alexiadou et al. (2023) approach the question of cross-linguistic transfer from a different vantage point: agreement on adjectives. They consider the case of Heritage Greek spoken in the US and in Germany, with English and German as the respective dominant languages. The focus of this study was on the expressions of agreement on adjectives. Greek Noun Phrases have a vibrant display of agreement relations; determiners and adjectives agree with the head noun in terms of number and gender. German differs from Greek, with only attributive adjectives exhibiting agreement. English has an even more reduced pattern: there is no agreement inside the NP in English. One of the research questions of this study is about the extent to which the majority language (either German or English) has an effect on agreement mismatches in heritage Greek adjectives. Based on a data elicitation task, they found that there was no significant difference between the two groups (in the US and in Germany) in terms of agreement relations. In other words, the presence or absence of agreement relations in the dominant language did not influence how adjective agreement developed in the heritage grammar. Alexiadou et al.'s study further fortifies the idea that cross-linguistic influence is minimal in the domain of adjectival agreement. Heritage speakers do not seem to rely on the adjective system in the dominant language to guide them in the face of reduced linguistic input in their L1.

The studies reviewed here have covered three major properties associated with adjectives: their position, interpretation and agreement. These studies have been insightful in setting some precedence for what to expect of adjectives in heritage contexts. And a major conclusion has been that cross-linguistic effects, i.e., transfer from the dominant language has been minimal in the domain of adjectives. Adjectives and their properties appear to be quite resilient in heritage grammars. It is in this context that we present the current study. Recall that this chapter began with the structure of adjectives in Tamil and how they are represented in the derivation. This was a particularly relevant exercise, especially considering that Tamil is a language in which the lexicon does not distinguish adjectives as a distinct category. The only way to create and use adjectives is syntactically, using derivational processes. Having covered this, the main question for the chapter is about heritage Tamil and its representation of adjectives and gender agreement, when developing in close contact with Hindi-Urdu as the L2.

In the next section, we shall explore the implications for heritage Tamil and its representation of adjectives and gender agreement.

3.5.2 The Current Study

The aim of the current study is to conduct an investigation into the structure of adjectives and gender agreement in heritage Tamil in New Delhi, and understand any variance in these representations. Essentially, earlier in the chapter, I proposed that adjectives are represented as a derived but syntactically distinct category in Tamil. There are designated derivational processes which lead to the generation of attributive as well as predicative adjectives.

Additionally, we also note a bifurcation in the derivational processes that lead to the generation of adjectives. The differences can be explained in terms of the base element that undergoes transformation to become an adjective. Another noteworthy point was the distinction between the attributive and predicative adjectives in terms of agreement processes. Attributive adjectives do not agree with the noun, whereas predicative adjectives agree with the number and gender features of the subject of the predication.

Given such a system of adjectives in Tamil, this study explores the variation that one might expect to see in heritage Tamil, where the primary linguistic input received by the speakers is restricted, and there is the possibility of potential transfer effects from the dominant language Hindi-Urdu.

3.5.2.1 Questionnaire and Methodology

The chief objective of the study was to elicit the production of attributive and predicative adjectives derived from both processes in Tamil. The speakers were shown images such as Fig. 3.1a and Fig. 3.1b below, and they had to construct sentences or phrases using the help of the pictures and words shown to them. Many heritage speakers do not have a high degree of literacy in their heritage languages, as they are not exposed to the language in formal settings. Hence, I used the roman script to transliterate the Tamil words that could act as prompts for the elicitation task.

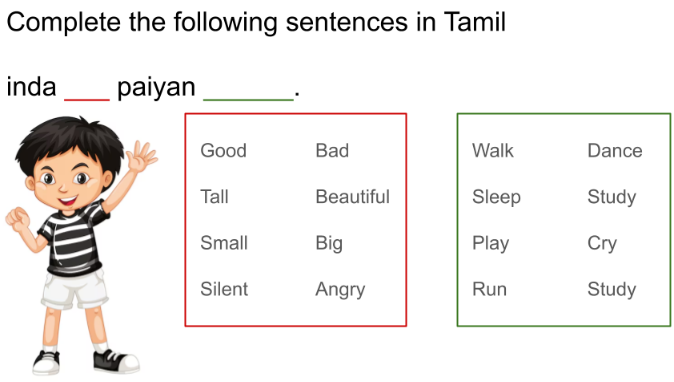
Data for baseline Tamil was collected from 8 native speakers of Tamil (age range=20-29). They were exposed to the same questionnaire, and asked to produce sentences that would elicit attributive and predicative adjectives derived using both strategies. The adjectives produced by them aligned with the structures laid out in Section 3.3.2

The prompts for this task were designed to be minimal but precise. In the case of Fig. 3.1a and Fig. 3.1b, the objective was to elicit the attributive form of the adjectives given below. Speakers were instructed to complete the sentence given above the pictures by using Tamil words from the red and green boxes. In order to successfully complete the sentence, they would have to use a modifier from the red box for the first blank in the sentence, and an action word from the green box for the second blank. The rationale behind using English lexical items in the task was that it would force the speakers to construct the equivalent Tamil adjectives on their own. Given a concept such as ‘good’ or ‘beautiful’ the onus is on them to choose one of the two derivational strategies at their disposal. This way, we can ensure minimal interference from the prompts of the task.

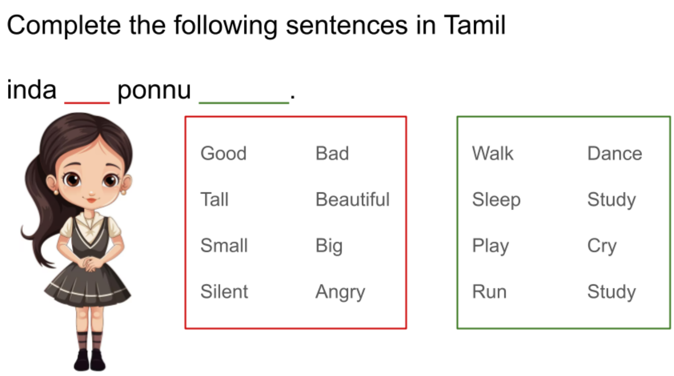
The adjectives in the red box consisted of a mix of both strategies. The expected sentences for the two sets of tasks are given in (38).

(38) a. *inda* ___ *paiyan* ___
this ___ boy ___

b. *inda* ___ *poNNu* ___
this ___ girl ___



(a) Attributive Adjective Task [MASC]

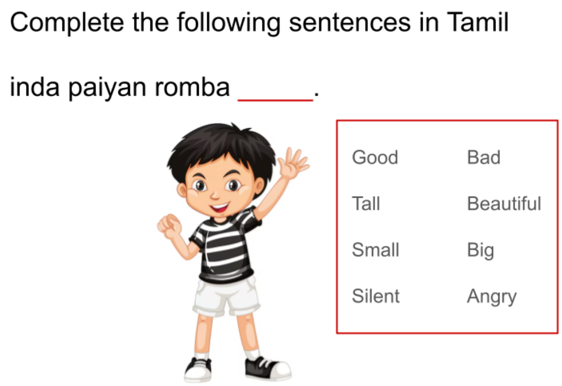


(b) Attributive adjective Task [FEM]

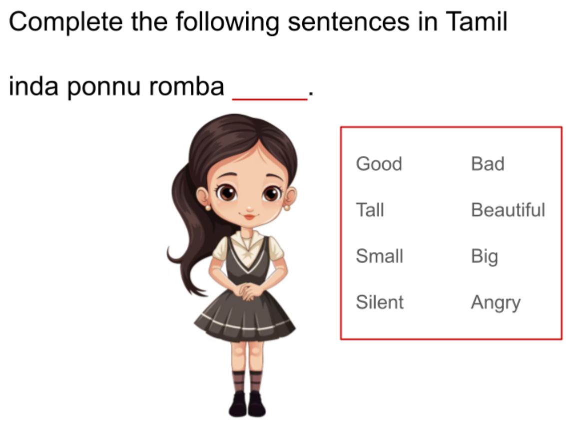
Figure 3.1: Prompts for Attributive Adjectives

The next task was aimed at eliciting predicative adjectives. The task was set up in a similar way to the previous one. In Fig. 3.2a and Fig. 3.2b , speakers were asked to complete sentences using the prompts given to them and the words in the box adjacent to the image. The box included adjectives in English, but the speakers were instructed to complete the sentence in Tamil. Here too, the adjectives in the box were a mix of both derivational strategies. There is agreement in the predicative domain in Tamil, and this task was set up to elicit those agreement morphemes as well. The sentences in this set of tasks is given in (39).

(39) a. *inda paiyan romba* __
this boy very __
'This boy is very __ '



(a) Predicative Adjective Task [MASC]



(b) Predicative adjective Task [FEM]

Figure 3.2: Prompts for Predicative Adjectives

b. *inda poNNu romba* __

this girl very __

'This girl is very __ '

The responses elicited from these tasks were codified as either 'target consistent' or 'target inconsistent'. In the case of attributive adjectives, structures formed using the appropriate strategy were marked as 'target consistent'. Any divergence from this was considered as inconsistent. Similarly, for the predicative domain, using the appropriate derivational strategy made an utterance target consistent. Additionally, we also expect agreement in this domain. The subject of the sentence is pre-determined for the speakers. Thus, production of congruent agreement morphemes, i.e., [MASC] agreement with [MASC] subjects, and [FEM] agreement with [FEM] subjects, was considered to be target consistent. Any divergence from these structures was codified as target inconsistent.

3.5.2.2 Hypotheses and Research Questions

A broad research question of the study is about how we can expect the system of adjectives in Tamil to be sustained in a heritage variety of the language, given that input in the language is restricted. The focus of our inquiry in this domain is two-fold:

- The derivational strategy used to generate adjectives (for attributive and predicative adjectives)
- Executing appropriate biological gender agreement (for predicative adjectives)

To elaborate, the objectives listed above can be divided as fundamentally different grammatical phenomena. The first objective laid emphasis on the derivational strategy used by Tamil speakers to generate adjectives. There are two strategies at disposal, and choosing one of the two is idiosyncratic to the entity that they want to adjectivalise. In other words, for each item that can be adjectivalised in Tamil, the speaker has to know if it is a root that can be directly relativised, or if it is a noun that has to undergo verbalization before further derivation. This information comes from case-by-case learning, and is stored as part of the lexical knowledge of speakers. The adjectives themselves do not need to be learned; it is only the process of their formation that requires learning. The second objective was to study agreement relations. It is well-established that agreement and its underlying operations are part of the derivational component, and do not require focused learning. This information is regular, rule-governed and part of structural knowledge. In this vein, we invoke two well-registered hypotheses in the field of heritage grammars. It is generally understood that grammatical phenomena that involve extensive learning, particularly on a case-by-case basis, tend to be affected and undergo attrition in heritage grammars, whereas phenomena that follow from the rule-governed component of syntactic rules are generally resilient to changes and remain stable (Polinsky and Scontras, 2020). This difference can be attributed to the fact that lexical learning is more dependent on primary linguistic input than syntactic phenomena are. Thus, reduced input found in heritage language acquisition contexts does not have a uniform impact on the two (Rinke et al., 2024;

Unsworth et al., 2014; Unsworth, 2013). For the purposes of the analysis, we made a three-way distinction in order to form suitable predictions that can comprehensively cover the entire domain of adjectives in Tamil.

The first order of analysis was at the level of identifying that there are two distinct strategies to form adjectives in Tamil, and that these two operate in a mutually exclusive manner. There are some roots in the lexicon which can only be relativised to form adjectives (40a). Nominalising them before attaching the relativiser is not a licit strategy (40b). Similarly, lexical nouns must be verbalised before relativisation (41a). Directly attaching the relativiser to these nouns is once again disallowed in this system (41b).

(40) a. *anda nall-a paiyan*
that $\sqrt{good}$-REL boy
'that good boy'

b. **anda nalla-du-aa-na paiyan*
that $\sqrt{good}$-REL-NMLZ-V-REL boy
Intended: 'that good boy'

(41) a. *anda uyaram-aa-na paiyan*
that height-V-REL boy
'that tall boy'

b. **anda uyaram-a paiyan*
that height-REL boy
Intended: 'that tall boy'

Essentially, there appears to be a parametric difference across languages: languages that mark adjectives as an independent lexical category, and languages where adjectives are syntactically derived using (sometimes more than one) specific mechanisms. The studies reviewed so far in this chapter focused on the first type; these studies were informed by languages with adjectives as a pre-defined category. Therefore, it was expected that the form of the adjectives would not be subject to change despite changing acquisition conditions (i.e., when acquired as a heritage language). Tamil, on the other hand, falls into the second category. Adjectives need to be derived syntactically - an additional step in order to have adjectives in the grammar. Hence, when Tamil is acquired with reduced PLD, we would also like to investigate if the form of the adjectives also undergoes variation in heritage Tamil.

Knowing that there are two distinct strategies of adjective formation, and that they apply in strictly complementary domains is an integral part of the mental representations of adjective formation in Tamil. This leads to the question: How does one identify which of the two strategy applies to a particular adjective? The problem is particularly relevant, given that there isn't a native word/ loanword distinction at play here. As explained earlier, both categories of words are found in both, (40) and (41). In the absence of such external cues, we infer that speakers of Tamil have to learn this distinction on a case by case basis. For each lexical item learnt, the speaker also needs to learn how to form adjectives out of it. It is generally hypothesised that such instances of intensive lexical learning cannot be supported fully by the limited linguistic input that heritage speakers are exposed to. Therefore, the prediction is that heritage speakers of Tamil may make considerable overlaps between the two strategies; often using a derivational strategy in environments where it is not felicitous in the baseline variety of Tamil.

The second set of predictions deal with the domain of the regular, rule governed component. Then we enter the domain of agreement on predicative adjectives. As outlined earlier, it is the presence of the head Pred that predicativises an adjective. Pred is a functional head that can act as a probe looking for phi features to agree with. In contrast, Pred is not present in the attributive domain; within the DP, adjectives are directly merged as adjunct specifiers to the head noun. The absence of functional heads in this structure automatically implies no agreement in the attributive domain. The steps laid out here are systematic and rule-governed, and do not involve much learning. This is entirely in the domain of syntax, and is therefore regular. Thus, we predict that heritage speakers will not face it challenging to maintain an agreement-based distinction between attributive and predicative adjectives.

Additionally, we find more backing for this prediction when we consider other studies on how agreement fares in heritage grammars. Vorobyeva et al. (2024) and Polinsky (2025a) report that (gender) feature values may be a problematic issue in heritage grammars: the array of phi features that is involved in agreement may be less rich in heritage grammars relative to the baseline. Similarly, fewer functional elements may probe for phi features in heritage grammars when compared to the baseline. However, the "loss" of agreement reported in such studies is typically observed when the L2 in the contact situation does not have as rich a feature array, or as prolific an agreement system as the L1. Note that the setting of the current study is different:

the features and agreement relations in L1 (Tamil) are actually a subset of those in the L2 (Hindi-Urdu), as explained in Chapters 2 and 3. As such, we do not expect any inhibitory transfer effects from L2 on agreement systems in L1. Therefore, the prediction is that heritage Tamil speakers, in line with the baseline grammar, will continue to have agreement in the predicative domain while not having it in the attributive domain.

Thirdly, we further zero in on the specific features involved in these agreement relations. As noted in Chapter 2, Tamil has only biological gender; all the gender values in this language are semantically interpretable ones. The study in Chapter 2 also provided empirical evidence to show that biological gender is stable in heritage Tamil. Based on these findings we predict that, in the case of predicative adjectives too, biological gender agreement will be intact. In other words, the prediction is that [MASC] nouns will trigger [MASC] agreement on predicative adjectives, and [FEM] nouns will trigger [FEM] agreement on predicative adjectives, with minimal deviance from the baseline.

With these predictions framing the objectives of the study, a set of sentence completion tasks was conducted with 23 heritage speakers of Tamil. The results of the study are discussed in the following section.

3.5.2.3 Results of the Study

The first objective of the study was to inquire into where heritage speakers make the structural distinction between the two strategies of adjective formation in Tamil. In order to ascertain this, the task was designed in such a manner that would evoke the usage of the derived form of adjectives in both, the attributive and predicative domains. Responses were coded as target-consistent if the adjectives were formed using the specified strategy, and as target inconsistent if they used an infelicitous strategy. As predicted, there were some attested deviations from the expected strategies. Table 3.2 and Fig. 3.3 present the number of deviations observed for each strategy of adjective formation, in the attributive as well as predicative domains.

The expected utterances for attributive adjectives are given below in (42). Adjectives resulting from relativising a root (Strategy A) are presented in (42a), and adjectives that were generated from lexical nouns (Strategy B) are given in (42b).

	Strategy A	Strategy B
Attributive	All 76 utterances were consistent	3 out of 53 showed deviation
Predicative	27 out of 74 showed deviation	14 out of 73 showed deviation

Table 3.2: Elicited Adjectives According to their Formation

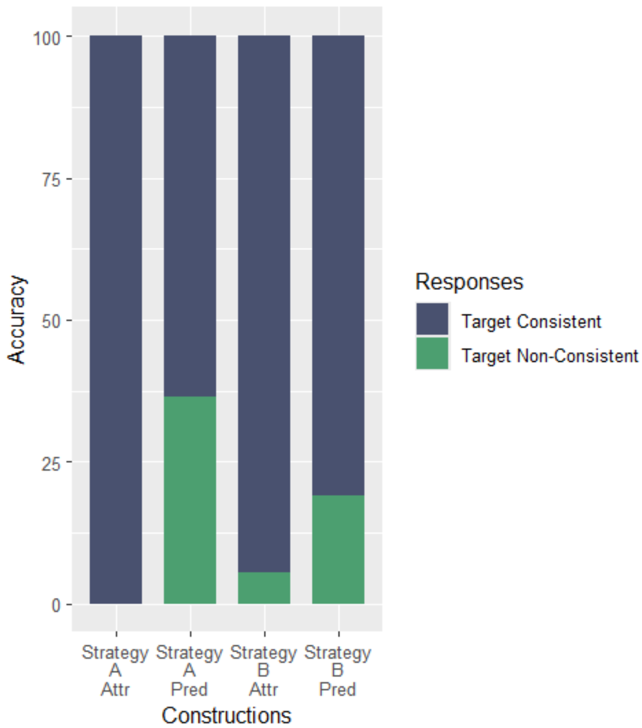


Figure 3.3: Distribution of Adjectives According to their Formation

(42) a. *inda* ***nall-a*** *paiyan thoongu-r-aan*
this $\sqrt{\textbf{good}}$**-REL** boy.MSG sleep-PRS-MSG
'This **good** boy is sleeping.'

b. *inda* ***uyaram-aa-na*** *paiyan thoongu-r-aan*
this **height-happen-REL** boy.MSG sleep-PRS-MSG
'This **tall** boy is sleeping.'

For the purposes of this analysis, we only consider the elicited adjective and its structural form. Agreement morphemes on the verb are not directly relevant to this study. While considering the shape of the adjectives, the responses elicited for the strategy used can be grouped into two types: Using the target-consistent strategy or a target-inconsistent strategy. The elicited data did not contain any target-inconsistent responses for Strategy A attributive adjectives. All the produced utterances were in accordance with the baseline grammar. However, for Strategy B attributive adjectives, there were a few target-inconsistent responses. They are listed below in (43). These utterances are labeled as target inconsistent, as (43a) contains a noun that is used as is; no verbalisation or relativisation processes are applied to the noun, rendering the utterance inconsistent with the baseline. (43b) is not consistent with the baseline, as here the speaker has innovated by adding another verb 'grow' with more functional projections, such as overt tense and aspect markers.

(43) a. *inda valati paiyan*
this height boy

b. *inda adarti-aa valand-iru-kir-a paiyan*
this thickness-happen grow-PRS-PERF-REL boy
'this boy who has grown big'

While there are exceptions as given in (43), it is possible to conclude that the attributive domain tends to remain stable in heritage Tamil. It is certainly worthwhile to note that there were no overlaps between the two strategies of adjective formation in the attributive domain. Heritage Tamil maintains the distinction between the two in a manner consistent with baseline Tamil.

The predicative domain fares differently. Target consistent utterances in the predicative domain are of the kind presented in (44). The two distinct strategies of deriving these adjectives are shown in (44a) - Strategy A and (44b) - Strategy B. Note that there is agreement with the subject in both the cases; on the adjective in (44a) and on the copula in (44b).

(44) a. *inda paiyan romba nall-a-van*
this boy.MSG very $\sqrt{good}$-REL-MSG
'This boy is very good.'

b. *inda paiyan romba uyaram-aa iru-k-aan*
this boy.MSG very height-happen be-PRS-MSG
'This boy is very tall.'

In this domain, we encounter a higher proportion of target-inconsistent responses. (45a)-(45f) cover Strategy A, and (45g)-(45h) cover Strategy B. We consider the responses in (45) to be target-inconsistent for the following reasons: there are no agreement morphemes on the predicative adjective (45a) and (45b), the speaker used the verbalisation strategy with a root, when the root should have been relativised directly (45c) and (45d), the speaker used both strategies simultaneously - direct relativisation of the root, as well using a copula, resulting in double instance of agreement (45e), the speaker repeated the subject lexical item instead of using agreement morphemes on the adjective (45f), the speaker did not apply any derivational process to the noun (45g), and the speaker added the agreement morpheme directly to a N+V complex instead of using a copula that can carry agreement morphemes (45h).

(45) a. *inda paiyan romba nall-a*
this boy.MSG very $\sqrt{good}$-REL
Intended: 'This boy is very good.'

b. *inda poNNu romba nall-a*
this girl.FSG very $\sqrt{good}$-REL
Intended: 'This girl is very good.'

c. *inda paiyan romba nall-aa iru-k-aan*
this boy.MSG very $\sqrt{good}$-REL-happen be-PRS-MSG
Intended: 'This boy is very good.'

d. *inda poNNu romba nall-aa iru-k-aa*
this girl.FSG very $\sqrt{good}$-happen be-PRS-FSG
Intended: 'This girl is very good.'

e. *inda paiyan romba kett-a-van-aa iru-k-aan*
this boy.MSG very $\sqrt{bad}$-REL-MSG-happen be-PRS-MSG
Intended: 'This boy is very bad.'

f. *inda poNNu romba nall-a poNNu*
this girl.FSG very $\sqrt{good}$-REL girl
'This girl is a very good girl.'

g. *inda paiyan romba uyaram*
this boy.MSG very height
Intended: 'This boy is very tall.'

h. *inda paiyan romba uyaram-aa-na-van*
this boy.MSG very height-happen-REL-MSG
Intended: 'This boy is very tall'

These are all instances where the elicited adjectival structures deviate from the baseline grammar. In Fig. 3.4 we provide the distribution of target-consistent and inconsistent responses across attributive and predicative adjectives. It is visually evident from Fig. 3.4 that while the attributive domain is largely target-consistent (95%), the predicative domain experiences considerable innovation and change on the part of heritage speakers (70%). The results are displayed in absolute terms in Table 3.3.

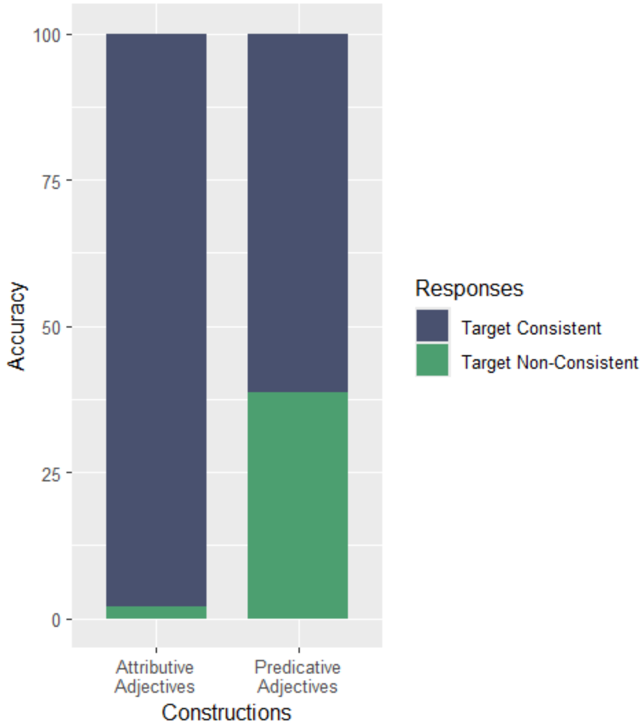


Figure 3.4: Distribution of Attributive and Predicative Adjectives

Construction type	Target-Consistent	Target-Inconsistent
Attributive Adjectives	134	3
Predicative Adjective	106	41

Table 3.3: Distribution of Attributive and Predicative Adjectives- Absolute Values

The results in Fig. 3.4 and Table 3.3 confirm the predictions of the first hypothesis: We expected there to be a considerable overlap between two adjective-formation strategies in Tamil. As can be seen, this is particularly prominent in the case of predicative adjectives, with the attributive domain being more resistant to change. We shall delve deeper into the differences exhibited by the two in the next section. For the moment, it is important to note that all the divergences registered here are in the domain of strategy selection/ application. Especially in the predicative domain, we encounter instances of misapplication of Strategy A to lexical nouns, or misapplication of Strategy B to roots. There are no deviations with regard to the internal structure of the strategy once selected. For instance, we do not find any utterances where Strategy

A has been selected and applied but with a copula (which is a composite part of Strategy B). From the absence of such data, we can infer that heritage Tamil deviates from the baseline only in regards to learning the domains of application of each strategy. Once selected, its structural form is intact.

The second hypothesis was about those parts of grammar that are not prone to effects of language contact. We had outlined that the syntactic, rule-governed component will be resilient to change in heritage Tamil. One predication that fell out of this hypothesis was: The primary differences between attributive and predicative adjectives will be maintained. In terms of agreement relations, the attributive domain will continue to not have agreement relations, while the predicative domain will facilitate subject-verb agreement. In other words, we predicted that heritage speakers of Tamil will not introduce agreement into the domain of attributive adjectives, and nor will they use predicative adjectives without agreement.

This prediction is mostly borne out. We do not find any instances of agreement morphemes appearing on attributive adjectives; that domain continues to marked by the lack of agreement relations. However, there are a few instances of predicative adjectives where the adjectives have been derived using the appropriate strategies, but they appear without overt agreement morphology. The relevant examples provided in (45a) and (45b) are repeated here as (46a) and (46b).

In (46a) and (46b), we see the root+REL form occurring in the predicative position, but it does not carry any agreement morphemes.

(46) a. *inda paiyan romba nall-a*
this boy.Msg very $\sqrt{good}$-REL
Intended: 'This boy is very good.'

b. *inda poNNu romba nall-a*
this girl.Msg very $\sqrt{good}$-REL
Intended: 'This girl is very good.'

Once again we are able to record an instance where the attributive domain was unaffected, but the predicative domain underwent some change. We discuss these findings in the following section.

Additionally, Section 3.3.4.1 explained the agreement system of Tamil predicative adjectives with two different agreeing heads. It is the highest functional head in a clause that exhibits morphological agreement. In Strategy A adjectives, the head is Pred, and in Strategy B it is T. According to the general principles of agreement in Tamil, a sentence may exhibit only one instance of agreement. Therefore, it can never be the case that both heads display agreement morphology simultaneously. As this is part of core syntactic structure, we do not expect heritage Tamil to diverge in this regard. The data elicited aligns with this expectation; there were no instances of hypothetical constructions such as (47) in heritage Tamil.

(47) **anda paiyan nall-a-van iru-k-aan*
that boy $\sqrt{good}$-REL-**MSG** be-PRS-**MSG**
Intended: 'that boy is good.'

The second prediction regarding the syntactic component narrowed in on predicative adjectives. The tasks presented to the speakers contained images of human characters, which would act as the subjects of predicative adjectives. As already established in Chapter 2, the gender values on these nouns are categorized as biological gender, with the values being semantically interpretable. We also outlined that predicative adjectives display number and gender agreement as a result of an Agree relation between Pred and the subject DP. In other words, biological gender agreement on predicative adjectives is part of the systematic, regular core-syntactic operation Agree, and is not contingent upon lexical learning. Because of such a configuration, we predicted that agreement on predicative adjectives will not be affected in heritage Tamil.

This prediction is fully attested in the results of the study. There are no cases of mismatched agreement in predicative adjectives. We do not see any instances of [MASC] nouns appearing with [FEM] agreement, or [FEM] nouns appearing with [MASC] agreement. These findings from adjectives concur with what has already been established using nouns in Chapter 2: Fixed biological gender does not undergo any change in heritage Tamil.

In the next section, we shall elaborate on the findings of this study and discuss their implications for a representational understanding of heritage grammars.

3.5.2.4 Discussion

The results of our current study on adjectives in heritage Tamil yield three significant conclusions:

- Heritage speakers of Tamil find it challenging to keep the two strategies of adjective formation distinct, as evidenced by the overlap between the two strategies

- Regular, rule-governed components do not undergo any change in heritage Tamil; heritage speakers maintain a distinction between attributive and predicative adjectives in terms of agreement

- Agreement in terms of fixed biological gender values does not undergo attrition in heritage Tamil; there are no instances of mismatched agreement

From the list given above, it immediately becomes clear that the discussion around adjectives and gender in heritage Tamil can broadly be divided into two categories: avenues where the speakers diverge from the baseline, and ones where they remain unchanged and display resistance to external conditions such as reduced input and transfer effects of the dominant language. In this section, we shall proceed to elaborate on the implications of these patterns for heritage Tamil.

Earlier in this chapter, it was established that there are two derivational strategies to form adjectives in Tamil. These two strategies exist in a complementary distribution; uncategorised roots ought to be directly relativised to form adjectives, whereas lexical nouns need to be verbalised before relativisation. This distinction is strictly maintained in the domain of adjective formation in Tamil, and must be learnt on a case-by-case basis; it cannot be garnered from a general rule. Case-by-case learning crucially hinges on the availability of sufficient linguistic input in the environment. Heritage speakers are generally not exposed to the language in the same way quantitatively as monolingual speakers are. The scarcity of input implies less experience in sorting items into one of the two adjective formation strategies. Hence, we see some innovation by the heritage speakers in this domain. They tend to blur the strict distinction between the two strategies, as evidenced by a significant overlap in the application of the two strategies when asked to form adjectives out of concepts. Such a tendency aligns with what has been reported about heritage grammars cross-linguistically.

The second part of the discussion revolves around domains of heritage Tamil where the heritage speakers did not show divergence from the baseline. While defining the objectives of the study, we hypothesized that grammatical phenomena that are within the ambit of regular, rule-governed syntactic phenomena are likely to be impervious to the effects of reduced input. Within this group, we identified two domains where heritage Tamil would not vary from the baseline: Not adding agreement to attributive adjectives, and fixed biological gender agreement in the predicative domain.

Firstly, we consider the domain of attributive adjectives. In baseline Tamil, attributive and predicative adjectives are differentiated by agreement relations. Attributive adjectives do not exhibit any agreement, while predicative adjectives agree with the subject of the predicate. Taking a step back, we explained agreement on adjectives as a result of a probe-goal relation. In Tamil, finite verbs act as probes and carry agreement morphemes. For the predicative adjectives, Pred acts as the probe, and initiates Agree with the subject DP. Consequently, number and gender agreement with the subject is overtly realised on the adjectival complex. In the attributive domain, there is no element that can act as a probe and facilitate agreement. Thus, we do not see any instance of agreement in the domain of attributive adjectives. The point most relevant for the purposes of the current discussion is that heritage speakers of Tamil recognise the absence of probes within the DP i.e., the domain of attributive adjectives. Heritage speakers do not force agreement in the attributive domain in their grammar of Tamil.

More interestingly, heritage speakers retain the system of attributive adjectives despite there being two potential motivators for restructuring the attributive domain: The dominant language, Hindi-Urdu, has agreement in the attributive domain. In Hindi-Urdu, adjectives are independent lexical categories, and are therefore merged directly as APs to the DP spine. As such, these are functional heads that are also active probes for agreement with the noun they modify; there is attributive adjective agreement in Hindi-Urdu. Thus, having such agreement in Tamil would assimilate their heritage language with the societal language. Secondly, adding agreement in the attributive domain would obliterate the distinction between the attributive and predicative domains, perhaps simplifying the system of adjectives in heritage Tamil. If both types of adjectives exhibited agreement with the noun, it would be possible to have a uniform understanding of adjectives in Tamil across the board. It is particularly relevant that despite these two motivations being present, heritage speakers of Tamil did not introduce agreement into the attributive domain. They maintain the structural distinction between attributive and predicative adjectives, despite the conditions of acquisition that characterise heritage grammars. Therefore, there is no evidence for contact-induced change in this domain.

Moving forward, we step into the domain of gender agreement found on predicative adjectives. As outlined previously, Tamil has only biological gender, either fixed or variable. Thus, gender agreement on predicative adjectives is also of the same type. Biological gender does not crucially depend on robust lexical learning for its realisation. These instances of gender are rooted in the gender value of real-world referents and are extracted from particular referents. Additionally, we also noted that biological gender is represented directly on the lexical head N. A confluence of these facts leads us to expect that biological gender will not undergo any attrition in heritage grammars, as the reduction of exposure/input is not a matter of concern for the acquisition of biological gender. This finding was empirically attested in Chapter 2, where we established that biological gender and its representations are not affected in heritage Tamil. In this chapter, we take this finding further, and move into a new domain: agreement relations involving biological gender. Under the hypothesis that rule-governed, syntactic phenomena are relatively stable in heritage grammars, we predicted that biological gender agreement on predicative adjectives in heritage Tamil will not exhibit any divergence. In other words, we predicted that there will be no instances of mismatched agreement: [FEM] nouns will invariably trigger [FEM] agreement, and [MASC] nouns will do so for [MASC] agreement. We did not expect there to be any instance of divergence from this pattern; we predicted that heritage speakers will not overgeneralise one gender value, and attempt to collapse the binary system into a unary one. These predictions were borne out in the empirical data we provided in this chapter. We do not observe any instance of mismatched agreement or overgeneralisation of one of the two values, or using a neutral agreement morpheme that would obliterate the distinction between the two. The system of gender agreement we report for baseline Tamil is sustained in the heritage variety too.

The findings of the current study and their implications bring us to the next topic of discussion: Stability in heritage grammars. At the beginning of this dissertation, core syntax was identified as a domain that remains invariant in heritage grammars. This point is illustrated by both the distinction between attributive and predicative adjectives, and the lack of mismatches in biological gender agreement. The fact that heritage Tamil structurally differentiates between attributive and predicative adjectives implies that the underlying system of probe-goal relations remains intact. Similarly, the presence of robust gender agreement too confirms that the underlying operation of Agree is at play. These are part of core syntax, or the structural representation, and do not require special learning mechanisms. They remain stable in the face of restricted PLD and potential transfer effects – both of which are a part of the acquisition context of heritage Tamil.

Another aspect of heritage Tamil that merits discussion here is the role played by cross-linguistic transfer effects. Given the asymmetric nature of heritage bilingualism, it is observed that the societal language and its grammatical features tend to have a bearing on the heritage language. In other words, certain grammatical properties of the L2 are transferred onto the L1, leading to its restructuring. Transfer effects do not affect all grammatical domains equally. The exact magnitude of these effects are unclear to the field as yet (Scontras et al., 2015; Lohndal, 2021). In the current study, we do not find evidence for transfer effects - either in the domain of adjectives or in gender agreement. In the domain of adjective formation, we observe a considerable overlap in the two strategies of adjective formation. This outcome can be explained as a consequence of inadequate PLD. The fact that there is an overlap between the two strategies, and not overgeneralization of one strategy strongly suggests that it is not an effect fo transfer. If the objective was to assimilate Tamil adjectives to those in Hindi-Urdu (a system where adjectives are differentiated structurally), heritage Tamil would have attempted to obliterate the distinction by using one of the two strategies for both kinds of adjectives. Since we do not report any of such instances in heritage Tamil, we can conclude that transfer effects are not at play here. Similarly for gender agreement, Tamil and Hindi-Urdu differ by not marking agreement and marking it, respectively. Cross-linguistic influence from Hindi-Urdu would have led to the emergence of attributive adjective agreement in heritage Tamil. However, there are no such cases in heritage Tamil. Based on this evidence, I conclude that cross-linguistic influence is not playing any role in heritage Tamil.

To summarise the discussion in this section, we can clearly demarcate where the effects of heritage grammar acquisition have a role to play and where they do not. Grammatical phenomena that involve considerable lexical learning witness the effects of heritage conditions, but anything that is part of core syntax does not, and exhibits resistance even in the face of reduced input and pressures from the societal language.

3.6 Conclusion

This chapter presented an account of adjectives and adjectival agreement in baseline and heritage varieties of Tamil. At the outset, it was established that adjectives are not distinct lexical categories in Tamil, and must be derived by means of some structural operations. We outlined the two ways of forming adjectives in the syntax, and inferred that speakers of Tamil need to learn the implementation of these two strategies on a case by case basis, which may present a challenge in contexts of acquisition where input is reduced, i.e., heritage Tamil. Another branch of inquiry in this chapter was the difference between the attributive and predicative forms of the adjective in Tamil, and its implications for adjectival agreement. With these formalisms in place, we conducted a study to understand their representation in heritage Tamil. Using novel empirical evidence from heritage Tamil, we were able to establish that grammatical phenomena that involve lexical learning tend to undergo restructuring in heritage grammars, whereas those that are part of core syntax tend to remain stable despite acquisition conditions that deviate from the norm. The findings of this study confirm that variation in heritage grammars is predictable based on the lexical and structural components of the phenomenon discussed.

Chapter 4

Gender in Possessives

4.1 Introduction

Chapter 4 presents yet another domain where core syntactic aspects of gender can be observed: possessives. Here an account of possessive DPs in Tamil is presented, where the presence of gender features on nouns determines whether or not the noun may occur in certain structural configurations. The focus of the chapter will be to test how these constraints develop in a heritage grammar. To elaborate, Chapter 2 presented the full picture of Tamil lexical items and their gender features. It was shown that Tamil has only biological gender, reflected on [+HUMAN] nouns. All other nouns in the lexicon, including [+ANIMATE -HUMAN] are completely devoid of gender features. This aspect of the gender system in Tamil is particularly relevant for the present discussion. Essentially, an investigation into the internal syntax of possessive DPs will shed light on the fact that gendered nouns are disallowed in predicative possession structures – a restriction that does not apply to non-gendered nouns. The expository part of this chapter is, therefore, devoted to presenting a complete picture of the structural representation of possessive DPs in Tamil. Following this, the text then delves into the particulars of gender features in possessive constructions. Alongside these technical points, the inquiry will also focus on how these representations and derivational constraints fare in a grammar acquired as a heritage language. With the structural representation of possessive DPs and the gender-based derivational constraints laid out, we then ask the questions: Is the representation of possessives in heritage

Tamil identical to the baseline grammar, or are there departures from the baseline? If so, how can we characterise these changes?

The current study on heritage Tamil reveals that structural constraints involving gender do find a place in heritage Tamil. In other words, we find evidence for the gender-based restriction in heritage Tamil predicative possessives as well. This finding confirms the idea that gender features develop in heritage grammars, if they are a part of core structural phenomena.

The chapter is organised as follows: Section 4.2 offers an overview into why the structure of possessives is relevant to the current discussion. With the context in place, the discussion turns to the empirical facts concerning possessors in Tamil and Hindi-Urdu. Section 4.3 follows with the structural representation of attributive and predicative possession in these languages, with an introduction to the role played by gender features in the latter, thus completing the initial technical discussion. Section 4.4 then ventures into the implications of such a system for language contact in heritage grammars. The details of the study conducted on heritage Tamil are provided in this section. Finally, Section 4.5 concludes that the gender-based restrictions in heritage Tamil possession remain stable, as they are part of core syntax.

4.2 Why Possessives? - The Gender Connection

Possessive structures have been at the centre of most inquiries into the structure of the noun phrase, as they express a structural relation between two nominal entities, both situated inside the DP (Aikhenwald, 2013; Stassen, 2009; Taylor, 1996). The two components of a possessive structure are the possessor and the possessum. The most general characterisation of the relationship between the two is in terms of the possessor acting as a restrictive modifier of the possessum. In (1), the possessor 'John' restricts the interpretation of the possessum 'house' to 'that house which belongs to John', to the exclusion of all other 'houses'.

(1) *John's house*

Such a relation of possession can be represented in the attributive domain, as in (1), or with the help of predication (2). In (2) the possessum and the possessor are related to each other as parts of the subject and the predicate of the construction, respectively.

(2) *This house is John's.*

A key reason why possessive structures have continued to receive attention in generative syntax is that they are complex structures. It is also this very characteristic of possessives that makes them ideal candidates for study in language contact situations such as heritage grammars: the derivational complexity of possessives makes these constructions potentially fertile grounds for innovation when a grammar develops under a constraint context of acquisition.

The second, empirical, motivation for choosing possessives as the focus of this chapter stems from the fact there is an interplay between biological gender features and the structural position of possessors in Tamil. As alluded to in the introductory section, nouns with gender features are not allowed in predicative positions. (3) illustrates that the non-gendered noun 'house' may appear as the possessum in a instance of predicative position (3a), whereas the gendered 'boy' may not (3b)[1].

(3) a. *inda viidu john-oda-du*
this house John-POSS-du
'This house is John's.'

b. **inda paiyan john-oda-du*
this boy John-POSS-du
Intended: 'This boy is John's.'

This particular fact will be taken up in greater detail in the following sections, once a fundamental characterisation of possessive structures has been established. The key difference between 'house' and 'boy' is that only the latter carries gender features. This chapter presents a case where the gender feature imposes certain derivational / structural restrictions. Studying the structure of possessives is an optimal way to observe how gender interacts with the structural organisation of DPs. The rest of this chapter is devoted to understanding the exact mechanisms by which gender plays such a role in possessive DPs. Once we paint a picture of the relationship between gender and possessives, the next step is to consider its implications for heritage Tamil.

Having stated the motivation behind studying possessors, we now proceed to review some basic empirical facts about possessives in Tamil and Hindi-Urdu. The structural properties exhibited by both the languages are relevant for the questions taken up in this chapter.

[1]For now, the morpheme *-du* will be glossed as is. An explanation of its nature and presence is provided in the forthcoming sections

4.2.1 Possessives in Tamil - Empirical Facts

One of the most prominent empirical facts considered when we look at possessive structures cross-linguistically is the relative order of the possessor and the possessum. In Tamil, the possessor linearly precedes the possessum within the DP - attributive possession (4). The possessor is overtly marked with the genitive marker *-oda* to indicate possession. The possession (POSS) marker *-oda* does not exhibit any agreement relations. It remains the same regardless of the number and gender features of the possessor and the possessum. (4) shows that *-oda* stays the same when the possessor/ possessum is [MASC], [FEM] or [PL].

(4) a. *ram-**oda** viidu*
Ram.MSG-POSS house
'Ram's house'

b. *sita-**oda** aNNa*
Sita.FSG-POSS brother
'Sita brother'

c. *koLandaingaL-**oda** kai*
children.PL-POSS hand
'children's hand'

d. *koLandaingaL-**oda** bommai*
children.PL-POSS toys
'children's toys'

Possession can also be expressed using pronominal forms. As explained in Chapter 2, III person pronouns are marked for number and gender features in Tamil. Thus, they can be used to indicate [MASC] (5a), [FEM] (5b) and [PL] (5c) (5d) possessors.

(5) a. *avan-**oda** viidu*
he-POSS house
'his house'

b. *avaL-**oda** aNNa*
she-POSS older brother
'her older brother'

c. *avangaL-**oda** kai*
they-POSS hand
'their hand(s)'

d. *avangaL-**oda** viidu*
they-POSS house
'their house'

There can be intervening modifiers such as numerals (6a) and adjectives (6b)-(6c) between the possessor and the possessum.

(6) a. *ram-**oda** **moonu** pasanga*
ram-POSS three boys
'Ram's three sons'

b. *ram-**oda** **cinn-a** paiyan*
ram-POSS $\sqrt{small}$-REL boy
'Ram's younger son'

c. *mira-**oda** **periy-a** viidu*

mira-POSS	$\sqrt{big}$-REL	house

'Mira's big house'

The examples mentioned above are all cases of possession within the DP. Apart from these, there is an option for the possessor to occur predicatively (7). In this instance, the possessum is the subject of the predicate within with the possessor is located. These are akin to equative copular sentences.

(7) a. *anda viidu ram-**oda**-du*

that	house	Ram-POSS-du

'That house is Ram's.'

b. *inda puttagam siita-**oda**-du*

this	book	Sita-POSS-du

'This book is Sita's.'

In (7) we can see that when the possessor occurs as a predicate, there is an added morpheme *-du* in the structure. This morpheme is absent from the possessives of the attributive kind. *-du* is a productive nominalizer in Tamil. The latter part of this subsection is devoted to the domain of nominalisation, so that the exact nature of *-du* can be established with certainty.

The third person non-human pronoun in Tamil is either *adu* 'it.distal' or *idu* 'it.proximal'. According to Krishnamurti (2003), this pronoun is a noun-forming affix in several Dravidian languages, including Tamil. krishnamurti shows that finite verbs can be nominalised by suffixing *-du* to them. Doing so strips the verb of its PNG markings completely. This process is exemplified by (8). (8a) and (8b) show the finite verb 'slept' agreeing with the phi features of the respective subjects. The nominalised form of the verb is present in (8c), where the nominalizer *-du* occurs with the tense-marked verb, effectively rendering it a noun. In its nominalised form, the verb does not inflect differently for [MASC] and [FEM] verbs anymore.

(8) a. *avan toongi-n-aan*

he	sleep-PST-3MSG

'He slept.'

b. *ava toongi-n-aa*

she	sleep-PST-3FSG

'She slept.'

c. *avan/ ava toongi-n-adu*

he/	she	sleep-PST-NMLZ

'his or her sleeping'

Consider another example (9) with the finite verb 'went' in (9a) and (9b) agreeing with the respective subjects, and its nominalised form in (9c). In (9c), the nominalised finite verb *po-n-adu* is part of the proposition that is being negated here.

(9) a. *kumar kovilu-ku po-n-aan*
Kumar.MSG temple-DAT go-PST-3MSG
'Kumar went to the temple.'

b. *radha kovilu-ku po-n-aa*
Radha.FSG temple-DAT go-PST-3FSG
'Radha went to the temple.'

c. *kumar/ radha kovil-ku po-n-adu illa*
Kumar.MSG/ Radha.FSG temple-DAT go-PST-NMLZ NEG
'Kumar/ Radha has never gone to the temple. (lit: Kumar/ Radha going to the temple has not happened.)'

Based on these empirical facts, we can establish *-du* as a nominalizer in Tamil. Additionally, Butt et al. (2020) illustrate that *-du* applies productively as a nominalizer across different construction types, such as relative clauses (10a), complementizers (10b), etc. The morpheme has been glossed as *-thu* 'neut (non-gendered) pronoun' in these examples.

(10) from Butt et al. (2020)

a. *[avan pizhai sey-tt-aan enp-a-thu-ai] ram*
he mistake do-past-3MSG COMP-REL-**PRON.3SN**-ACC Ram.NOM
nirupi-tt-aan
prove-PST-3MSG
'Ram proved (the fact) that he made mistakes.'

b. *[avan pizhai sey-t-a-thu-ai] ram nirupi-tt-aan*
he mistake do-PAST-REL-**PRON.3SN**-ACC Ram.NOM prove-PST-3MSG
'Ram proved (it) that he made mistakes.'

The prevalence of the nominalizer across different constructions encourages us to posit that it is the same morpheme that occurs with predicative possessors. The structural implications of having a nominalizer as part of a possessive DP will be explored in subsequent sections. For the purposes of the current discussion, we return to our characterisation of possessive DPs in light of the presence of a nominalizer in predicative possession.

The picture that emerges of Tamil is then that possessives can be in the attributive or predicative domain. A key difference between the two is the appearance of a nominalizer. The nominalizing morpheme *-du* is disallowed in the attributive domain (11), and obligatory in the predicative domain (12).

(11) Attributive domain: [$\text{DP}_{\text{Possessor}}$-POSS $\text{NP}_{\text{Possessum}}$]

a. *peter-oda viidu*
Peter-POSS house
'Peter's house'

b. **peter-oda-du viidu*
Peter-POSS-NMLZ house

(12) Predicative domain: [$\text{NP}_{\text{Possessum}}$] [$\text{DP}_{\text{Possessor}}$-POSS-NMLZ]

a. *inda viidu peter-oda-du*
this house Peter-POSS-NMLZ
'This house is Peter's.'

b. **inda viidu peter-oda*
this house Peter-POSS

With these observations in place, we now proceed to review the empirical facts of possession in Hindi-Urdu.

4.2.2 Possessives in Hindi-Urdu - Empirical Facts

Possessives in Hindi-Urdu occur in both, attributive and predicative, forms. The internal word-order of possessives is such that the possessor linearly precedes the possessum in the attributive domain. Possession is marked by suffixal attachment to the possessor. An important fact of possession in Hindi-Urdu is that the POSS marker agrees with the phi features of the possessum. (13) provides an illustration of these agreement patterns with the possession morpheme agreeing with the [MASC] (13a, 13d), [FEM] (13b, 13e) and [PL] (13c, 13f) features of the possessum. The features of the possessor are not transferred onto the POSS marker.

(13) a. *ram k-aa ghar*
Ram.MSG POSS-MSG house.MSG
'Ram's house'

b. *ram k-ii ghaDi*
Ram.MSG POSS-FSG watch.FSG
'Ram's watch'

c. *ram k-e kapde*
Ram.MSG POSS-PL clothes.PL
'Ram's clothes'

d. *mira k-aa ghar*
Mira.FSG POSS-MSG house.MSG
'Mira's house'

e. *mira k-ii ghaDi*
Mira.FSG POSS-FSG watch.FSG
'Mira's watch'

f. *mira k-e kapde*
Mira.FSG POSS-PL clothes.PL
'Mira's clothes'

Predicative possession is expressed using similar morphological mechanisms. In the predicative domain, the possessum is represented as the subject of a predicate containing the possessor. An overt copula mediates the relation between the two (14). Identical to the attributive domain, the POSS morpheme inflects for the features of the possessum in the predicative instance as well.

(14) a. *yeh ghar ram k-aa hai*
this house.MSG Ram POSS-MSG COP
'This book is Ram's.'

b. *yeh ghaDi ram k-ii hai*
this watch.FSG Ram POSS-FSG COP
'This watch is Ram's.'

c. *yeh kapde ram k-e hain*
this clothes.PL Ram POSS-PL COP.PL
'These clothes are Ram's.'

Additionally, Hindi-Urdu allows all nouns in the predicative position. All nouns are gendered in Hindi-Urdu, regardless of their status vis-a-vis [+/-HUMAN]. [-HUMAN] (14) and [+HUMAN] (15) nouns occur in predicative possession. The distinction found in Tamil does not apply here.

(15) a. *yeh student us teacher k-aa hai*
this student.MSG that teacher POSS-MSG COP
'This student [pointing] is that teacher's.'

b. *yeh patient us doctor k-e hain*
this patient.MSG.HON that doctor POSS-MSG.HON COP.HON
'This patient [pointing] is that doctor's.'

It is the presence of the copula that mainly differentiates between the attributive and predicative domains in Hindi-Urdu.

4.2.3 Generalisations

Based on the facts surveyed in this section we can arrive at certain key differences between Tamil and Hindi-Urdu in terms of how possessive DPs are built. In Tamil, there is a morpho-syntactic difference between the attributive and predicative domains. The predicative domain includes an additional morpheme - the nominaliser - which is absent in the attributive domain. In Hindi-Urdu, the distinguishing factor between the predicative domain and the attributive domain is the presence of a copula in the former. While both languages have a way of marking the presence of a predicative element in predicative possession, they differ in how they execute the strategy. A second difference observed between Tamil and Hindi-Urdu is in terms of agreement: Tamil does not have agreement in the attributive or the predicative domains, whereas Hindi-Urdu has the POSS morpheme agreeing with the phi features of the possessum. With the basic empirical observations laid out, we shall delve into the structural representations underlying these facts.

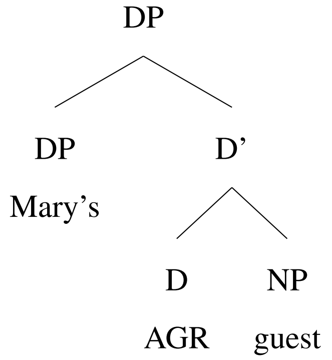
4.3 Structural Representation of Possessives

This subsection addresses the derivational structure underlying possessives. At the outset it needs to be clarified that attributive and predicative possession are different manifestations of a common underlying structure. Heine (1997) and Stolz et al. (2008) develop the idea that the linguistic distinction between different kinds of possession is essentially a semantic one. The relations between possessor and possessum remain invariant across the two. The two domains differ only in terms of how the relation is mediated. In the attributive domain, the possessor is a direct modifier of the possessum and the relation is mediated NP-internally. Contrastingly, in the predicative domain, the possessum is an argument of the predicate containing the possessor. The relation between possessor and possessum remains the same across both the types of possession. Nevertheless we shall address the attributive and predicative domains independently in order to elaborate on these relations with clarity.

4.3.1 Attributive Possession

Abney (1987) provides one of the earliest systematic and unified accounts of noun-phrase internal syntax. A key hallmark of Abney's proposal is the presence of a functional projection DP as an overseer of all the internal constituents of the noun phrase. His account of possessives also capitalises on this functional projection: Possessors are placed at SpecDP to emphasise their status as modifiers of head nouns (16). The possessor, along with the POSS morpheme, is situated at SpecDP, the head of which is reserved for AGR - a functional head responsible for facilitating agreement between the possessor and the possessum (if necessary).

(16)



The relevance of AGR becomes apparent when we consider Hungarian - a language which has made significant empirical contributions to the study of possessive structures. An important property of Hungarian possessive DPs is that the POSS morpheme exhibits agreement with the phi features of the possessor. The examples in (17) show number agreement between the possessor and the POSS morpheme.

(17) Hungarian (Abney, 1987)

a. *az en vendeg-e-m*
the **I-NOM** guest-POSS-**1SG**
'my guest'

b. *a te vendeg-e-d*
the **you-NOM** guest-POSS-**2SG**
'your guest'

c. *(a) mari vendeg-e-∅*
(the) **Mary-NOM** guest-POSS-**3SG**
'Mary's guest'

Abney posits that AGR housed at D is the nominal counterpart to INFL in the clausal domain. As such, it facilitates agreement between the possessor and the possessum. However, a deeper look into the Hungarian facts renders Abney's proposal unsustainable. Szabolcsi (1983, 1994) presents an elaborate account of Hungarian empirical facts. The key points of Szabolcsi's characterisation of possession are: Hungarian possessors occur in two variants - NOM and DAT marked. These two types of possessors occur in distinct structural positions: post- and pre-article, (18a) and (18b) respectively. The two constructions do not differ in meaning.

(18) Hungarian (Szabolcsi, 1994)

a. *(a) mari kalap-ja-i*
(the) Mary-NOM hat-POSS.PL-3SG

‘Mary’s hats’

b. *mari-nak a kalap-ja-i*
Mari-DAT the hat-POSS.PL-3SG
‘Mari’s hats’

The two possessors also have different extraction properties: the NOM-marked possessor is not allowed to undergo certain movements that the DAT-marked possessor is. (19a) shows that a NOM-marked possessor may not raise to a higher position, whereas (19b) shows that a DAT-marked possessor may undergo the same movement.

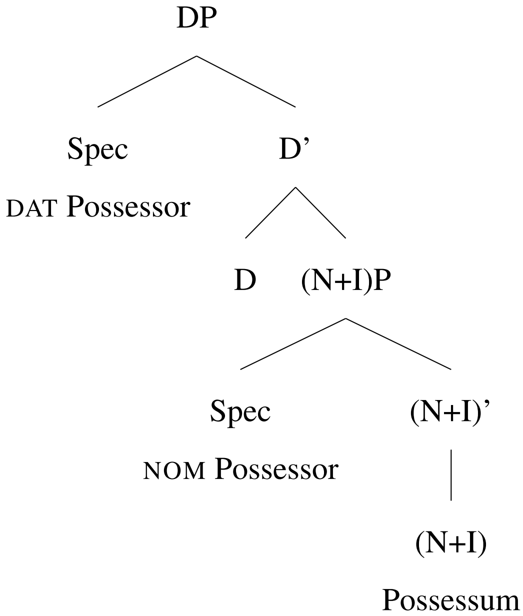
(19) Hungarian (Szabolcsi, 1994)

a. **mari$_i$ fekete volt a t_i kalap-ja*
Mary-NOM$_i$ black was the t_i hat-POSS.3SG
‘Mary’s hat was black.’

b. *mari-nak$_i$ fekete volt a t_i kalap-ja*
Mary-DAT$_i$ black was the t_i hat-POSS.3SG
‘Mary’s hat was black.’

These facts add considerable complexity to the theory of possessors. In light of these facts, we can infer that it is not only possible, but also necessary, to conceptualise more than one structural position for possessors. Szabolcsi’s proposal is an attempt to address this very question. Her solution to the problem is to posit two structural positions for possessors within the DP - the specifier of a functional projection connected to NP (N+I), and the specifier of the higher DP (20). Out of the two, the former is the default position; all possessors are base generated at Spec(N+I)P, and some move to SpecDP. Within this framework, (N+I)P and DP correspond to IP and CP in the clausal domain, respectively. Based on these stipulations, (N+I)P is capable of assigning NOM case to any element located in its specifier, resulting in the possessor in that position being NOM-marked. As for the DAT variant, Szabolcsi proposes that movement of the possessor from Spec(N+I)P to SpecDP is what awards it the DAT marker. Szabolcsi adds that identical to the CP layer, SpecDP too is an operator position with additional semantic properties, enabling it to encase the possessor situated there in DAT.

(20)



These arguments straightforwardly account for the word order observed in Hungarian. As stated above, the NOM-marked possessor follows the article (at D) because it remains in situ in a position lower than D. The DAT-marked possessor linearly precedes the article as it has moved to a position above the article. As for the extraction differences pointed out in (19), the ability of the DAT-marked possessor to displace to the exclusion of the NOM-marked possessor is also explained by (20). The DAT-marked possessor is eligible for further movement by virtue of having moved from its base-generated position to an "escape-hatch" in SpecDP. The NOM-marked possessor, on the other hand, is not allowed to move from its base-generated position.

For the purposes of the current discussion, the crux of Szabolcsi's proposal is that possessors are base-generated at a position lower than SpecDP but higher than NP. She arrives at this conclusion by eliminating certain impossibilities: SpecDP or any position higher than that is ruled out because generating the possessor in either of those positions will not derive the word order found in Hungarian (D Possessor Possessum). Any position lower than NP is ruled out because, if the possessor was generated much lower and then moved to SpecNP, the movement should have had repercussions in terms of scope interpretation. However, such effects are not observed in Hungarian[2] . Based on the absence of ambiguity, Szabolcsi concludes that the option

[2]Szabolcsi (1994) provides a detailed account of how the purported movement of the possessor from a lower position to SpecNP should have lead to scope interpretation ambiguity. Consider (i) below, where out of the two plausible interpretations of (i), only A is allowed:

of raising the possessor from a lower position is unavailable. Therefore, she is logically left with the only option of positing the possessor at an intermediate functional projection between DP and NP - ergo, the possessor is said to be base-generated at Spec(N+I)P. Szabolcsi's account gained significant acceptance and following. Future accounts of possessives differed on the nomenclature of the functional projection - AgrP (Cardinaletti, 1998), PosP (Schoorlemmer, 1998) - but they preserve the essence of Szabolcsi's account of possession.

While the idea of base-generating possessors at Spec(N+I)P seems appealing, the picture of structural representation of possessors is not complete yet. Recall that the main thrust behind Szabolcsi's proposal comes from the domain of raising and scope interpretation effects. den Dikken (1995, 1998, 1999, 2006) evaluates Szabolcsi's proposal by directly addressing the notion of raising. Den Dikken's fundamental claim is that the connection between raising and scope ambiguity is not obligatory, contradicting the arguments in Szabolcsi (1994). In other words, raising of an NP from a lower position to a higher position does not guarantee that it would have scopal properties at the lower as well as the higher positions. In order to illustrate this point, Den Dikken takes up the example of ditransitive verbs in English, which can surface in two different forms: the prepositional dative (21a) or the double object construction (21b). The prepositional dative construction (21a) is ambiguous between a narrow-scope and wide-scope reading for *some*. However, the corresponding double object construction in (21b) does not face such ambiguity: *some* is allowed to take wide scope only.

(21) a. *He sent every letter to some student.*

(i) *keves ember minden lepes-e-t tudtuk megfigyelni*

few men every step-3SG-ACC could watch

'I could watch few men's every step.'

Interpretation A: For few men y, for every x, if x is a step of y, we could watch x

Interpretation B: *For every x, for few men y, if x is a step of y, we could watch x

Szabolcsi's argument is that if we assume that the possessor 'few men's' was base generated at a position above SpecNP, the validity of A and invalidity of B are natural conclusions. However, under the assumption that 'few men's' was base generated at a much lower position and then moved up to SpecNP, both interpretations should have been valid, as at some point in the derivational history, the quantifier 'every' was above 'few men'. Szabolcsi argues that, since B is not a valid interpretation, it must mean than the possessor 'few men' was never below the quantifier at any point

b. *He sent some student every letter.*

The crucial point to note in the case of (21) is that (21b) does not result in ambiguous scope interpretation despite the fact that it was derived via movement[3]. This is taken to be an instance of raising movement that does not alter the scope interpretation properties of the moved element. With the help of these examples, Den Dikken builds the argument that raising does not automatically entail changes in the interpretive properties of the moved element. This line of thinking is particularly relevant considering the fact that Szabolcsi's rejection of a raising analysis for possessors was chiefly built on the absence of ambiguous scope effects. Thus, by disbanding the connection between raising and scope ambiguity, Den Dikken significantly reduces the strength of Szabolcsi's proposal. With that in place, he proposes that possessors are base-generated lower in the DP, and it is some kind of raising that brings them up to SpecFP - a functional projection higher up the nominal cline.

To elaborate, Den Dikken's claim is that Predicate Inversion is the raising movement that drives the indirect object 'some student' over the direct object 'every letter' in (21). The dative marker on the indirect object in (21b) is a null allomorph (with the other allomorph being the dative-marked variant)- Den Dikken's claim is that the null allomorph is licensed in (21b) precisely because the PP has inverted to a position closer to the verb (so that it can be incorporated into the verb). The movement of the entire PP is a prerequisite for allowing the null allomorph of dative to appear. The null allomorph cannot be licensed at the lower position (Complement of P). An added benefit of such a proposal is that Predicate Inversion allows us to derive the two instantiations of ditransitive verbs from a common underlying source.

Returning to the notion of possessors, Den Dikken proposes that Predicate Inversion as an operation should be extended to the syntax of possession as well. Possessive DPs are taken to be underlyingly small clause constructions where the possessor is the internal argument of the dative proposition and the possessum is the subject of the PP (22). A key assumption relevant to this approach is that the possessor is taken to be underlyingly DAT case-marked[4]. This idea follows

[3]Den Dikken alludes to Baker (1988) and adopts the stance that the double object construction is derived via raising the dative PP predicate of the ditransitive sentence

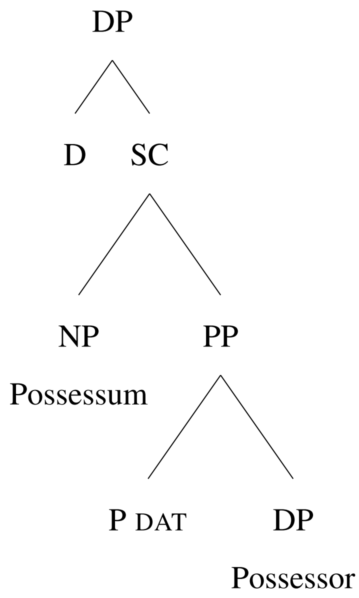
[4]The overt GEN marker seen on possessors in some languages is a mere genitive marker, and not an indicator of the case of the possessor. This is clear from the fact that it occurs on the entire possessor rather than on its head (the man standing over there's hat /*the man's standing over there hat)

from the fact that DAT is an indicator of the idea of possession, and that relations of possession are generally expressed via dative markers in many languages across the world (Herslund and Baron, 2008; Allen, 1964).

(22) [$_{SC}$ the car [$_{PP}$ P $_{DAT}$ John]]

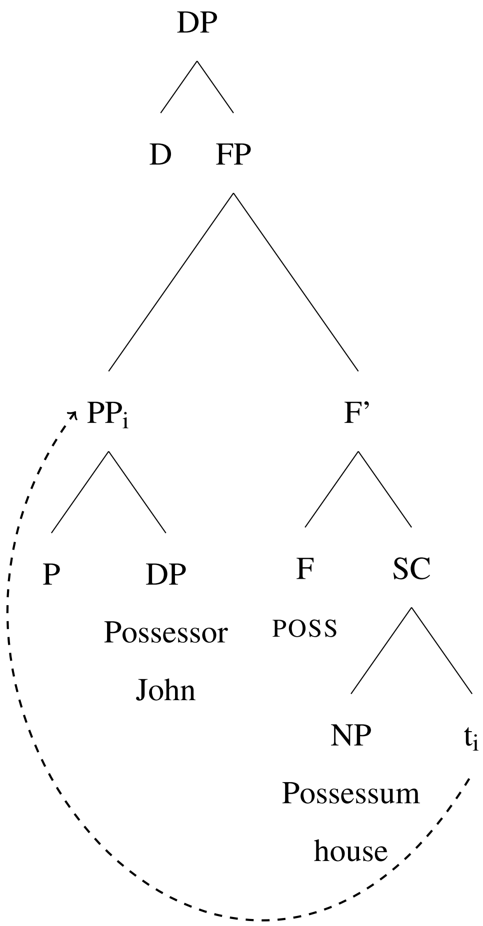
Den Dikken then proposes that the possessor is base-generated inside a dative PP (23). The dative PP is responsible for endowing the possessor with its thematic role of possession.

(23)



Next, the entire PP undergoes Predicate Inversion, and ultimately ends up to the left of the possessum (24). Movement of the possessor is explained as follows: Whenever the dative P is silent and not licensable (identical to the case of double object constructions), PP must raise so that the null P is maneuvered into the ambit of a licensing head. Morphologically, it is clear that the so-called genitival morpheme pops up only when the possessor is to the left of the possessum (as a result of PP-movement). Den Dikken therefore posits that the genitive morpheme ultimately appears on the possessor for three reasons: (i) as an acknowledgement of the possessor having moved into this position (ii) to license the null P, and (iii) to serve as a relator between two nominal elements - the possessor and the possessum. The movement of the possessor and its subsequent taking up of the genitive morpheme results in the observed linear order of [Possessor-GEN Possessum]. The structure is ultimately realized as (25).

(24)



(25) *john-oda viidu*
John-POSS house
'John's house'

To reiterate, the approach I adopt for the syntax of possession is the one proposed by Den Dikken. This proposal is more compelling than the other accounts surveyed so far in this chapter, as it makes use of existing theoretical machinery to explain the syntax of possession. By invoking the idea of Predicate Inversion to explain possessors, Den Dikken's account finds yet another organic parallelism between the nominal (possession) and the clausal (ditransitive verbs) domains. The account also manages to give satisfactory explanations of the source of case for the Possessor DP. Owing to these factors, I proceed to take up this account and venture into the domain of predicative possession.

4.3.2 Predicative Possession

In this section, we take up the case of one particular type of predicative possession, definite predicate possession, and present its structural representation. Broadly, there are two ways to express possession with the syntax of predication, with ‘have’ (26a) and with ‘be’ (26b).

Possession with ‘have’, or indefinite possession (26a), has received considerable attention in the literature (Chappell and Creissels, 2019; Stassen, 2009) and various analyses have been proposed for these, in accordance with the cross-linguistic variation found in them. I focus on (26b), which is relatively underexplored in the literature (Stassen, 2009; Zribi-Hertz, 1997). It has been termed as definite predicative possession or inverse predicative possession, but most accounts of possessive syntax do not elaborate on their structural representation. This section is concerned with possessive structures of type (26b), where the possessum is the grammatical subject, and is the topic of the sentence as well (Stassen, 2009).

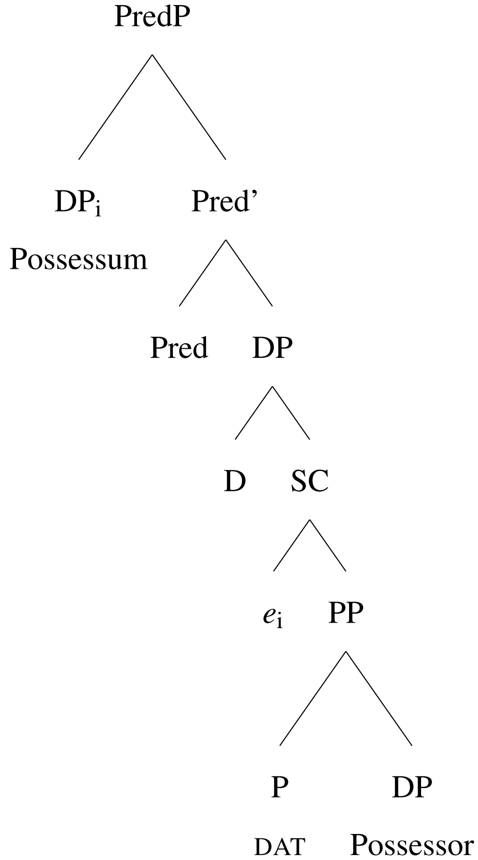
(26) a. *John has a house.*

b. *This house is John’s.*

Although existing literature does not expand on the syntax of (26b), there is consensus that it shares a common underlying representation with attributive possession (Stolz et al., 2008; Stassen, 2009). The sentence in (26b) has also been termed ‘predication of belonging’, where the possessum is the topic of the sentence. The possessor merely adds new, modificational information to the possessum (Carnesale, 2022). The primary relation between the two entities remains the same as in attributive possession. Therefore, we proceed with the assumption that (24) is the underlying structure for both, attributive as well as predicative possession.

In order to understand the structural organisation of predicative possession, we need to take a step back and revisit the fundamental relationship between a possessor and a possessum in a possessive construction. A possessor is a possessor only by virtue of the structural configuration within which it exists. In other words, a DP is interpreted as a possessor only when it appears inside a DAT PP predicate of another NP. Although Den Dikken does not make explicit mention of this notion, it is possible to infer this from his proposal. The structural representation of possessives posited in (24) expresses the idea that, of the two elements in the dyad, the possessum is the core one; possessors are structurally dependent on the presence of a possessum. A rudimentary inference that immediately emerges at this point is that, even when the possessum is present at the the subject of the sentence, (26b), there must be some element inside the predicative domain that marks the presence of the possessum and is bound by it. We can characterise this element as a placeholder for the possessum, and the entity that allows the possessor to surface. For the moment, we shall name this placeholder *e*, and posit it within the predicative domain. (27) illustrates that in case of predicative possession, the possessor is part of a DAT PP, the specifier of which is occupied by *e*. This is the structural position of the possessum in attributive possession.

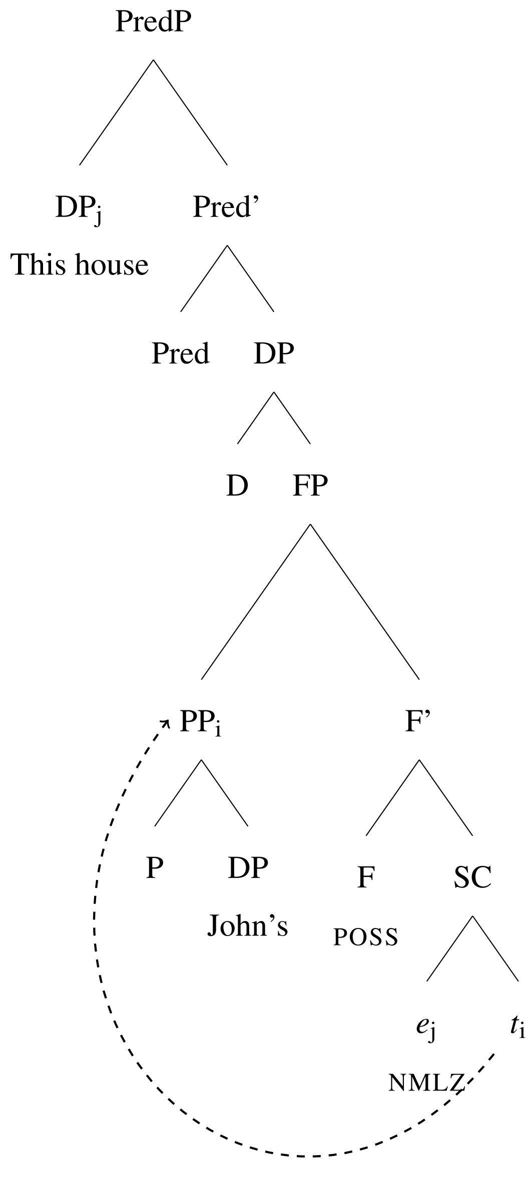
(27)



The conditions outlined above for attributive possession extend to predicative possession as well. To repeat, the presence of null DAT forces the possessor to raise to a higher position, where it can be licensed. The possessor raises to a functional projection (FP) above the small clause. Movement of the possessor also endows it with the GEN marker, resulting in (26b). The entire process is illustrated in (28) below, where the possessum is present as the subject of the predicate containing the possessor. The possessor, which was base-generated in a lower position, now appears with a possession marker.

In (28), the predicate contains the possessor (with possession marking) and a null possessum. As is the case with all null elements, they require formal licensing - an aspect that shall be addressed in the forthcoming paragraphs.

(28)



With the assumption that the there is a structural connection between the possessum and *e*, we shall now revisit the empirical facts of Tamil laid out earlier. In Section 4.2.1, we stated that predicative possessors contain an obligatory nominalizer. (3a) is repeated here as (29).

(29) *inda viidu john-oda-du*
this house John-POSS-NMLZ
'This house is John's.'

In connection to (29), we can now phrase the assumption in these terms: the null possessum in (28) is formally licensed by the nominalizer in Tamil[5]. The proposal that the nominalizer is the licenser of the null possessum in predicative possession is strengthened when we consider attributive and predicative possession in tandem. Attributive possession (30) does not contain any null elements that may require licensing, and therefore NMLZ is disallowed there (30b). Predicative possession (31) contains the null possessum, and obligatorily requires NMLZ (31a).

(30) a. *john-oda viidu*
John-POSS house
'John's house'

b. **john-oda-**du** viidu*
John-POSS-**NMLZ** house
Intended: 'John's house'

(31) a. **inda viidu john-oda-_*
this house John-POSS-_
Intended: 'This house is John's.'

b. *inda viidu john-oda-**du***
this house John-POSS-**NMLZ**
'This house is John's.'

While this helps us establish the nominalizer as the formal licenser in predicative possession, it also raises the question of why a nominalizer is present in such a derivation. In order to address this question, we take a step back to revisit the idea of verbless clauses in Tamil presented in Chapter 3. Section 3.3.4.1 noted that [NP1] [NP2] in a subject-predicate configuration can form a complete sentence with equative copular interpretation. Predicative possession (31) is an example of such a construction which expresses the relation between the subject 'this house' and the predicate 'John's' without invoking an overt copula. Such constructions are allowed in Tamil only when both, the subject and the predicate, are NPs. Thus, in order to realise [John-POSS] as an NP with an index, the predicate is supplied with a nominalizer. The resultant structure [John-POSS-**NMLZ**] now obtains a nominal flavour, thus rendering it a suitable candidate for an [NP1]-[NP2] configuration.

In order to fully understand the nominalizer as the licenser of the null possessum, we begin with the hypothesis that there must be a structural relation between the possessum and the

[5]Lehmann (1989) briefly notes that certain forms of possession, such as the ones in (29) occur with a nominalizer in Tamil

nominalizer. We also hypothesise that it is through this relation that the nominalizer obtains reference or co-indexation with the possessum, which ultimately enables it to license the possessor structurally. In other words, the nominalizer is merely a reflex of the relationship between the possessum and *e*.

In the following paragraphs, we shall explore ways to confirm the hypothesis laid out above. Hypothesising a connection between the possessum and the nominalizer makes the following prediction: there should be phi feature correlation between the nominalizer and the possessum. Turning our focus to Tamil, we shall now investigate if the prediction is borne out or not. It was established earlier that the nominalizer in Tamil *-du* is derived from III person non-gendered pronouns. Thus, our prediction can be restated in more precise terms: The nominalizer does not have gender features. Therefore, it can only establish connections with those possessum DPs that do not have a gender value themselves. Conversely, the nominalizer will never occur with a gendered DP as the possessum.

Before testing this prediction out, we need to clarify certain facts about gender features in Tamil. As explained in Chapter 2, Tamil has a gender system where the only kind of gender available is biological gender; there are no other types of gender. Thus, when we predict that the nominalizer will not be compatible with gender features, what it means in the context of Tamil is the nominalizer is not compatible with biological gender. The inability of nominalisation, as a derivational process, to access biological gender is explored in Chapter 2. Section 2.4 illustrates that the only way for nouns to obtain biological gender was through inherent lexical specification. Derived (derivationally formed) nominals can never carry biological gender features, as the nominalizing element does not have access to it. Thus, all nominalised elements in Tamil appear without gender features. This fact becomes immediately evident when we look at (32) (repeated from Chapter 2). We can use verbal agreement morphemes as empirical evidence for the presence/ absence of gender features in the subject. (32a) shows that ‘woman’ carries FEM features. However, the same noun with the addition of a nominalizing affix to form a derived nominal ‘woman-hood’, it loses its gender features, as shown in (32b).

(32) a. *anda peN kannadi-la teri-nj-aa*
that woman.FSG mirror-LOC visible-PST-**FSG**
‘That woman was visible in the mirror.’

b. *avaL-oda peN-mai kannadi-la teri-nj-udu/*aa*
she-POSS woman-hood mirror-LOC visible-**PST-NEUT**(non-gendered) / ***FSG**
'Her woman-hood (femininity) was visible in the mirror.'

These examples further bolster the idea that nominalised elements (and the nominalizer *n*) are not compatible with gender features in Tamil. Recall that, as the formal licenser of the silent possessum in predicative possession constructions, the nominalizer is obligatory in these. Additionally, the nominalizer is incompatible with gender features. Consequently, it is unable to license any possessum that is [+GENDER].

With these observations in place, we can now return to our main prediction: In predicative possession, the nominalizer *-du* can only occur with non-gendered NPs. When we look at the empirical data in Tamil, we can immediately see that the prediction is borne out. (33) contains instances of nouns which are allowed to occur as possessa of predicative possession. These are all devoid of gender values in Tamil.

(33) Non-Gendered Nouns: Nominaliser is allowed

a. *inda puttagam avan-oda-du*
this book he-POSS-NMLZ
'This book is his.'

b. *inda naai avan-oda-du*
this dog he-POSS-NMLZ
'This dog is his.'

c. *inda koLandai avan-oda-du*
this child he-POSS-NMLZ
'This child is his.'

Conversely, (34) shows that nouns endowed with biological gender, such as 'boy [MASC]', 'girl [FEM]', and 'grandmother [FEM]', in Tamil are not allowed in this structural configuration.

(34) Gendered Nouns: Nominalizer is disallowed

a. **inda paiyan avan-oda-**du***
this boy he-POSS-NMLZ
'This boy is his.'

b. **inda poNNu avan-oda-**du***
this girl he-POSS-NMLZ
'This girl is his.'

c. **inda paaTi avan-oda-**du***
this grandmother he-POSS-NMLZ
'This grandmother is his.'

From (33) and (34), it is safe to conclude that only gendered nouns are excluded from this particular structural configuration. With our prediction being met, we can confirm the hypothesis that the possessum of predicative possession must be structurally connected to a placeholder, which acts as its substitute and licenses the possessor in the absence of an overt possessum. The following subsection elaborates on the structural constraints that could yield such a gender-based distinction in Tamil.

4.3.2.1 Biological Gender, *n*, and Predicative Possession

To recapitulate, two facts clearly emerge from the insights gained so far:

- In predicative possession, the possessor contains a null (or overt) element that stands in for the possessum.
- Biological gender in Tamil presents a derivational constraint: presence or absence of the feature determines whether or not a noun can occur in a structural position.

These two points can be explained structurally once we revisit the gender system of Tamil in light of possession data. Tamil has solely biological gender features. Only [+HUMAN] nouns are assigned gender features in the lexicon - structurally represented as a feature on N, shown in (35a) and (35b).

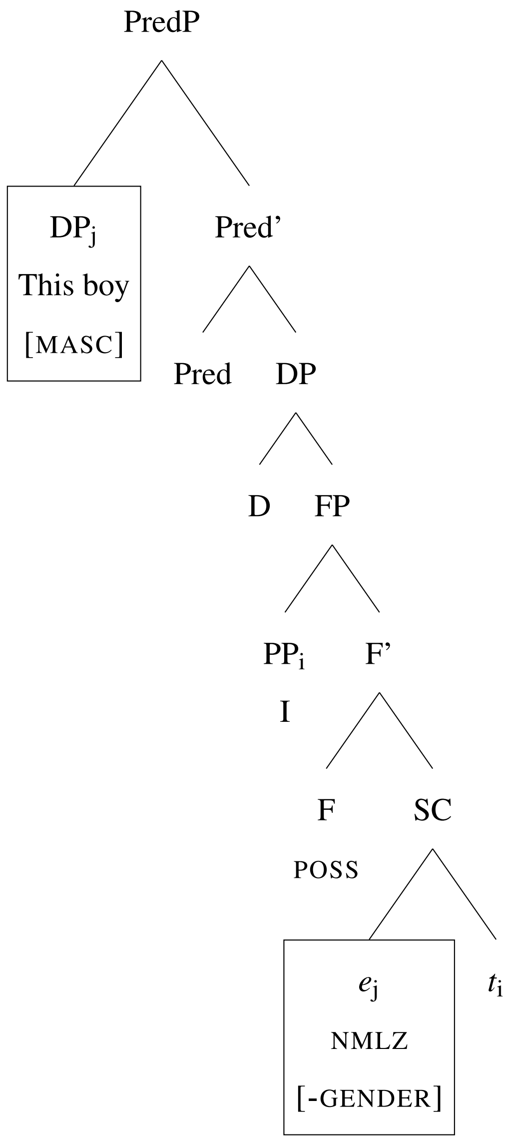
(35) a. [*n*P [*n* [NP [N boy.MASC]]]]
b. [*n*P [*n* [NP [N girl.FEM]]]]

The nominal domain also contains a functional head *n* – nodal for two phenomena: (i) Grammatical gender (ii) Nominalization (Kramer, 2015). Since Tamil lacks grammatical gender altogether, the only function of *n* in Tamil is nominalization. It has been established that in languages where *n* carries grammatical gender features, nominalized structures too are gendered (Kramer, 2015). Conversely, I propose that when *n* is devoid of gender features, as in Tamil, nominalized entities cannot be gendered, as the process of nominalization cannot access gender features at all. Essentially, *n* in Tamil can nominalize but it does not have access to gender features.

Combining this with the fact that the nominalizer morpheme *-du* is obligatorily present in predicative possession, we now have a way to explain why predicative possession is incompatible with [+GENDER] nouns: *-du* is licensed via *n*, a head that can only represent non-gendered nouns. Thus (36a) is not a grammatical derivation in Tamil. In (36b), the nominalizer cannot license *e*, as *e* has gender features, and the *-du*, *n*, lacks gender features. In other words, *n* cannot license *e*, as their specification of [-GENDER] and [+GENDER], respectively, are incompatible with each other.

(36) a. **inda paiyan enn-oda-du*
this boy I-POSS-NMLZ
'This boy is mine.'

b.



To conclude, the syntax of predicative possession in Tamil presents an added layer of structural constraints - one with gender features at the midst of it. This restriction is part of the derivational structure. As a structural constraint it will form a key aspect of the investigation into heritage Tamil.

4.3.2.2 Semantic Constraint on Possession?

The previous section shed light on a sharp bifurcation present within predicative possession. To reiterate the point, nouns such as 'boy' are disallowed in predication (34), whereas nouns such as 'book' are allowed (33). The disparity was explained in structural terms by invoking the nominalizer in Tamil, and its inability to be co-indexed with gendered nouns. Even so, there exists an alternative analysis of the same facts; the difference between (33) and (34) could also be interpreted as a semantic constraint. Essentially, the [+HUMAN] encoding on the nouns in (34) could prevent them from occurring in contexts of possession. On the other hand, the [-HUMAN] nouns in (33) are objects, and thus are not curtailed by such semantic constraints.

The current subsection explores this possibility in detail, and provides empirical arguments against the notion of a semantic constraint operating in predicative possession. There are two major counterexamples to the claim; both of them reveal the inconsistencies in a semantics-based classification. These examples illustrate that [+HUMAN] features are neither necessary nor sufficient conditions to explain the restriction on predicative possession in Tamil.

The first example is the noun 'child' - a [+HUMAN] noun. Going by the purported semantic restriction, 'child' should be disallowed from occurring in predicative possession. However, that is not the case; (37) is a grammatical construction in Tamil.

(37) *inda kuLandai avan-oda-du*
this child 3MS-POSS-NMLZ
'This child is his.'

Conversely, there are certain [+HUMAN] nouns which are allowed in predicative possession in other languages. Nouns such as 'student', 'teacher', 'patient' can occur in contexts of predicative possession. All the sentences in (38) are licit constructions, as they convey the semantics of relatedness by making use of possession morphology (Partee and Borschev, 2000).

(38) a. *That [pointing] student is mine.*

b. *That [pointing] teacher is hers.*

c. *This [pointing] patient is John's.*

However, such cases are not found in Tamil. While Tamil does allow the usage of possession morphology to convey relatedness in the attributive position (39a), it is not allowed in the predicative domain (39b).

(39) a. *avar john-oda student/ teacher/ patient*
he John-POSS student/ teacher/ patient
'He is John's student/ teacher/ patient.'

b. **anda student/ teacher/ patient john-oda-du*
that student/ teacher/ patient John-POSS-NMLZ
'That student/ teacher/ patient is John's.'

Recall that Chapter 2 of this dissertation takes up the case of common-gendered nouns such as 'student', 'teacher', etc. in Tamil. These nouns are not lexically specified with a gender value. But they are endowed with Biological Gender features by virtue of Edge-Computation and discourse elements. Common-gendered nouns such as 'student' are deictic in nature; they (typically) refer to an individual referent in the discourse. When these nouns are used in the derivation, the functional head D interfaces with the discourse layer in order to locate a referent and obtain the biological gender value of the referent. When the referent is male, the noun adopts a [MASC] affix (40a) and when female, a [FEM] affix (40b).

(40) a. *maana-van*
student-MSG
'male student'

b. *maana-vii*
student-FSG
'female student'

Ultimately, 'student' and 'teacher' are gendered nouns in Tamil. The difference between (39a) and (39b) strongly suggests that [+GENDER] nouns are fundamentally incompatible with predicative possession morphology in Tamil. A semantic distinction based on [+/-HUMAN] will not be able to capture the empirical facts laid out here. Based on this, we conclude that gendered nouns (not all +HUMAN nouns) are disallowed from predicative possession owing to structural constraints relating to the nominalizer.

4.3.3 Interim Summary

The current section provided an overview of possessives in Tamil and Hindi-Urdu. Both the languages discussed here have attributive as well as predicative possession. A major difference between the two systems is expressed in terms of gender agreement. Hindi-Urdu possessive constructions display phi agreement between the possessum and the possessor. This is executed in the attributive and the predicative domains. Tamil, on the other hand, does not have gender agreement in the domain of possession. Apart from this difference, there are some additional structural constraints that distinguish Tamil possessives system from that of Hindi-Urdu. Despite not having gender agreement, predicative possession in Tamil employs a gender-based distinction as the determinant of whether a noun is allowed in the predicative position or not.

This is another instance of DP vs clause level syntax behaving differently with respect to possessors. In the DP level, all nouns are treated the same. The [+GENDER] feature of a noun is not paid heed to. At the clausal level, the feature matters. Any noun with a [+GENDER] specification is not allowed to occur. The gender-based constraint in predicative possession is explained via *n*. With that in place, we now turn our attention to how possession develops in heritage Tamil in New Delhi.

4.4 Current Proposal: Implications for Language Contact

This section is divided into two parts. The first part covers existing literature on the syntax of possession in heritage grammars cross-linguistically. Based on this, we formulate and discuss the possible ways in which heritage Tamil in contact with Hindi-Urdu could be expected to fare in the domain of possession. The second part of this section then presents the study conducted. It lays out the methods, predictions, and discusses the results of the study.

4.4.1 Existing Accounts of Possession in Heritage Grammars

Existing work on the syntax of possession in heritage grammars primarily focuses on two major aspects of possession: the morphology employed to indicate possession, and the relative word order of the internal constituents of possessive structures. This section presents a brief outline of four studies that focus on the possessives in heritage languages.

Cuza and Solano-Escobar (2025) and Solano-Escobar and Cuza (2024) present a large-scale study on the morpho-syntax of possession in heritage Spanish spoken in the US. At the outset they shed light on a key difference between Spanish (the L1) and English (the L2) in terms of how possession is expressed in the two languages. English makes use of possessive determiners such as *his, her, my* etc. Spanish, on the other hand, uses definite determiners to mark possession. The sentences in (41) indicate the use of the definite determiner *el* to mark possession. There is, however, one key difference between (41a) and (41b) - in terms of alienable vs inalienable possession. (41a) expresses alienable possession of the object 'mirror', and is marked using the definite determiner. (41b) contains the inalienable possession of a body part - one's arm - and this is marked using the definite determiner along with a verbal clitic *se*. Both these elements are obligatorily marked in Spanish possessive DPs.

(41) a. *Alex rompió el espejo*
Alex broke the mirror.

b. *Juan se rompió el brazo*
John CLITIC broke his arm.
(lit: John broke the arm)

Essentially, Spanish differs from English in two major ways - by using the definite determiners instead of possessive determiners, and by differentiating between alienable and inalienable possession. Based on these differences, Cuza and Solano-Escobar aimed to investigate if cross-linguistic influence from English would make heritage Spanish assimilate to the system found in the L2, English. Over a series of Elicited Production and Acceptability Judgment tasks, they found evidence for cross-linguistic influence on heritage Spanish. They report that heritage speakers of Spanish - both, adults and children - extend the usage of possessive determiners to mark possession in Spanish (42).

(42) *#se rompió su brazo*
He broke his arm.

The results of this study show evidence for considerable divergence in the structures produced by the heritage speakers. The study recorded several instances of the possessive determiner being used in place of the definite determiner. Alongside this variation, it was also observed that the differences between alienable and inalienable possession were not maintained strictly in heritage Spanish. While this particular study does not delve into the derivational constraints underlying the morphology of possession in Spanish, it offers insights into one aspect of possession that can be unexpected to undergo change in a heritage bilingual context.

Another aspect of possession that has been explored is the relative word order of the two internal constituents of possessive DPs - the possessor and the possessum. Cardullo and Costea (2024) and Anderssen et al. (2018) provide two case studies that shed light on this particular phenomenon. Cardullo and Costea take up the case of two Romance languages, Megleno-Romanian and Eolian as heritage languages in contact with Macedonian and Sicilian, respectively. Both pairs of languages examined in the study exhibit a gradual shift from post-nominal possession (N POSS) to prenominal possession (POSS N), as reflected by the word order found in Modern Megleno-Romanian and Modern Eolian. The two possible orders are a function of N movement within the DP; high movement of N results in the N POSS order, whereas low movement of N leads to the POSS N order, where N has not risen above DET.

The conclusion of the study is that both, Megleno-Romanian in contact with Macedonian, and Eolian in contact with Sicilian, show a change in the word order of possessives. In other words, cross-linguistic influence from their respective L2 has led to a change in the position of the possessor with respect to DP-internal movement.

Anderssen et al.'s study presents another investigation into how the relative word order of the possessor and possessum could fare in a heritage contact situation. The empirical ground of this study is formed by heritage Norwegian in the US. Possession in baseline Norwegian is expressed in two ways: prenominally (43a) as well as postnominally (43b). Both the orders are attested in the language, with the postnominal occurrence of the possessor being the more frequent one of the two. The construction with the postnominal possessor includes an additional meaning of definiteness.

(43) Norwegian (Anderssen et al., 2018)

a. *min bil*
POSS car
'my car'

b. *bilen min*
car.DEF POSS
'my car'

In contrast, the L2 (English) only has prenominal possession. Anderssen et al.'s study found that the heritage speakers in their study could be effectively grouped into two: Group A with a higher preference for the POSS-N order, and Group B with a higher preference for the N-POSS order. These results are interpreted to mean that Group A is more accepting of cross-linguistic influence from the L2, whereas Group B is more resistant to change. Overall, they acknowledge the variability that exists in the domain of possession.

A common theme found across all the studies recounted here is that they investigate those domains of possession that present differences in the two grammars in a contact situation, and find evidence for convergence. In other words, whenever the L1 and L2 differ in terms of a parameter (morphology, word order, etc.), the L1 is likely to assimilate to the system found in the L2. For the purposes of this discussion, a key takeaway is that the domain of possession is certainly susceptible to change. With this conclusion obtained from a survey of the existing literature, we shall now form the research questions for the current study.

4.4.2 The Current Study

This section discusses the current study in light of the inferences drawn from existing accounts of possession in heritage grammars, and the possessive systems of Tamil and Hindi-Urdu introduced previously. Essentially, what we have is a configuration where the L1 and L2 crucially differ in two aspects: gender agreement (absent in L1 and present in L2) and a gender-based distinction in predicative possession (present in L1 and absent in L2). Alongside this set up, we can infer from other studies that in domains where the L1 differs from the L2 we can expect cross-linguistic influence from the L2 and subsequent convergence.

Thus, the objective of the current study is to investigate how possession in heritage Tamil is likely to fare. The possibilities are for it to either remain stable or assimilate to the system in Hindi-Urdu. The first-order objective of the the study is to verify whether heritage Tamil retains attributive and predicative instantiations of possession. Upon establishing that, the next line of inquiry will be into whether it preserves the gender-based constraint in predicative possession, or converges to Hindi-Urdu by obliterating such a distinction. Another aspect to be examined is whether heritage Tamil adds agreement in possessives, showing yet another instance of departure from baseline Tamil.

4.4.2.1 Questionnaire and Methodology

A Picture-based Elicitation Task was conducted in order to test the possessive structures in heritage Tamil. Participants (N=14) were shown picture-based prompts and asked to construct sentences on their own in Tamil.

Data for baseline Tamil was collected from 12 native speakers of Tamil (age range=20-38). They were exposed to the same questionnaire, and asked to produce sentences containing possessives. Baseline Tamil data shows a uniform adherence to the constraints laid out in Section 4.3.2.1. Attributive possession features both gendered as well as non-gendered nouns, while predicative possession only features non-gendered nouns. There were no exceptions to this.

The prompts used to elicit attributive possession are displayed in Fig. 4.1. The screen shown to the participants was divided into two parts. The left side of the screen consisted of a group of named people, any of whom could be selected as the possessor of the sentence. The right side of the screen consisted of a set of images (shown one by one) of potential possessum of the construction. For each item on the right side, the participants were asked to decide who among the possessors the object belonged to, or is related to. Fig. 4.1a is an example of non-gendered items, such as bag, as the possessum, and Fig. 4.1b is an example of a gendered noun, such as girl [FEM] as the possessum[6].

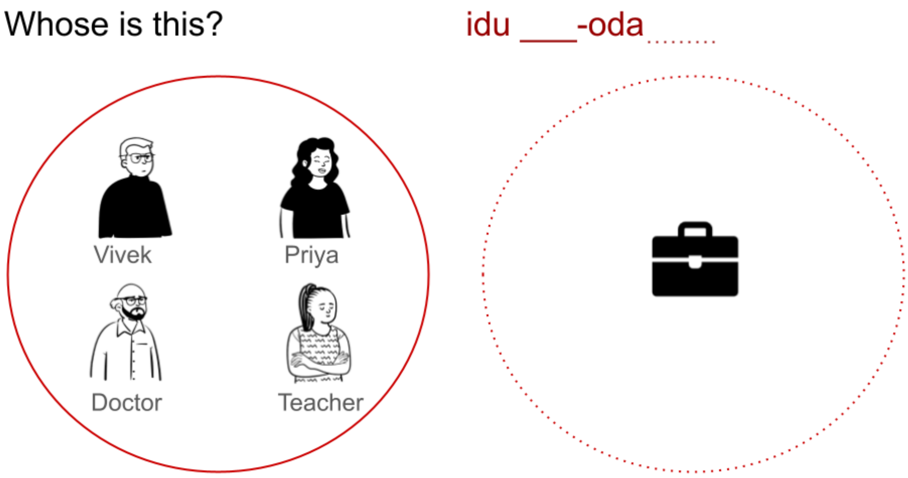
The participants were provided with a basic set of instructions on how to build the sentences for the task. They were asked to choose any one entity from the solid circle to fill the solid blank, and use the item in the dotted circle to fill the dotted blank, in order to complete the sentence. Thus, the expected responses for prompts such as Fig. 4.1a are (44), and those for Fig. 4.1b are (45). (44) and (45) contain an expression of attributive possession 'doctor's bag', or 'teacher's daughter' in the nominal predicate.

(44) a. *idu doctor-oda pai*
this doctor-POSS bag
'This is the doctor's bag.'

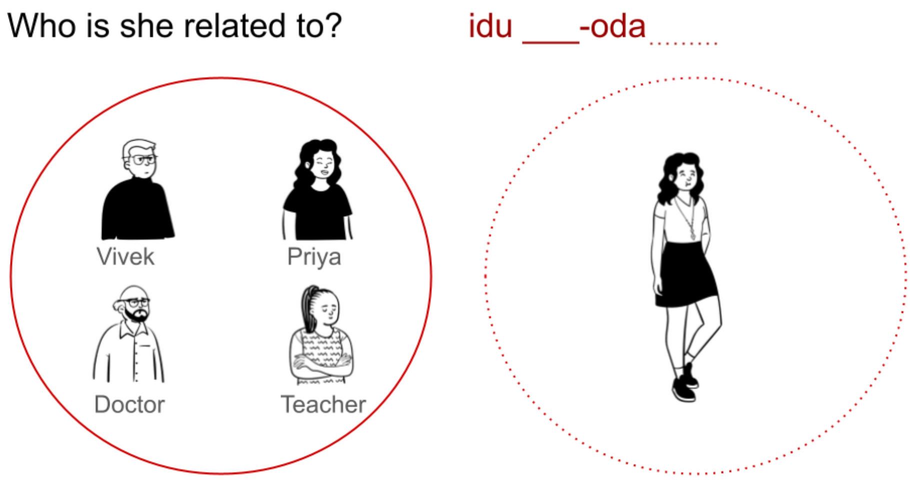
b. *idu vivek-oda pai*
this Vivek-POSS bag
'This is Vivek's bag.'

(45)

[6]The decision to keep the POSS morpheme as part of the displayed prompt followed from the fact that the POSS morpheme in Tamil *-oda* does not participate in any agreement relations, either with the possessor or the possessum. As such, it could not possibly lead to any priming effects on the task.



(a) Attributive Possession - Non-gendered Nouns



(b) Attributive Possession - Gendered Nouns

Figure 4.1: Prompts for Attributive Possession

a. *idu teacher-oda poNNu*
this teacher-POSS girl
'This is the teacher's daughter.'

b. *idu priya-oda akka*
this Priya-POSS sister
'This is Priya's older sister.'

As explained in Section 4.3.1, attributive possession in Tamil does not differentiate between gendered and non-gendered nouns, Both types of nouns occur in this position, and the prompts were designed to elicit both.

The second part of the study dealt with predicative possession. The prompts for the elicitation of predicative possession are shown in Fig. 4.2. Section 4.3.2 underscored the relevance of the gender feature of nouns in predicative possession in Tamil. The questionnaire was designed mindful of these nuances. Fig. 4.2a is an example of an inanimate non-gendered noun, such as 'house'. Fig. 4.2b is an example of an animate, non-human and non-gendered noun, 'dog'. Fig. 4.2c shows examples of animate, human but non-gendered nouns, 'child'. Finally, Fig. 4.3a is an example of a gendered noun, 'girl' [FEM].

Similar to the previous task, here too, the participants were provided with some basic instructions. They were asked to form sentences for each picture in the solid box, and complete the sentence by choosing one of the four entities below as the possessor/ a related entity. As already explained in Section 4.3.2, there is a key difference between the nouns in Fig. 4.2 and Fig. 4.3. The nouns in Fig. 4.2 are non-gendered, and are therefore allowed in predicative possession. This is illustrated in the examples below with (46a), (46b) and (46c) corresponding to Fig. 4.2a, 4.2b and 4.2c, respectively.

(46) a. *inda viidu teacher-oda-du*
this house teacher-POSS-NMLZ
'This house is the teacher's.'

b. *inda naai sheela-oda-du*
this dog Sheela-POSS-NMLZ
'This dog is Sheela's.'

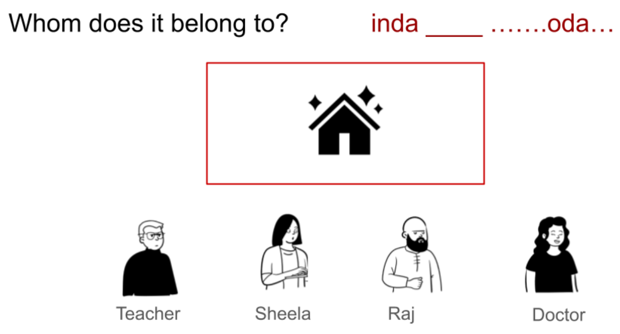
c. *inda kuLandai raj-oda-du*
this child Raj-POSS-NMLZ
'This child is Raj's.'

On the other hand, gendered nouns are not allowed in this configuration. The prompt in Fig. 4.3 cannot be constructed as a sentence the same way (47).

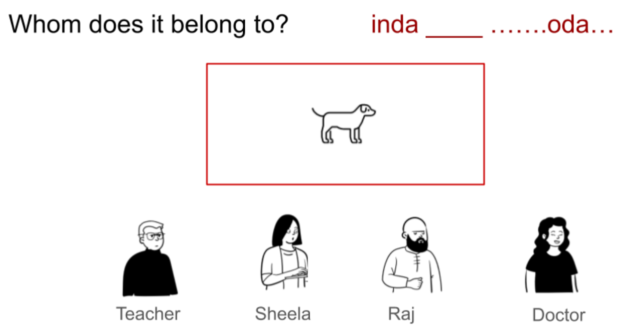
(47) **inda poNNu sheela-oda-du*
this girl Sheela-POSS-NMLZ
'This girl is Sheela's.'

Since there is no possibility of realizing [+HUMAN] nouns in a predicatively possessed configuration, Tamil resorts to attributive possession to convey this concept (48), with either deictic demonstratives or gendered pronouns.

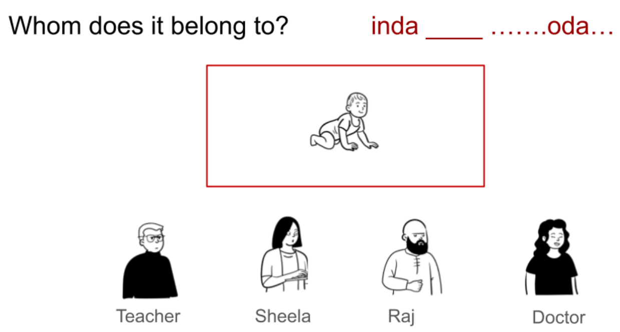
(48)



(a) Predicative Possession - Non-gendered Nouns [-ANIMATE]



(b) Predicative Possession - Non-Gendered Nouns [+ANIMATE -HUMAN]

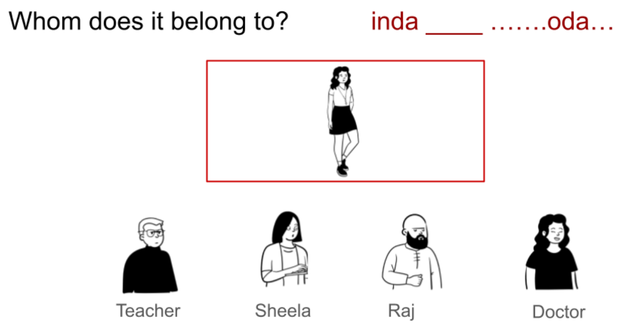


(c) Predicative Possession - Non-gendered Nouns [+HUMAN]

Figure 4.2: Prompts for Predicative Possession: Non-Gendered Nouns

a. *idu raj-oda poNNu*
this Raj-POSS girl
'This is Raj's daughter.'

b. *iva raj-oda poNNu*
she Raj-POSS girl
'She is Raj's daughter.'



(a) Predicative Possession - Gendered Nouns

Figure 4.3: Prompts for Predicative Possession: Gendered Nouns

The prompts in Fig. 4.2 were interspersed with those in Fig. 4.3 in order to test if the participants maintained or obliterated the crucial difference between gendered and non-gendered nouns, or if they retained them. The responses produced by the participants were coded as either 'Target-Consistent' or 'Target-Non-Consistent'. All instances of non-gendered nouns occurring in predicative possession were coded as target-consistent, whereas the presence of gendered nouns in the same configuration rendered the responses target non-consistent.

The picture-based elicitation method with some prompting formed the basis for the current inquiry into possessives in heritage Tamil.

4.4.2.2 Hypotheses and Research Questions

The broad goal of the current study is to investigate how the syntax of possession develops in heritage Tamil, given that the conditions of its acquisition are marked by two potentially unfavorable factors: Quantitatively and/or qualitatively restricted Primary Linguistic Data in Tamil in the formative years, and possible cross-linguistic effects from the more dominant language in the immediate environment, Hindi-Urdu. This objective can be refined further by narrowing our focus onto those aspects of possession where Tamil differs from Hindi-Urdu - empirical patterns observed in the former to the exclusion of the latter - as these are the domains for which primary linguistic data is expected to be particularly scarce. From this perspective, I identify two aspects of possession in Tamil:

(i) **Nominalizer and the gender-based constraint:** In Tamil predicative possession is marked by a nominalizer morpheme, and that results in a gender-based constraint. Both, the morpheme and the resulting constraint, are absent in the attributive domain.
(ii) **The absence of agreement:** Neither attributive nor predicative possession exhibits any form of agreement between possessor and possessum.

The question of the gender-based constraint in (i) is primarily concerned with whether or not the constraint will be present in heritage Tamil. There are two major aspects to it: the biological gender values of the particular lexical items, and the structural underpinnings that prevent them from occurring in predicative possession. We have evidence coming from the studies in Chapters 2 and 3 which are suggestive of biological gender features being retained in heritage Tamil. As for the structural constraints, there are claims in the literature (Unsworth, 2013; Unsworth et al., 2014) as well as novel empirical evidence from Chapters 2 and 3 of this dissertation which posit that grammatical phenomena that do not involve much learning—that are rather part of core syntax—tend to be resilient in heritage grammars. Here too, the position is that the structural constraint underlying the gender-based restrictions do not need to be learnt by the language acquirer; they are not dependent on a lot of PLD. Putting all this together, I predict that the gender-based restriction will persevere in heritage Tamil.

The agreement based difference outlined in (ii) can be seen with the lens of Cross-linguistic Influence (CLI) - an effect prevalent in heritage grammars. In this particular language dyad, Tamil does not have agreement in the domain of possessives (neither attributively nor predicatively). Hindi-Urdu provides a contrast by having gender agreement in both types of possession. Our question then, is whether heritage Tamil would add agreement to its possessives. Cross-linguistic influence often affects those areas of grammar where the L2 lacks a feature present in the L1. With L2 being more prevalent in the environment than L1, it follows that the phenomena gets affected due to cross-linguistic influence. However, in this case, the situation is reversed; agreement in present in the L2 but not in L1. Therefore, the prediction is that agreement is unlikely to come about in heritage Tamil.

Additionally, evidence from Chapter 3 fortifies this prediction further. In Chapter 3, it was laid out that attributive adjectives are another domain where Hindi-Urdu has agreement and Tamil does not. The attributive adjectives produced by heritage Tamil speakers did not feature agreement in a bid to assimilate to Hindi-Urdu. We consider these data as evidence for lack of cross linguistic influence in the domain of adjectives.

Based on all these factors, I predict that heritage Tamil will not assimilate to Hindi-Urdu by adding agreement relations in the domain of possession.

4.4.2.3 Results of the Study

The first objective of the study dealt with the structural representation of possessives in heritage Tamil. There are two parts to this objective. The first question is regarding the difference between attributive and predicative possession: is it maintained or obliterated in heritage Tamil? As explained in Section 4.4.2.2, pictorial prompts were designed to elicit both kinds of possession from the participants of the study. The responses collected include 190 tokens of attributive possession, and 151 tokens of predicative possession. The target-consistent responses for attributive possession are given in (49a)-(49c)where the possessor linearly precedes the possessum, and there are no nominalizers present in this environment. On the other hand, nominalizers are obligatory in the environment of predicative possession (50a), and the utterances without the NMLZ morpheme (50b) are coded as target non-consistent.

(49) a. *raj-oda puttagam*
raj-POSS book
'Raj's book'

b. *avan-oda aNNa*
he-POSS older brother
'his older brother'

c. *avaL-oda akka*
she-POSS older sister
'her older sister'

(50) a. *inda puttagam raj-oda-du*
this book Raj-POSS-NMLZ
'This book is Raj's.'

b. *inda puttagam raj-oda*
this book Raj-POSS
Intended: 'This book is Raj's.'

The responses collected show us that the distinction between attributive and predicative possession continues to be made in heritage Tamil. There were no instances of either attributive possession occurring with the nominalizer, or predicative possession occurring without it. This finding sets the stage for the next level of inquiry: Given that heritage speakers of Tamil can tell attributive possession apart from predicative possession, how do they treat the finer layers of complexity within predicative possession?

It was established that gendered nouns are disallowed in contexts of predicative possession, whereas non-gendered nouns do not face this restriction in Tamil. In order to test whether heritage Tamil too maintains this distinction, the prompts included an equal representation of gendered and non-gendered nouns. For non-gendered nouns, the target-consistent responses are of the kind in (51) - they appear in the predicative position with the nominalizer.

(51) *inda puttagam raj-oda-du*
this book Raj-POSS-NMLZ
'This book is Raj's.'

Gendered nouns require further explanation. Since gendered nouns are not allowed in the position taken by 'book' in (51) in Tamil, there are alternative ways of expressing the same content. One of the strategies adopted is the usage of deictic pronouns at the subject position with an attributively possessed noun as the predicate (52). The deictic pronoun in the subject position can be marked for gender features (52a), (52b), honorificity status (52c) or devoid of any features at all (52d), (52e).

(52) a. *ivan raj-oda paiyan*
he raj-POSS boy
'He is Raj's son.'

b. *iva raj-oda tangachi*
she raj-POSS younger sister
'She is Raj's younger sister.'

c. *ivanga anda doctor-oda patient*
he/she.HON that doctor-POSS patient
'He/she is that doctor's patient.'

d. *idu enn-oda poNNu*
this my-POSS girl
'This is my daughter.' (used only while introducing one's daughter to someone)

e. *idu enn-oda kanavar*
this my-POSS husband
'This is my husband.' (used only while introducing one's husband to someone)

Recall that, in the current task, speakers were prompted to begin their responses with ‘This boy...’ or ‘This girl....’, thus not allowing them to exercise the options listed in (52). Therefore, they found alternate ways to complete the sentences by producing structures such as (53). Essentially, the speakers retained ‘this girl/boy’ as the subject, but modified the predicate to convey a relation of kinship using the possessive marker.

(53) a. *inda poNNu raj-oda tangachi*
this girl Raj-POSS younger sister
‘This girl is Raj’s younger sister.’

b. *inda paiyan sheela-oda tambi*
this boy sheela-POSS younger brother
‘This boy is Sheels’s younger brother.’

The results of the elicitation task are presented in Table 4.1. The production of attributive possession was entirely target-consistent; there were no instances of divergence. Predicative possession, too, was at ceiling, as it has an accuracy of 98.67%. It is important to note that out of

Construction type	Target Consistent	Target Non-consistent
Attributive Possession	190	0
Predicative Possession	149	2

Table 4.1: Distribution of Attributive and Predicative Possession- Absolute Values

the 151 tokens of predicative possession collected in this study, only 2 were marked as target non-consistent. They featured gendered nouns with the nominalizer (54). All other occurrences avoided using the nominalizer with gendered nouns. For the purposes of this study I treat the tokens in (54) as outliers that are not truly reflective of the representations underlying them.

(54) a. *inda poNNu sheela-oda-du*
this girl sheela-POSS-NMLZ
‘This girl is Sheela’s.’

b. *inda paiyan teacher-oda-du*
this boy teacher-POSS-NMLZ
‘This boy is the teacher’s.’

Another aspect of possession tested in this study was whether cross-linguistic influence from the L2 would induce agreement to possession in heritage Tamil. This possibility can be ruled out entirely, as in the 339 tokens of possessives collected, agreement was not present in any of them. The possessive morpheme *-oda* remained invariant for phi features.

The key findings of the study are repeated below:

- Heritage Tamil maintains the morphological distinction between attributive and predicative possession by marking only the latter with the nominalizer.
- Heritage Tamil preserves the gender-based structural constraint found in predicative possession.
- There are no effects of cross-linguistic influence in the domain of agreement; heritage Tamil does not assimilate to Hindi-Urdu by adding agreement relations in possessive structures.

The findings and their implications are discussed in the following subsection.

4.4.2.4 Discussion

The current study and its findings open up three major strands of discussion around the syntax of possession in heritage Tamil. The main ideas discussed here are the differences between attributive and predicative possession, the gender-based constraint in predicative possession, and the absence of cross-linguistic influence in this domain.

As stated in previous sections, attributive and predicative possession are morphologically differentiated in Tamil. Structurally they share a common underlying representation, as explained in Section 4.3. To recap, both attributive and predicative possessions are represented within DAT PPs that undergo obligatory movement for licensing requirements. The morphological difference observed between them is a function of whether the possessum is realized overtly adjacent to the possessor or not. The particulars of the morphological differences between the two is taken up in the following paragraphs.

Predicative possession - to the exclusion of attributive possession - is marked by the presence of a nominalizer. The nominalizer *-du* is necessary in this environment for structural reasons. Predicative possession such as 'This book is mine' contains two parts: the subject, which is the possessum, and the predicate, which contains the possessor and an empty placeholder for the possessum. The null expression of the possessum requires structural licensing - as all null elements do. This dissertation argues that the licensing is executed by the functional head *n*, and often phonetically realized as the nominalizer. As such, the nominalizer is obligatory in predicative possession in Tamil. In the attributive position, the same morpheme is disallowed, as there are no licensing requirements in that domain, making the presence of the nominalizer superfluous.

The fact that heritage Tamil maintains the morphological distinction between attributive and predicative possession is indicative of the idea that the underlying structural representations of both remain unaltered despite the constrained conditions of acquisition under which heritage Tamil developed. The restricted nature of the PLD available to heritage Tamil speakers did not have an effect on the development of possession in this grammar. This finding is in line with the conclusions of the previous chapters. In Chapter 3, we found evidence for the hypothesis that core syntax is impervious to reduced PLD effects. Grammatical phenomena that are within the realm of regular, rule-governed syntax rarely have idiosyncrasies that require intensive learning on the part of the language acquirer. Thus, they tend to develop unhindered even in the face of reduced PLD. The findings of this study too corroborate the conclusions drawn in Chapter 3. The difference between attributive and predicative possession is a structural one, and is retained in heritage Tamil.

The second strand of discussion surrounds the role of gender features in the domain of possession. As noted in the previous sections, nouns with gender features are disallowed in predicative position - a restriction that does not apply to attributive possession. This disparity between the two can be explained as a structural constraint. The null possessum in the predicate of predicative possession necessitates a licensing functional head *n*. Ordinarily, *n* is associated with grammatical gender features. However, by virtue of having a solely biological gender system, Tamil has no derivational way of assigning gender, thus cutting off *n*'s access to gender features. Consequently, *n* is merely the nominalizer in Tamil. The [+GENDER] feature of certain nouns and the [-GENDER] nature of *n* renders them structurally incompatible with each other. An extended consequence of its inability to access gender features is that *n* cannot license the null placeholder of a gendered noun. Without such licensing, predicative possession is not be expressed. These structural constraints ultimately surface as a restriction where gendered nouns are not allowed to occur in predicative possessives.

Identical to the previous point, here too, we have a case of a core syntactic phenomenon. The difference between gendered and non-gendered nouns is not based on the idiosyncratic features of each noun. It emerges from the interaction between the gender features of a noun and functional heads such as *n* in the structure. In Tamil, the biological gender feature on certain nouns precludes them from appearing in predicative possession. Chapter 2 provides evidence for the stability of biological gender features in heritage Tamil. [+HUMAN] nouns which are specified with gender features in Tamil continue to reflect the same in heritage Tamil. Chapter 3 extends the discussion further by establishing the prevalence of biological gender features in predicative adjectives in heritage Tamil. Adjectival agreement was intact with no instances of mismatched agreement between [MASC] and [FEM] features, pointing towards their stability in the grammar. The findings of these two chapters allow us to conclusively establish that gender features are present in heritage Tamil.

This chapter takes a step forward, and illustrates biological gender features interacting with other parts of the structure in a systematic manner. A major contribution of this study is that it highlights one aspect of gender features - their role as derivational constraints. The restriction seen with predicative possession is one that is imposed by gender features. In other words, this is an instance where features drive computations; the (gender) features of a noun decide whether or not it can occur in certain structural configurations (predicative possession). This property of features has been defined as one of the core tenets of the Minimalist Program and its interest in syntax (Adger, 2003; Svenonius, 2017). To see evidence of such an arrangement between features and structure in a heritage grammar is added confirmation of the fact that core syntax develops undeterred even in contexts of heritage bilingualism, where the reduced nature of linguistic experience makes language acquisition more challenging than the norm.

The third part of this discussion revolves around the notion of cross-linguistic influence from the dominant language. The main difference between the possession systems of Tamil and Hindi-Urdu are in terms of agreement relations. The latter executes phi agreement between possessor and possessum in both configurations of possession, where Tamil does so in neither of them. The possessive morpheme in Tamil *-oda* does not inflect for any phi features of either the possessor or the possessum; it remains invariant. This study found that heritage Tamil did not deviate from the baseline in this regard. Cross-linguistic effects of transfer did not lead to the emergence of phi agreement in heritage Tamil possessives.

What is particularly relevant is that Tamil is a language which has agreement in the predicative domain. Verbal, and even adjectival predicates exhibit agreement with the subject of the sentence. It is only nominal predicates that do not participate in any agreement relations. The fact that predicative possession does not facilitate agreement lends further support to the proposal outlined in this chapter: the underlying representation of predicative possession contains a nominal function head, namely *n*. The nominal nature of this predicate is what prevents it from agreeing with the subject, the possessum. Continuous contact with Hindi-Urdu did not motivate heritage Tamil to modify the underlying structure of its predicative possession, and allow agreement. This is especially interesting, given that such a route is indeed possible[7]. Heritage Tamil could have extended the underlying structure of other predicates, adjectival for instance, to possessives in order to allow agreement and at least partly assimilate to the system in Hindi-Urdu. An example of this possible but unselected route is given in (55) – the agreement morphemes used in the domain of predicative adjectives (55a) could, in principle, be used for predicative possession (55b), but are not.

(55) a. *anda paiyan nall-a-**van***
that boy.MSG $\sqrt{good}$-REL-MSG
'That boy is good.'

b. **anda paiyan enn-oda-**van***
that boy.MSG my-DAT-MSG
'That boy is mine.'

Essentially, we find another instance in heritage Tamil that is not affected by cross-linguistic influence.

To summarize, we have uncovered three aspects of heritage Tamil that have not deviated from the baseline, despite restricted PLD and potential cross-linguistic influence from Hindi-Urdu. These findings fortify the the role of structure in grammatical phenomena, and how structural constraints tend to be resilient in heritage bilingual contexts.

[7]There are some recorded instance of such a construction in poetry, but it is ungrammatical in the literary as well as vernacular standard Tamil. (55b) is considered to be acceptable in a few varieties of Tamil spoken outside of India, but it is typically dispreferred in standard Tamil.

4.5 Conclusion

This chapter dealt with the syntax of possession in baseline and heritage Tamil. The domain of possession is relevant for the on-going discussion on gender, as we find empirical evidence for gender-based constraints in predicative possession. At the outset, the chapter provided the structural representation of both, attributive as well as predicative possession, along with an explanation for the role of gender features in the latter. Subsequently, a study on the syntax of possession and gender was conducted in heritage Tamil. The results of the study confirm that structural constraints prevail in a grammar despite the lack of sufficient primary linguistic data and potential cross-linguistic influence at the time of acquisition.

Chapter 5

Conclusion

5.1 Summary of the Dissertation

This dissertation opened with two noteworthy points. Firstly, we presented the idea that core syntax is an exceptionally resilient grammatical phenomenon in heritage language syntax, which is otherwise characterized by constant grammatical change and innovation. Alongside this observation, we also registered the caveat that the exact bounds of what constitutes core syntax is not yet fully uncovered. Operations such as Merge, Move and Agree lie at its heart (Lohndal, 2021), but a search for its other constituents is an on-going exercise in the field of heritage language syntax. The second major point in consideration is the apparent vulnerability of gender features in heritage grammars. Irrespective of whether the two languages in the dyad contained similar systems of gender, it was reported that the category of gender would invariably undergo change in the heritage grammar (Polinsky, 2018). This manifests as instability in gender values ([MASC]/[FEM]/[NEUT]), overgeneralization of one over all the others, etc.

By juxtaposing the stability of core syntax and the variability of gender features in heritage grammars, this dissertation raised questions about the outcome when we combine the two. Essentially, the central research question of this dissertation is: are there aspects of gender that are part of core syntax, and if so, do they exhibit stability or change in heritage Tamil? This question was approached from three different dimensions of gender: (i) gender features assigned to nouns, (ii) gender features participating in agreement relations with predicative adjectives, and (iii) gender features interacting with structural configurations such as possessive DPs, as explored in Chapters 2,3 and 4 respectively. Below I present a distilled version of each study focusing mainly on the results.

Chapter 2 explored gender assignment in heritage Tamil. Having laid out that Tamil only has Biological/ semantic gender, the questions were about which aspects of it undergo change and which remain stable in heritage Tamil. The findings of the study report that inherently specified biological gender was stable in heritage Tamil, while discourse-dependent biological gender showed some variation. This disparity was then explained as a function of the interface hypothesis, according to which grammatical phenomena lying at the interface of syntax and discourse tend to show variability in heritage grammars. In terms of the structural representations, Tamil has a gender system where only [+HUMAN] nouns are endowed with a gender value. Heritage Tamil retained this configuration; it maintained the connection [HUMAN] and [GENDER] features. The L2 in this setting is Hindi-Urdu - a language with semantic as well as grammatical gender. In other words, the presence or absence of the [HUMAN] feature has no implications for gender assignment in Hindi-Urdu. What we can then say about heritage Tamil is that it does not obliterate the [HUMAN]-[GENDER] connection in an effort to assimilate to Hindi-Urdu. By resisting the effects of transfer, heritage Tamil maintains the structural representation of its gender assignment system.

Chapter 3 took the discussion forward by examining gender as a feature in agreement relations. This takes us one step forward into the discussion on the relationship between gender and core syntax; agreement, executed by the operation Agree, is definitively considered to be a part of core syntax. Hence, we are poised to ask the question: how do the (semantic) gender features in heritage Tamil fare in agreement relations? Tamil has agreement only predicates - both adjectival and verbal. We chose the former for the current chapter, as it allows to explore gender in connection with yet another aspect of structural complexity. Adjectives in Tamil are configurational in nature. There are two derivational strategies of adjective formation, and it is the task of the language acquirer to learn the unique applications of each strategy. One consequence of the structural representation of adjectives in Tamil is an agreement-based difference. Predicative adjectives contain the functional head PRED that acts as a probe for phi features, resulting in agreement with the subject. Attributive adjectives, on the other hand, lack probing functional heads, resulting in no agreement in that domain. This difference is an inherent part of the structure, of core syntax. The lack of sufficient PLD in heritage Tamil translates as heritage speakers showing a considerable overlap between the two strategies of adjective formation - a departure from the baseline. However, the difference between attributive and predicative adjectives is sustained in heritage Tamil. Essentially, the internal structural configuration of the adjectives remains unchanged despite reduced PLD. Hindi-Urdu, the L2, does not distinguish between attributive and predicative adjectives; both have phi agreement. However, this did not prompt heritage Tamil to restructure its attributive adjectives to incorporate agreement. Thus, both gender agreement as well as the core syntactic structure of adjectives are retained in heritage Tamil.

Chapter 4 examined how gender features interact with the structural configurations in which nouns occur. The empirical domain used for this purpose is possessives. Possessives are relevant to our discussion on gender in heritage Tamil, as they come with an additional constraint in Tamil. Attributive and predicative possessors behave differently because of gender features. Essentially, the structure of predicative possession contains a null element that requires licensing - a task performed by nominalizers here. As nominalizers are non-gendered in Tamil, they cannot license biological gender features. Without such licensing, predicative possession cannot be realized. Therefore, nouns with gender features are not allowed to occur in predicative possessives in Tamil. This can be analysed as an instance of features driving syntactic computations - a part of core syntax. Attributive possessives do not contain null elements or the consequent licensing requirements. Hence, all nouns are allowed in that configuration. Our study finds that heritage Tamil retains this gender-based constraint. This constraint is notably absent from the grammar of the L2 as well, making it an ideal candidate for erasure through cross-linguistic transfer. However, that does not take place. Heritage Tamil retains the structural configuration of possessives where the presence of gender features on a noun determines whether it can occur in a given structural configuration. There is no restructuring, either due to reduced PLD or cross-linguistic transfer.

To summarize, this dissertation has identified three different aspects of gender that remain stable in heritage Tamil despite potential cross-linguistic transfer due to dissimilarity with the L2. As such, we now have three components of gender features that could be enumerated among core syntactic properties of grammar, because of which they remain unchanged in a heritage grammar.

5.2 Key Takeaways

This section discusses the two major takeaways from this dissertation: (i) the absence of cross-linguistic transfer in all the domains covered so far, and (ii) what we can say about core syntax based on this study of gender in heritage Tamil.

5.2.1 Absence of Cross-Linguistic Transfer

The first point under discussion here is the role of cross-linguistic transfer in shaping the grammar of heritage Tamil, or the lack thereof. The three sub-studies conducted in this dissertation all point towards the L2 (Hindi-Urdu) not having a major role in determining the outcome of the grammar of heritage Tamil. In this section, we first review how transfer effects have been understood globally in the domain of heritage language syntax, and then narrow our focus to how it can be applied and understood in terms of this study.

Within the field of heritage language syntax, transfer from the dominant language(s) - also understood as Cross-linguistic Interference (CLI) - is considered to be one of the many factors that have a role to play in the ultimate development of heritage grammars.

However, it is also a highly divisive one. Polinsky and Scontras (2020) raise some pertinent questions about the prevalence of transfer effects as a determinant of heritage grammars. According to their vast review, it is not a robust factor, or a unifying explanation for observed grammatical change in heritage grammars. Muysken (2020) differs from this perspective. His account of grammatical change under language contact ascribes more prominence to transfer effects. However, he too registers a caveat here: Transfer effects are not robust in their role as determinants, as they are gradual and slow to manifest. Thus, it might not be immediately observable in just one generation of heritage bilingual speakers. According to Muysken, transfer effects are more likely to emerge when the linguistic context contains "intensive bilingual usage", i.e., when multiple generations of speakers have been actively bilingual for a considerable period of time. This setting is conducive for there to be transfer from one language to another. In terms of heritage Tamil, this could be possible factor behind the absence of transfer effects. The Tamil speaking community in New Delhi does not go beyond two generations in most cases, thus not creating the right context for abundant transfer[1]. It is plausible that with more time (and more intensive contact), grammatical features of Hindi-Urdu may start to bleed into heritage Tamil.

Further arguments against transfer effects as an essential determinant of heritage grammars can be found. Montrul (2023), in her review of heritage languages, notes that the societal language's contribution to the outcome of heritage grammars remains "poorly understood" within the field of heritage language syntax. Despite the vast empirical ground covered by these studies, the results are inconclusive about the exact ways in which L2 can impact L1. Similar views are echoed in Polinsky (2025a) who notes that transfer effects are not consistent. They emerge more prominently when the previous generations (parents and other caregivers of the heritage speakers considered) have low proficiency in the societal language, setting the context for transfer to take place. Further, Polinsky (2025b) also reminds us of another aspect of transfer effects: transfer need not always be from the societal language to the heritage language. It can also be bidirectional; it is, in theory, possible for features of the heritage language to lead to grammatical change in the grammar of the L2.

Yet another dimension to consider in this regard is structural salience and its connection to transfer effects. It is understood that structurally salient grammatical phenomena such as tense, determiners, and other parts of core syntax (Lohndal and van Baal, 2025) tend to be impervious to effects of cross-linguistic transfer (Živojinović et al., 2025). While this observation yields some clarity to the discussion on transfer, it is not a generally applicable property, as we shall examine in the forthcoming paragraphs.

With this context in place, we are now equipped to focus on the lack of transfer effects in heritage Tamil in contact with Hindi-Urdu. The literature survey conducted in this dissertation also reflects the ambivalence of transfer effects. In the domain of adjectives Camacho (2018); Sánchez and Camacho (2021) and Alexiadou et al. (2023) study the structural position, interpretation and agreement relating to adjectives, respectively. A sweeping conclusion that applies to all three is the lack of transfer effects from the dominant languages onto the respective heritage grammars. We can contrast this with the domain of possessives studied in Chapter 4. Cuza and Solano-Escobar (2025) and Solano-Escobar and Cuza (2024) both study the morphosyntax of

[1]While migration from Tamil Nadu has been happening since the 1960s, the participants of the current study report that their parents migrated in the 1990s, thus having only two generations of migrants.

possession in heritage Spanish. Cardullo and Costea (2024) and Anderssen et al. (2018) take up the internal word order of possessives. In both the sets of studies, they report significant grammatical change in the heritage grammars owing to cross-linguistic transfer from the respective dominant languages. In other words, they found evidence of the possessives in L1 converging to those in L2.

The disparity between the two grammatical domains is indicative of the lack of consistency observed in transfer effects across studies on HL syntax. The effects are more pronounced in certain language dyads, and certain grammatical phenomena. It is certainly not a uniform effect, and we know by now not to expect it to manifest everywhere.

Additionally, the current study possesses certain limitations by way of having a small sample and no longitudinally collected data. Thus, it is plausible that the context is not sufficient to detect the presence of transfer effects, if any. A more robust, empirically richer study will be better equipped to study cross-linguistic transfer between heritage Tamil and L2 Hindi-Urdu.

5.2.2 The Resilience of Core Syntax in Heritage Languages

We began this dissertation with the notion that core syntax is an exceptionally stable component in an otherwise volatile setting, such as heritage grammars. Alongside this observation, we also noted that gender is a particularly vulnerable category across different types of language contact settings. By placing these two ideas adjacent to each other, this dissertation aimed to investigate if there are aspects of core syntax present within the category of gender, and if they exhibited stability or change in the given heritage language contact situation.

As summarized in this chapter, this objective was addressed by unpacking the structural mechanisms through which gender is expressed in the grammar of Tamil. The main objective was then to understand how these fare in heritage Tamil in contact with Hindi-Urdu. Studying gender in Tamil was trifurcated into gender assignment on nouns, gender as a feature participating in predicative adjective agreement relations, and gender feature on nouns interacting with the structural configuration of possession.

In Chapter 2, we found that gender assignment strategies underwent differential treatment in heritage Tamil based on their structural representation. Inherently specified gender has a tight connection to the [HUMAN] feature in Tamil, a core part of the representation of nouns.

Discourse-based gender involves additional derivational steps, such as locating a referent in the discourse, edge computation, etc. The syntax-discourse interface has been recognized as problematic for bilingual, especially through heritage, grammars.

Chapter 3 placed gender features in the midst of another aspect of core syntax - agreement. Agreement, executed by the operation Agree, is considered to be a part of core syntax. While studies on heritage language syntax have reported that agreement shows variation in heritage grammars, the observed variation is only in terms of gender features, and not the underlying operations themselves. Chapter 3 takes biological gender features with an added layer of structural complexity - Tamil has agreement with attributive but not predicative adjectives. This disparity is a result of the respective structural configurations of attributive and predicative adjectives. The former lacks functional heads that could probe and result in agreement. The latter has PRED for this purpose. What we found in heritage Tamil was evidence for biological gender values and the distinction between attributive and predicative adjectives staying intact.

In Chapter 4, we consider the interaction of gender features with other structural configurations. Here, we covered a case of gender features on nouns preventing the nouns from occurring in certain structural positions - predicative possession in Tamil. We covered the case of predicative possession in Tamil where gendered nouns were not allowed in that configuration. This was explained as a function of the structural organization of possessives. Predicative possession includes a nominalizer (which is also specified as [-GENDER]). The presence of the non-gendered nominalizer is what precludes gendered nouns from occurring here. This arrangement is a part of the core structure building principles of the grammar, and do not require explicit learning. As such, it is maintained in heritage Tamil.

From these three sub-studies, the main point that we can garner is that gender is not a monolithic category in grammar. Rather than treat gender as a vulnerable category, as has been done in the literature (Polinsky, 2018), this dissertation uncovered layers to its representation. This way, we are able to identify aspects that are part of its core structural representation, and can develop unhindered regardless of the acquisition conditions. In this quest, we have been able to identify the following as part of core syntax and hence remaining stable in heritage Tamil:

(i) Assignment of biological gender in connection to [+/-HUMAN] feature

(ii) Gender agreement expressed via biological gender features

(iii) The incompatibility of gender features on functional heads, and their connection to lexical nouns.
All these are part of core syntax and stay stable in a grammar that is otherwise prone to change.

Another important factor that is characteristic of heritage Tamil syntax is to do with the DP-clause distinction found in the baseline variety. This property emerges clearly from the results of the sub-studies in Chapters 3 and 4. To recap, Tamil adjectives behave differently when inside the DP and when in the predicate. In the former, there is no agreement whereas the latter expresses phi agreement with the subject of the sentence. Possessives are differentiated on a similar basis in Tamil. Attributive possession allows the occurrence of all nouns indiscriminately, whereas predicative possession only allows nouns lacking gender specification. All these properties are explained as structural constraints in the expository parts of Chapters 3 and 4. Empirical data from heritage Tamil gives conclusive evidence for them being maintained in the heritage grammar as well. The clause-DP differential treatment, stemming from core structural properties, is maintained in heritage Tamil.

5.3 Suggestions for the Way Forward

Based on the summary and key takeaways presented in this chapter, this section introduces a few suggestions for future research in this direction. There are two major ways in which this can be executed: conceptually in terms of developing a better understanding of core syntax in heritage grammars, and empirically by increasing the domains where gender can be observed in Tamil. Both are explained in the following subsections.

5.3.1 More Dimensions of Core Syntax

A fundamental premise explored by this dissertation dealt with core syntax and its stability in heritage grammars. Having established this point using agreement systems and derivational constraints, we now have the opportunity to the point further. We can now take structurally more complex configurations, and explore their development in heritage Tamil. This can be done by considering long distance dependencies, movement, etc. D'Alessandro et al. (2025, p. 8) notes that long-distance anaphor binding, and movement with intermediate gaps are particularly useful domains to study in heritage language syntax, as they are computationally complex domains and might reveal insights on how syntax fares in heritage grammars.

tann is the anaphor in Tamil, denoting 'self' (1). Sundaresan (2018) notes that it has certain peculiar properties, deviating from our conventional understanding of lexical anaphors. Sundaresan presents that the anaphor violates locality conditions, preferring local and long-distance antecedents in equal measure.

(1) *raman* ***tann-a*** *kannadi-la paa-tt-aan*
Raman **self-ACC** mirror-LOC see-PST-3MSG
'Raman saw himself in the mirror.'

In (2), the anaphor *tann* is bound to the antecedent *Krishnan* across several other DPs that are structurally closer to the anaphor, at least one of which (namely *Raman*) also readily qualifies as a potential antecedent to it. Being two clauses higher, *Krishnan* is also clearly non-local to the clause containing *tann*.

(2) *[raman [anand-kitta [sita* ***tann-a$_i$*** *kaapa-tti-na-nu] so-nn-aan-nu]*
Raman Anand-LOC sita **self-ACC$_i$** rescue-PST-3FSG-COMP say-PST-3MSG-COMP
krishnan$_i$ paa-tt-aan]
Krishnan$_i$ see-PST-3MSG
'Krishnan$_i$ saw that Ram told Anand that Sita rescued self$_i$'.

Sundaresan (2018) explains (2) as a result of complex structural relations about the binding tendencies of the anaphor *tann*. The specifics of the analysis are not presented here, but it is evident that investigating phenomena such as these in heritage Tamil will go a long way in informing us about how core syntax fares when Tamil is acquired under restricted input conditions.

Studying these would help fortify the idea that core syntactic principles remain stable in heritage grammars. These do not pose a learnability problem to heritage speakers, as they do not require a large amount of input data to be present in the environment.

5.3.2 More Empirical Domains

Another way forward is by increasing the empirical coverage of the study. The primary focus of this dissertation was to understand gender features in core syntax, and we chose three domains to do it. We can delve further into the question of gender by studying more domains where gender marking appears in Tamil.

As noted in Chapter 2, pronouns are gender-marked in Tamil. Thus, one approach to studying gender further is when we look at a wider distribution of where pronouns occur in the grammar. One way to do it is through the domain of binding. The pre-test questionnaire mentioned in Chapter 1 revealed some insights regarding this. (3) and (4) are examples of gendered possessive pronominals produced by heritage Tamil speakers. Both present an interesting divergence from the baseline. In (3), the morphological shape of the pronominal should be [FEM] in accordance with it being bound to the subject 'that girl'. However, it is inflected for [MASC]. The reverse can be observed in (4). It is worth investigating if such inconsistencies are due to binding to a different antecendent ('brother' or 'sister') from the expected one.

(3) *anda poNNu* ***avan*** *tambi-oda vilaya-dar-aa*
that girl **his** brother-COMM play-PRS-FSG
'That girl is playing with **his** brother.' (Intended: that girl is playing with **her** brother)

(4) *anda paiyan* ***ava*** *tangachi-oda vilaya-dar-aan*
that boy **her** sister-COMM play-PRS-MSG
'That boy is playing with **her** sister.' (Intended: that boy is playing with **his** sister)

Such constructions have to be investigated further. They might reveal more about how different elements of a sentence interact with each other in terms of morphological marking of gender, whether there are intervention effects, etc.

Yet another strand of analysis to consider is the variation itself. This dissertation was focused on understanding gender as a broad phenomenon in heritage Tamil, and therefore did not go into details about variation that exists within the grammar. It is possible to do deeper analysis into both inter-speaker and intra-speaker variation to be in a better position to characterise heritage Tamil. It will also help us determine if factors such as self-perceived proficiency, schooling experience, etc. have a correlation with the amount of variation found in a heritage speakers' grammar.

5.4 General Conclusions

This dissertation addressed the issue of grammatical stability and change in heritage grammars. The question was mainly approached by contrasting core syntax with other domains of grammar. Across three domains of gender features in heritage Tamil we were able to record that the core syntactic aspects of gender remained stable while others, such as those involving lexical learning or interface with the discourse domain, undergo variation. The broad implication of these findings is that we can argue for syntax emerging as a resilient grammatical phenomenon despite the presence of reduced/ restricted primary linguistic data and potential transfer effects from other languages in the environment.

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Publications

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Conference Proceedings

Venkatesan, M. **Deriving Adjectives in Heritage Tamil: Stability and Change** in the *Proceedings of GLOW 47. 1-14.*
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Presentations and Talks

Clausal Properties as Determinants of NEG Position: The Case of Odia with Biswanath Dash at the *Nothing But Negation Young Researchers' Conference*, Goethe University, Frankfurt am Main, May 2026

Structural Height of Nominalization: A Cross-Linguistic Comparison of Gerundive Nominals in Three South Asian Languages with Biswanath Dash at the *16th Formal Analysis of South Asian Languages (FASAL-16)*, University of South Carolina (Online), April 2026

Exploring Stability and Change in Heritage Grammars: Derived Adjectives in Heritage Tamil at the *Syntax Interface Lectures*, Utrecht University [Invited Talk], March 2026

Biological Gender, *n*P and the Syntax of Predicative Possession in Tamil at the *GLOW in Asia 2nd Workshop for Young Scholars (GLOW in Asia WYS-2)*, Nanzan University, Nagoya, Japan, March 2026

Core Syntactic Aspects of Gender Show Stability in Heritage Grammars: Evidence from Heritage Tamil in New Delhi at the *Workshop on Migration and Linguistic Heritage: Theoretical Frameworks and Methodological Approaches*, Lausanne University, Switzerland (Online), February 2026

Gender as a Derivational Constraint in Predicative Possession: The Case of Heritage Tamil at the *6th Heritage Language Syntax Conference (HLS-6)*, Boğaziçi University, İstanbul, Türkiye (Online), November 2025

Syntactic Doubling as Innovation in Heritage Tamil with Pritha Chandra at the *6th Heritage*

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Deriving Adjectives in Heritage Tamil: Stability and Change (Poster and Lightning Talk) at the *15th Formal Approaches to South Asian Languages (FASAL-15)*, University of Texas, Austin (Online), April 2025

Deriving Adjectives in Heritage Tamil: Stability and Change (Poster and Lightning Talk) at the *47th Generative Linguistics in the Old World (GLOW-47)*, Goethe University, Frankfurt am Main, March 2025

Attrition of Discourse-based Biological Gender: Evidence from Heritage Tamil at the *15th Students' Conference of Linguistics in India (SCONLI-15)*, University of Delhi, New Delhi, January 2025

Keeping the Syntax of Modification and Predication Distinct: The View from Tamil at the *International Workshop on the Syntax of Predication and Modification*, Nihon University, Tokyo, November 2024

The Syntax of Attributive and Predicative Adjectives in Tamil at the *Western Conference on Linguistics*, California State University, Fresno (Online), November 2024

An Agree-based Analysis of Nominal Agreement: Evidence from Hindi-Urdu at the *14th Formal Approaches to South Asian Languages (FASAL-14)*, Stony Brook University, New York (Online), April 2024

Grammatical Gender in a Tamil-Hindi Bilingual Situation: The Role of Valuation at the *13th (Formal) Approaches to South Asian Languages (FASAL-13)*, University of Michigan, Ann Arbor (Online), April 2023

Point of View: An Important Feature of Reflexivity in Tamil with Abishek Stephen at the *42nd International Conference of the Linguistic Society of India (ICOLSI-42)*, GLA University, Mathura (Online), December 2020